

数学物理方程



3 Bessel 函数

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3.0 为什么要有Bessel函数

物质世界 **三维空间**系统在**时域**内的单向演化→详实描述变量时空演化→必须拓展至高维

PDE向高维拓展的新问题

$$u = u(x, t)$$

↓ 向高维拓展

- ① 时间维度不变→空间维度增多→变量**总数增多**

$$u = u(x, y, t) = u(\rho, \theta, t) = u(\rho, \theta, h) = u(x, y, z)$$

- ② 变量空间分布的表达→强烈依赖于**空间维度的组织形式**→即坐标系的选取

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \Delta u = \frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \theta^2}$$

平面直角系

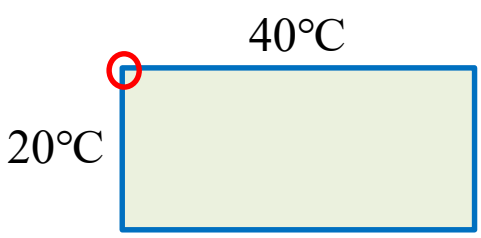
极坐标系

$$\frac{\partial^2 u}{\partial t^2} - a^2 \Delta u = 0$$

$$\frac{\partial^2 u}{\partial t^2} - a^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) = 0$$

$$\frac{\partial^2 u}{\partial t^2} - a^2 \left(\frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \theta^2} \right) = 0$$

- ③ 边界条件可能出现奇点



二维导热

分离变量法能否直接使用？

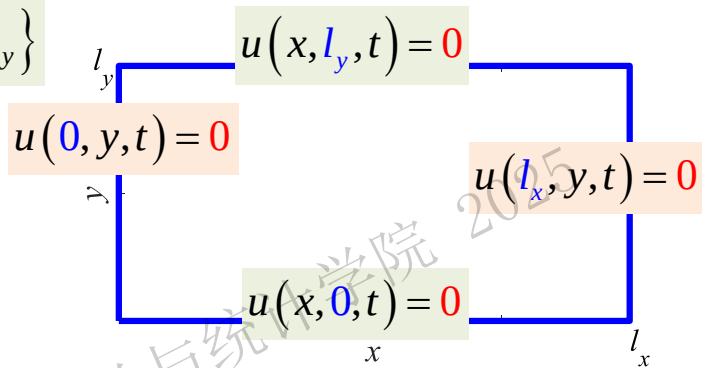
3.0 为什么要有Bessel函数

平面直角系上的波动方程

例1
$$\begin{cases} u_{tt} - a^2(u_{xx} + u_{yy}) = 0, & (x, y) \in \Omega, t > 0 \\ u(0, y, t) = u(l_x, y, t) = 0, & 0 \leq y \leq l_y \\ u(x, 0, t) = u(x, l_y, t) = 0, & 0 \leq x \leq l_x \\ u(x, y, 0) = \varphi(x, y), u_t(x, y, 0) = \psi(x, y), & (x, y) \in \bar{\Omega} \end{cases}$$

波动/二维/齐次	
1齐	1齐
1齐	1齐

$$\Omega = \{(x, y) | 0 < x < l_x, 0 < y < l_y\}$$



解 第1步 时间/空间分离

设 $u(x, y, t) = T(t)w(x, y)$ 代入原PDE

分离变量有 $T''(t)w(x, y) - a^2T(t)\Delta w(x, y) = 0$

$$\frac{T''(t)}{a^2T(t)} = \frac{\Delta w(x, y)}{w(x, y)} = -\mu$$

时空分离常数

初值问题

线性二阶常系数齐次ODE

$$T''(t) + \mu a^2 T(t) = 0$$

边界条件

$$\begin{cases} u(0, y, t) = T(t)w(0, y) = 0 \\ u(l_x, y, t) = T(t)w(l_x, y) = 0 \\ u(x, 0, t) = T(t)w(x, 0) = 0 \\ u(x, l_y, t) = T(t)w(x, l_y) = 0 \end{cases}$$

原PDE齐次边界
由边值问题实现

$T(t)$ 不恒为0

二维边值问题

$$\begin{cases} -\Delta w(x, y) = \mu w(x, y), & (x, y) \in \Omega \\ w|_{\partial\Omega} = 0 \end{cases}$$

$$\text{运算}(\text{元素}_i) = \lambda \cdot \text{元素}$$

二维边值问题=算子 $-\Delta$ 的特征值问题

定义运算

$$A = -\frac{d^2}{dx^2}$$

$$-\Delta = -\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)$$

3.0 为什么要有Bessel函数

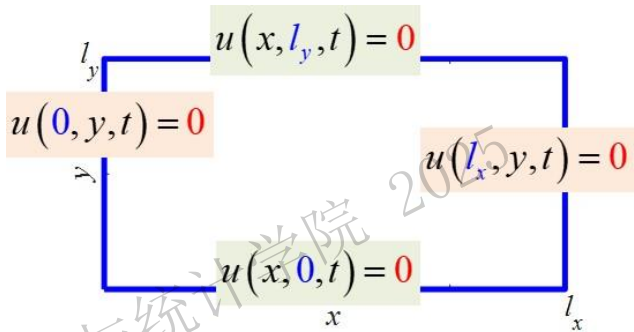
平面直角系上的波动方程

例1

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第2步 空间各维度分离

$$u(x, y, t) = T(t)w(x, y)$$

$$\begin{cases} \Delta w(x, y) + \mu w(x, y) = 0, & (x, y) \in \Omega \\ w|_{\partial\Omega} = 0 \end{cases}$$

设 $w(x, y) = X(x)Y(y)$ 代入二维边值问题, 按直角系展开Laplace算子 Δ

$$X''(x)Y(y) + X(x)Y''(y) + \mu X(x)Y(y) = 0$$

$$-\frac{Y''(y) + \mu Y(y)}{Y(y)} = \frac{X''(x)}{X(x)} = -\lambda$$

空间分离常数

y方向边值问题 $= \lambda_y$

$$\begin{cases} Y''(y) + (\mu - \lambda_x)Y(y) = 0 \\ Y(0) = Y(l_y) = 0 \end{cases}$$

x方向边值问题

特征值问题

$$A_1 = -\frac{d^2}{dx^2}$$

$$\begin{cases} X''(x) + \lambda_x X(x) = 0 \\ X(0) = X(l_x) = 0 \end{cases}$$

$$\lambda_y = \mu - \lambda_x \rightarrow \mu = \lambda_x + \lambda_y$$

直角系:时空分离常数=各维特征值之和= $A_2 = -\Delta$ 的特征值

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平面直角系上的波动方程

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算子 $A = -\frac{d^2}{dx^2}$ 不同齐次边界条件下的特征值/特征函数		
齐次边界类型	特征值问题	特征值/特征函数
[1,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right), (n=1, 2, 3, \dots)$
[1,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \sin\left[\frac{(2n+1)\pi}{2l}x\right], (n=0, 1, 2, 3, \dots)$
[2,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \cos\left[\frac{(2n+1)\pi}{2l}x\right], (n=0, 1, 2, 3, \dots)$
[2,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \cos\left(\frac{n\pi}{l}x\right), (n=0, 1, 2, 3, \dots)$
周期	$\begin{cases} \Phi'(\theta) + \lambda \Phi(\theta) = 0 \\ \Phi(0) = \Phi(2\pi), \\ \Phi'(0) = \Phi'(2\pi) \end{cases}$	$\lambda_0 = 0, \Phi_0(\theta) = 1, (n=0)$ $\lambda_n = n^2, \Phi_n(\theta) = C_1 \cos n\theta + C_2 \sin n\theta, (n=1, 2, 3, \dots)$

第3步 空间维度上的特征值

$w(x, y) = X(x)Y(y)$

$$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l_x) = 0 \end{cases}$$

$$\begin{cases} Y''(y) + (\mu - \lambda)Y(y) = 0 \\ Y(0) = Y(l_y) = 0 \end{cases}$$

$$\lambda_{nx} = \left(\frac{n_x \pi}{l_x}\right)^2, X_{nx}(x) = \sin\left(\frac{n_x \pi}{l_x}x\right), (n_x = 1, 2, 3, \dots)$$

$$\mu - \lambda_{nx} = \left(\frac{n_y \pi}{l_y}\right)^2, Y_{ny}(y) = \sin\left(\frac{n_y \pi}{l_y}y\right), (n_y = 1, 2, 3, \dots)$$

二维特征值函数与各维函数

$$\begin{aligned} -\Delta w(x, y) &= -\Delta[X(x)Y(y)] \\ &= -X''(x)Y(y) - X(x)Y''(y) \\ &= \lambda_{nx} X(x)Y(y) + (\mu - \lambda_{nx}) X(x)Y(y) \\ &= \mu X(x)Y(y) \end{aligned}$$

$$\begin{cases} \Delta w(x, y) + \mu w(x, y) = 0, & (x, y) \in \Omega \\ w|_{\partial\Omega} = 0 \end{cases}$$

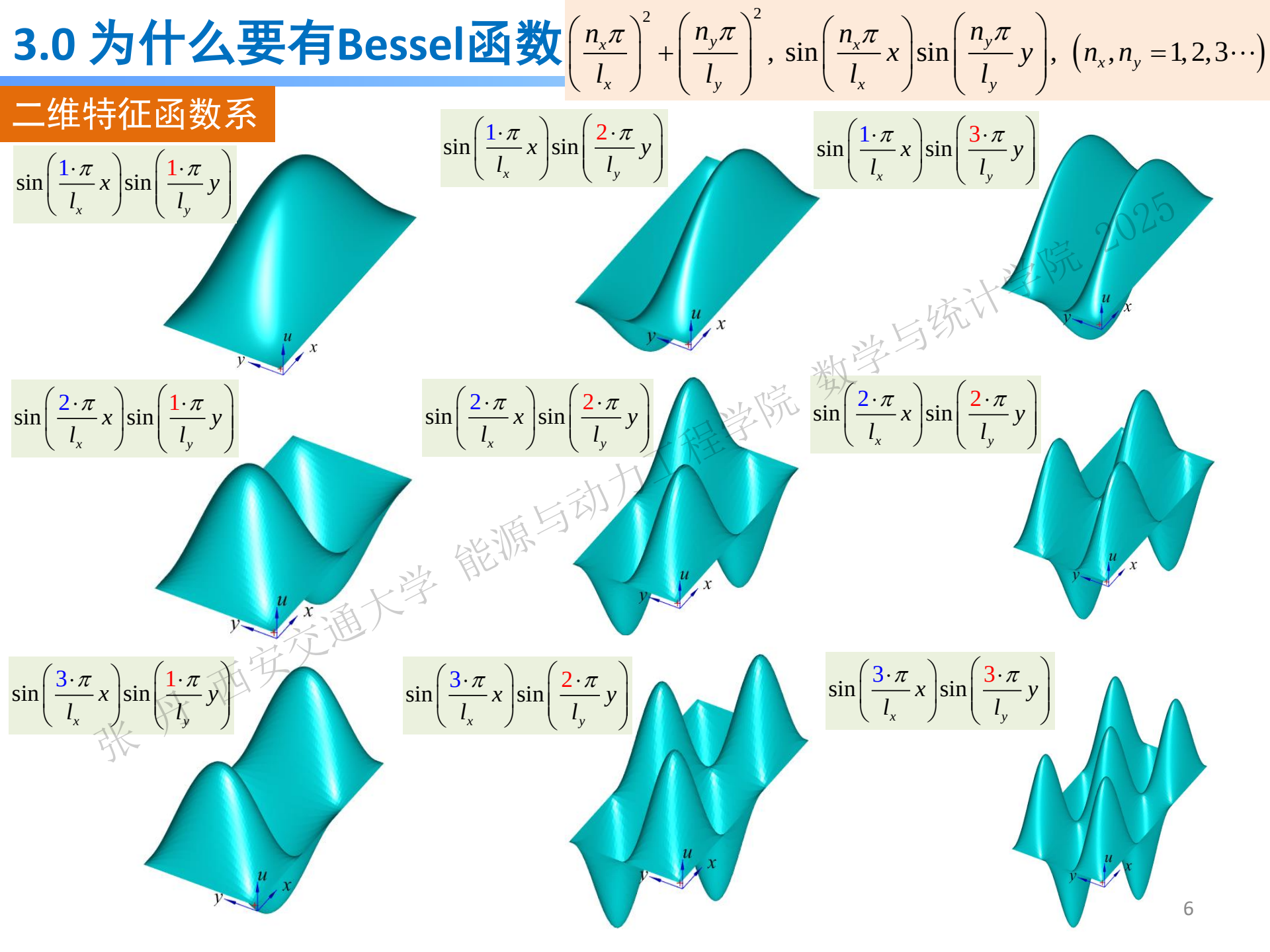
二维边值问题特征值/特征函数

两组序号

$$\mu_{nx,ny} = \lambda_{nx} + \lambda_{ny} = \left(\frac{n_x \pi}{l_x}\right)^2 + \left(\frac{n_y \pi}{l_y}\right)^2, (n_x, n_y = 1, 2, 3, \dots)$$
$$W_{nx,ny}(x, y) = \sin\left(\frac{n_x \pi}{l_x}x\right) \sin\left(\frac{n_y \pi}{l_y}y\right)$$

直角系

二维边值问题特征值=各维度特征值之和
二维边值问题特征函数=各维特征函数之积



3.0 为什么要有Bessel函数

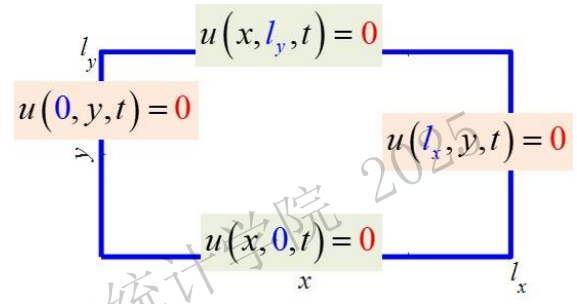
平面直角系上的波动方程

$$\Omega = \{(x, y) | 0 < x < l_x, 0 < y < l_y\}$$

例1

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时空分离常数

第4步 求解初值ODE $T''(t) + \mu_{nx,ny} a^2 T(t) = 0$

通解为 $T_{nx,ny}(t) = C_{1n} \cos(a\sqrt{\mu_{nx,ny}}t) + C_{2n} \sin(a\sqrt{\mu_{nx,ny}}t)$

由叠加原理, 原定解问题的形式解为

$$u(x, y, t) = \sum_{ny=1}^{\infty} \sum_{nx=1}^{\infty} [T_{nx,ny}(t) \cdot W_{nx,ny}(x, y)]$$

序号二重累加

$$= \sum_{ny=1}^{\infty} \sum_{nx=1}^{\infty} [C_{1n} \cos(a\sqrt{\mu_{nx,ny}}t) + C_{2n} \sin(a\sqrt{\mu_{nx,ny}}t)] \cdot \sin\left(\frac{n_x \pi}{l_x} x\right) \sin\left(\frac{n_y \pi}{l_y} y\right)$$

对时间偏导为

$$u_t(x, y, t) = \sum_{ny=1}^{\infty} \sum_{nx=1}^{\infty} [-a\sqrt{\mu_{nx,ny}} C_{1n} \sin(a\sqrt{\mu_{nx,ny}}t) + a\sqrt{\mu_{nx,ny}} C_{2n} \cos(a\sqrt{\mu_{nx,ny}}t)] \cdot \sin\left(\frac{n_x \pi}{l_x} x\right) \sin\left(\frac{n_y \pi}{l_y} y\right)$$

利用形式解实现初始条件→确定系数 C_{1n}, C_{2n}

$$\mu_{nx,ny} = \left(\frac{n_x \pi}{l_x}\right)^2 + \left(\frac{n_y \pi}{l_y}\right)^2, (n_x, n_y = 1, 2, 3 \dots)$$
$$W_{nx,ny}(x, y) = \sin\left(\frac{n_x \pi}{l_x} x\right) \sin\left(\frac{n_y \pi}{l_y} y\right)$$

3.0 为什么要有Bessel函数

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波动/二维/齐次	
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$$\mu_{nx,ny} = \left(\frac{n_y \pi}{l_y}\right)^2 + \left(\frac{n_x \pi}{l_x}\right)^2, (n_y, n_x = 1, 2, 3 \dots)$$

$$W_{nx,ny}(x, y) = \sin\left(\frac{n_x \pi}{l_x} x\right) \sin\left(\frac{n_y \pi}{l_y} y\right)$$

第4步 解初值ODE

$$u(x, y, 0) = \sum_{ny=1}^{\infty} \sum_{nx=1}^{\infty} C_{1n} \cdot \sin\left(\frac{n_x \pi}{l_x} x\right) \sin\left(\frac{n_y \pi}{l_y} y\right) = \varphi(x, y)$$

$$u(x, y, t) = \sum_{ny=1}^{\infty} \sum_{nx=1}^{\infty} \left[C_{1n} \cos(a\sqrt{\mu_{nx,ny}} t) + C_{2n} \sin(a\sqrt{\mu_{nx,ny}} t) \right] \cdot \sin\left(\frac{n_x \pi}{l_x} x\right) \sin\left(\frac{n_y \pi}{l_y} y\right)$$

$$u_t(x, y, t) = \sum_{ny=1}^{\infty} \sum_{nx=1}^{\infty} \left[-a\sqrt{\mu_{nx,ny}} C_{1n} \sin(a\sqrt{\mu_{nx,ny}} t) + a\sqrt{\mu_{nx,ny}} C_{2n} \cos(a\sqrt{\mu_{nx,ny}} t) \right] \cdot \sin\left(\frac{n_x \pi}{l_x} x\right) \sin\left(\frac{n_y \pi}{l_y} y\right)$$

$$u_t(x, y, 0) = \sum_{ny=1}^{\infty} \sum_{nx=1}^{\infty} \left[a\sqrt{\mu_{nx,ny}} C_{2n} \sin\left(\frac{n_x \pi}{l_x} x\right) \sin\left(\frac{n_y \pi}{l_y} y\right) \right] = \psi(x, y)$$

$$C_{1n} = \frac{4}{l_x l_y} \int_0^{l_y} \int_0^{l_x} \varphi(x, y) \sin\left(\frac{n_x \pi}{l_x} x\right) \sin\left(\frac{n_y \pi}{l_y} y\right) dx dy$$

二元函数Fourier正交分解

原定解问题的解
$$C_{2n} = \frac{4}{l_x l_y a \sqrt{\mu_{nx,ny}}} \int_0^{l_y} \int_0^{l_x} \psi(x, y) \sin\left(\frac{n_x \pi}{l_x} x\right) \sin\left(\frac{n_y \pi}{l_y} y\right) dx dy$$

$$u(x, y, t) = \sum_{ny=1}^{\infty} \sum_{nx=1}^{\infty} \left[C_{1n} \cos(a\sqrt{\mu_{nx,ny}} t) + C_{2n} \sin(a\sqrt{\mu_{nx,ny}} t) \right] \cdot \sin\left(\frac{n_x \pi}{l_x} x\right) \sin\left(\frac{n_y \pi}{l_y} y\right)$$

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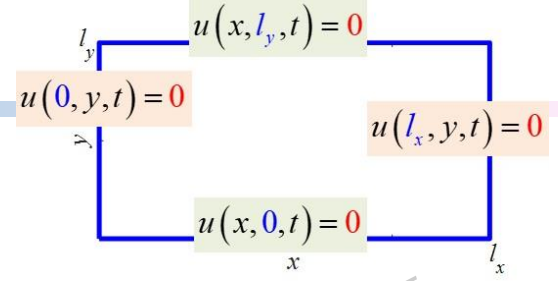
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$$\mu_{n_x, n_y} = \left(\frac{n_y \pi}{l_y}\right)^2 + \left(\frac{n_x \pi}{l_x}\right)^2, \quad (n_x, n_y = 1, 2, 3, \dots)$$
$$W_{n_x, n_y}(x, y) = \sin\left(\frac{n_x \pi}{l_x} x\right) \sin\left(\frac{n_y \pi}{l_y} y\right)$$

讨论

初始位移 $u(x, y, 0) = \varphi(x, y) = Ax(x-l_x)y(y-l_y)$

初始速度 $u_t(x, y, 0) = \psi(x, y) = 0$

由Fourier展开得 $C_{2n} = 0$

$$C_{1n} = \frac{4}{l_x l_y} \int_0^{l_y} \int_0^{l_x} \varphi(x, y) \sin\left(\frac{n_x \pi}{l_x} x\right) \sin\left(\frac{n_y \pi}{l_y} y\right) dx dy$$
$$= \frac{4A}{l_x l_y} \int_0^{l_x} x(x-l_x) \sin\left(\frac{n_x \pi}{l_x} x\right) dx \cdot \int_0^{l_y} y(y-l_y) \sin\left(\frac{n_y \pi}{l_y} y\right) dy$$

1

$$\int_0^{l_x} x(x-l_x) \sin\left(\frac{n_x \pi}{l_x} x\right) dx = \left[-\frac{l_x}{n_x \pi} x^2 \cos \frac{n_x \pi}{l_x} x + \frac{l_x^2}{n_x \pi} x \cos \frac{n_x \pi}{l_x} x + \frac{2l_x^3}{(n_x \pi)^3} \cos \frac{n_x \pi}{l_x} x \right]_0^{l_x} = \frac{2l_x^3}{n_x^3 \pi^3} [(-1)^{n_x} - 1]$$

2

$$\int_0^{l_y} y(y-l_y) \sin\left(\frac{n_y \pi}{l_y} y\right) dy = \left[-\frac{l_y}{n_y \pi} y^2 \cos \frac{n_y \pi}{l_y} y + \frac{l_y^2}{n_y \pi} y \cos \frac{n_y \pi}{l_y} y + \frac{2l_y^3}{(n_y \pi)^3} \cos \frac{n_y \pi}{l_y} y \right]_0^{l_y} = \frac{2l_y^3}{n_y^3 \pi^3} [(-1)^{n_y} - 1]$$

得展开系数

$$C_{1n} = \frac{16A l_x^2 l_y^2}{\pi^6} \cdot \frac{[(-1)^{n_x} - 1][(-1)^{n_y} - 1]}{n_x^3 n_y^3}$$

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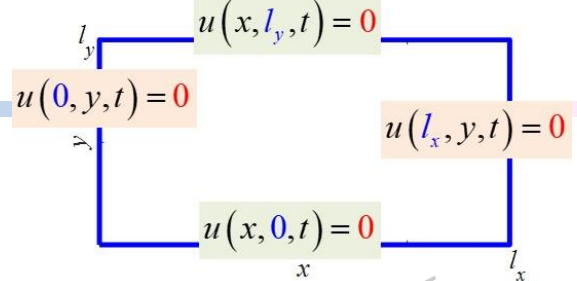
平面直角系上的波动方程

例1

$$u_{tt} - a^2(u_{xx} + u_{yy}) = 0, \quad (x, y) \in \Omega, \quad t > 0$$
$$u(0, y, t) = u(l_x, y, t) = 0, \quad 0 \leq y \leq l_y$$
$$u(x, 0, t) = u(x, l_y, t) = 0, \quad 0 \leq x \leq l_x$$
$$u(x, y, 0) = \varphi(x, y), u_t(x, y, 0) = \psi(x, y), \quad (x, y) \in \bar{\Omega}$$

波动/二维/齐次

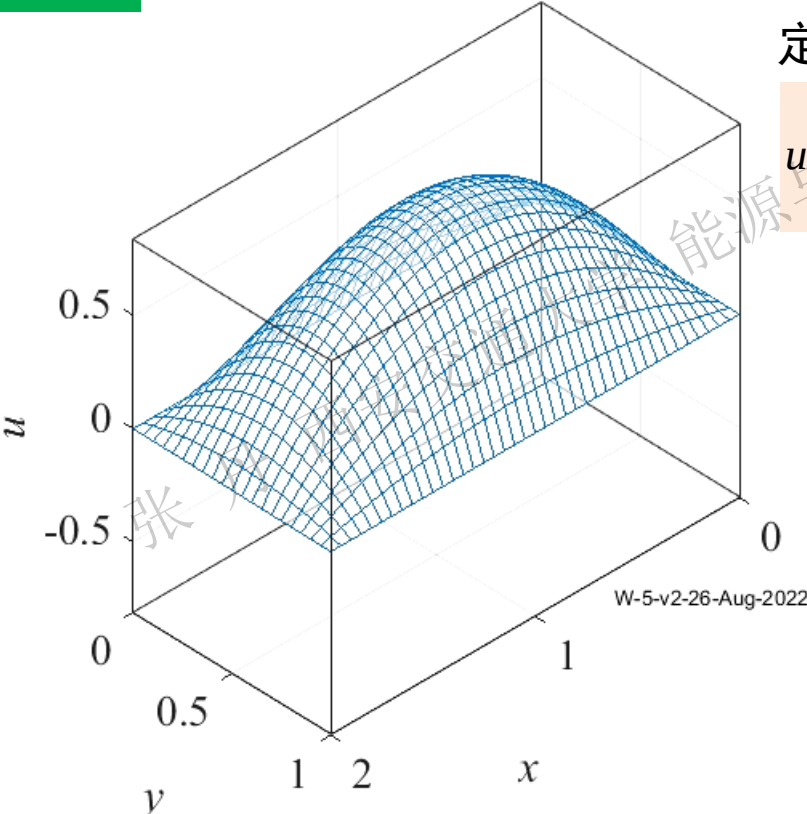
1齐	1齐
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$$\mu_{nx,ny} = \left(\frac{n_x \pi}{l_x}\right)^2 + \left(\frac{n_y \pi}{l_y}\right)^2, \quad (n_x, n_y = 1, 2, 3 \dots)$$
$$W_{nx,ny}(x, y) = \sin\left(\frac{n_x \pi}{l_x} x\right) \sin\left(\frac{n_y \pi}{l_y} y\right)$$

讨论

t=0.00, a=1, A=3



定解问题的解 $\varphi(x, y) = Ax(x-l_x)y(y-l_y), \quad \psi(x, y) = 0$

$$u(x, y, t) = \sum_{ny=1}^{\infty} \sum_{nx=1}^{\infty} C_{1n} \cos(a\sqrt{\mu_{nx,ny}} t) \cdot \sin\left(\frac{n_x \pi}{l_x} x\right) \sin\left(\frac{n_y \pi}{l_y} y\right)$$

$$C_{1n} = \frac{16Al_x^2 l_y^2}{\pi^6} \cdot \frac{[(-1)^{n_x} - 1][(-1)^{n_y} - 1]}{n_x^3 n_y^3}$$

分离变量法圆满解决了定义在平面直角系的二维波动/导热方程定解问题

3.0 为什么要有Bessel函数

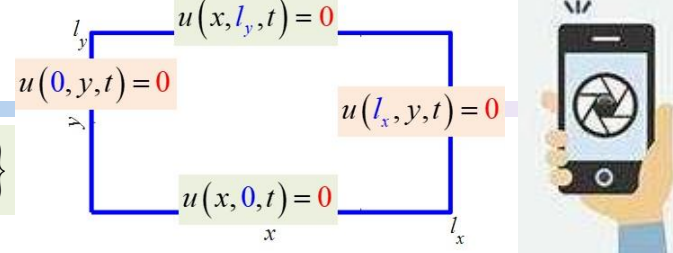
平面直角系上的波动方程

例1

$$\begin{cases} u_{tt} - a^2(u_{xx} + u_{yy}) = 0, & (x, y) \in \Omega, t > 0 \\ u(0, y, t) = u(l_x, y, t) = 0, & 0 \leq y \leq l_y \\ u(x, 0, t) = u(x, l_y, t) = 0, & 0 \leq x \leq l_x \\ u(x, y, 0) = \varphi(x, y), u_t(x, y, 0) = \psi(x, y), & (x, y) \in \bar{\Omega} \end{cases}$$

波动/二维/齐次

1齐	1齐
1齐	1齐



$$\mu_{n_x, n_y} = \left(\frac{n_x \pi}{l_x}\right)^2 + \left(\frac{n_y \pi}{l_y}\right)^2, \quad (n_x, n_y = 1, 2, 3 \dots)$$
$$W_{n_x, n_y}(x, y) = \sin\left(\frac{n_x \pi}{l_x} x\right) \sin\left(\frac{n_y \pi}{l_y} y\right)$$

小结

$u(x, y, t) = T(t)w(x, y) \rightarrow \frac{T''(t)}{a^2 T(t)} = \frac{\Delta w(x, y)}{w(x, y)} = -\mu$

$T''(t)w(x, y) - a^2 T(t) \Delta w(x, y) = 0$

初值问题 $T''(t) + \mu a^2 T(t) = 0$

时空分离常数

二维边值问题

$$\begin{cases} \Delta w(x, y) + \mu w(x, y) = 0, & (x, y) \in \Omega \\ w|_{\partial\Omega} = 0 \end{cases}$$

空间分离常数

$$w(x, y) = X(x)Y(y) \rightarrow \frac{X''(x)}{X(x)} = -\frac{Y''(y) + \mu Y(y)}{Y(y)} = -\lambda$$
$$X''(x)Y(y) + X(x)Y''(y) + \mu X(x)Y(y) = 0$$

- 1. 两级分离→分离变量法可用
- 2. 时空分离常数 μ ; 空间分离常数 λ
- 3. 二维特征值由2个维度联合确定
- 4. 二次分离后的2个边值问题/特征值问题都是常系数ODE

x方向边值问题

y方向边值问题

$$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l_x) = 0 \end{cases} \quad \begin{matrix} \text{特征值} \\ \text{问题} \end{matrix}$$

$$\begin{cases} Y''(y) + (\mu - \lambda)Y(y) = 0 \\ Y(0) = Y(l_y) = 0 \end{cases}$$

二维空间用直角系描述→分离变量法求解没有障碍

3.0 为什么要有Bessel函数 → 表达径向Bessel方程的解



极坐标上的导热方程

$$\frac{\partial u}{\partial t} - a^2 \Delta u = 0 \quad u(\rho, \theta, t) = T(t)w(\rho, \theta)$$

$$T'(t)w(\rho, \theta) - a^2 T(t) \Delta w(\rho, \theta) = 0$$

时空分离常数 ② 尚未确定

$$\frac{T'(t)}{a^2 T(t)} = \frac{\Delta w(\rho, \theta)}{w(\rho, \theta)} = -\mu$$

二维边值问题/特征值问题 $\lambda_2 = -\Delta$

初值问题

$$T'(t) + \mu a^2 T(t) = 0$$
$$\begin{cases} -\Delta w(\rho, \theta) = \mu w(\rho, \theta), (\rho, \theta) \in \Omega \\ w|_{\partial\Omega} = 0 \end{cases}$$

$$w(\rho, \theta) = R(\rho)\Phi(\theta)$$
$$\Delta w = \frac{\partial^2 w}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial w}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 w}{\partial \theta^2}$$

$$R''(\rho)\Phi(\theta) + \frac{1}{\rho} R'(\rho)\Phi(\theta) + \frac{1}{\rho^2} R(\rho)\Phi''(\theta) = -\mu R(\rho)\Phi(\theta)$$

空间分离常数

$$\frac{\Phi''(\theta)}{\Phi(\theta)} = -\frac{\mu R(\rho) + R''(\rho) + \frac{1}{\rho} R'(\rho)}{\frac{1}{\rho^2} R(\rho)} = -\lambda$$

周向 径向

$$\lambda_n = n^2, \quad \Phi_n(\theta) = \{1, \cos n\theta, \sin n\theta\}$$

$(n = 0, 1, 2, 3, \dots)$

相当于原方程是Poisson方程

$$\begin{cases} \Phi''(\theta) + \lambda \Phi(\theta) = 0 \\ \Phi(\theta) = \Phi(\theta + 2\pi) \end{cases} \quad \begin{cases} |R(0)| < +\infty, R(\rho_0) = 0 \\ \rho^2 R''(\rho) + \rho R'(\rho) + (\mu \rho^2 - \lambda) R(\rho) = 0 \end{cases}$$

Euler方程 $\mu = 0$

$$\rho^2 R''(\rho) + \rho R'(\rho) - n^2 R(\rho) = 0$$

Bessel方程 $\mu > 0$ 1824

$$\rho^2 R''(\rho) + \rho R'(\rho) + (\mu \rho^2 - n^2) R(\rho) = 0$$

线性二阶变系数齐次ODE ① 如何求解



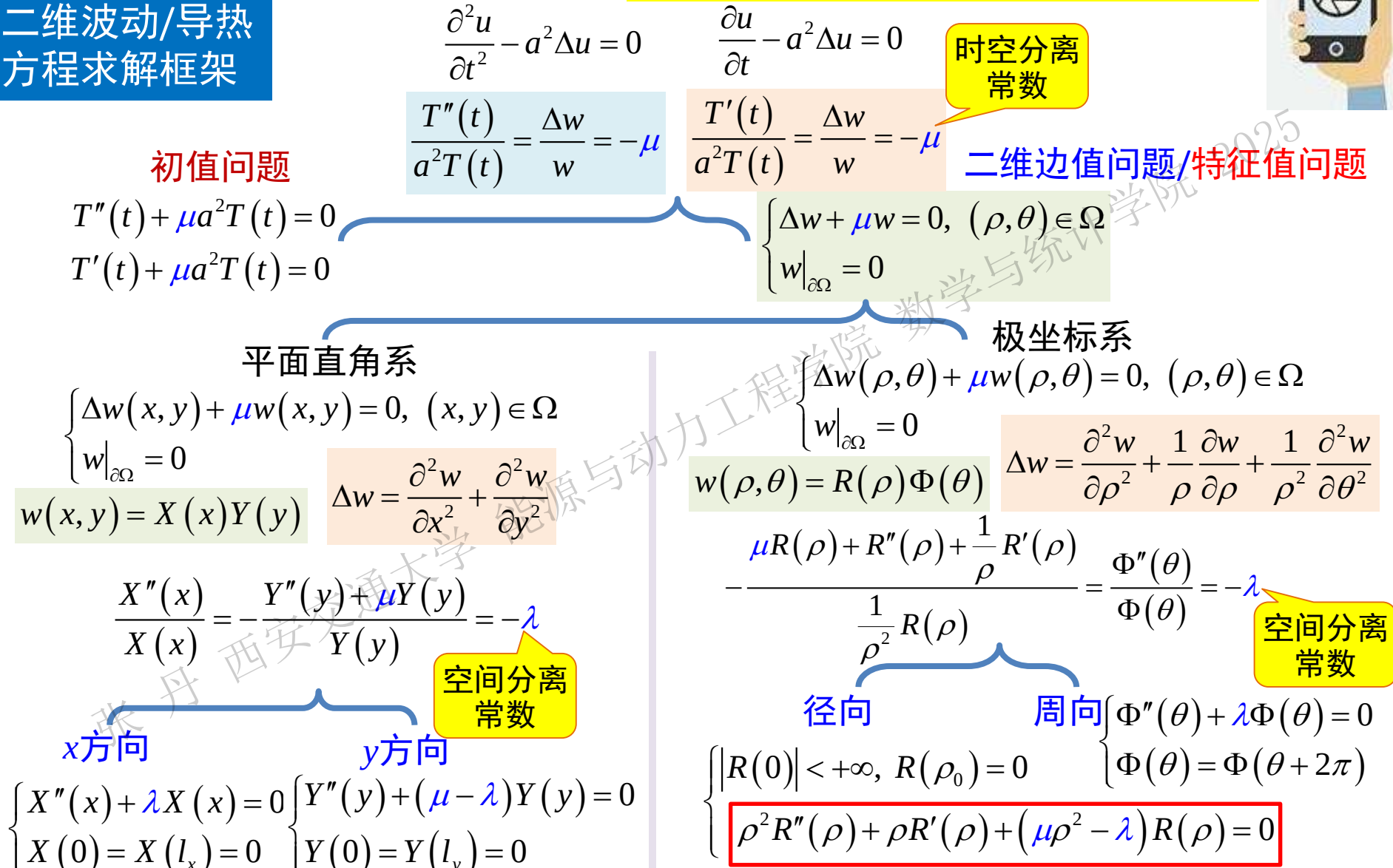
F.W.Bessel
1784-1846
German Math.
Astronomer



3.0 为什么要有Bessel函数

求解Bessel方程是求解极坐标下波动/导热方程/柱系下Poisson方程的技术瓶颈

二维波动/导热方程求解框架



本章重点解决Bessel方程

1. 数学建模和基本原理介绍

2. 分离变量法

3. Bessel函数

4. 积分变换法

5. Green函数法

6. 特征线法

7. Legendre多项式

3.1 二阶线性常微分方程的幂级数解法

3.2 Bessel函数

3.3 多个自变量分离变量法举例

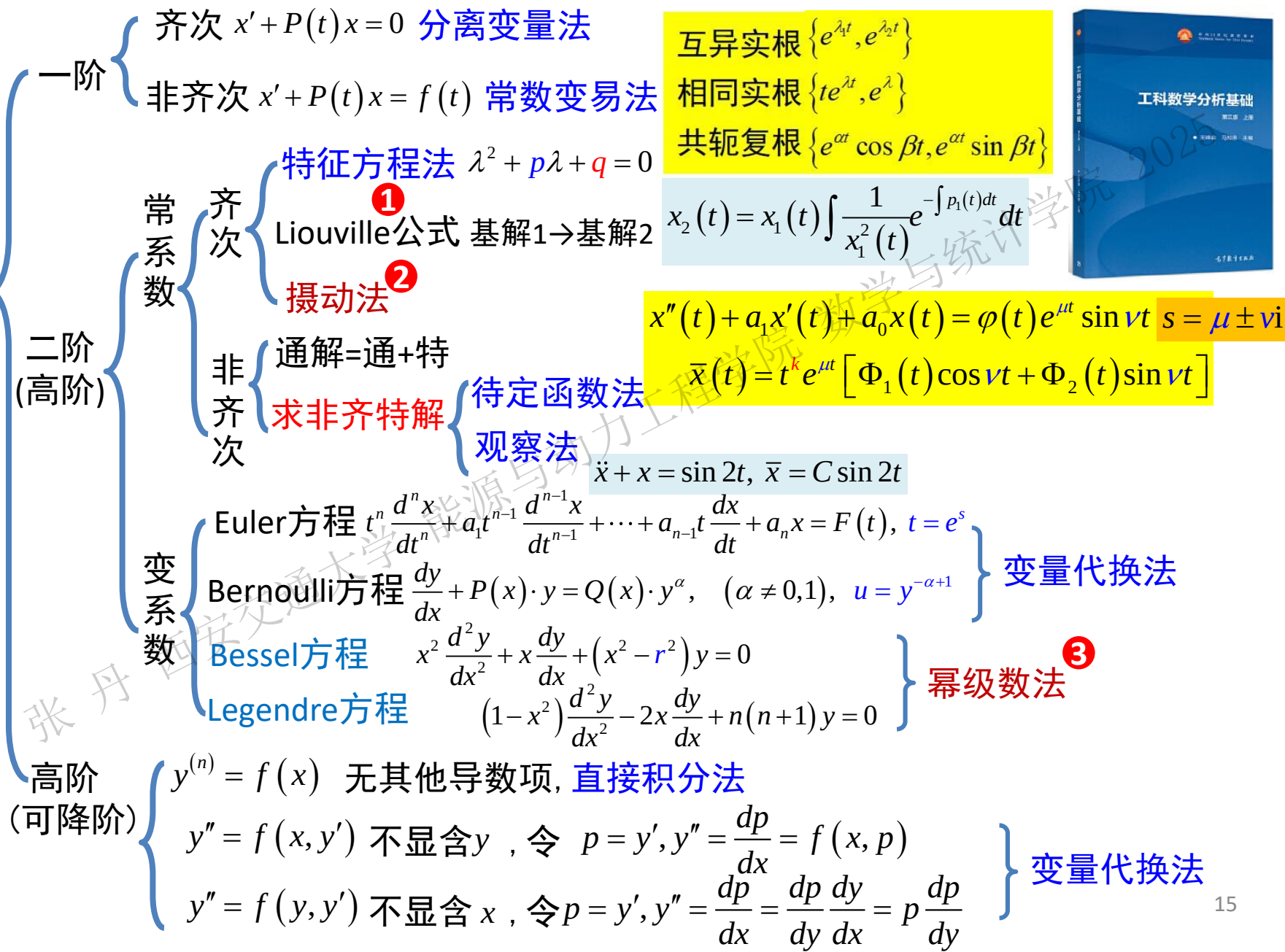
常系数二阶线性ODE
的基解组

变系数二阶线性ODE
的幂级数解法

线性ODE的解法

$$x^{<n>} + P_1 x^{<n-1>} + P_2 x^{<n-2>} + \cdots + P_{n-1} x^{<1>} + P_n x^{<0>} = F$$

线性常微分方程解法



3.1 二阶线性ODE幂级数解法

常系数二阶线性ODE的基解组

例1 用特征方程法求解下列齐次ODE

(1) $y'' - 3y' + 2y = 0$

解 二阶线性常系数ODE

特征方程 $\lambda^2 - 3\lambda + 2 = 0$

特征根 $\lambda_1 = 1, \lambda_2 = 2$

互异实根, 基解组 $\{e^x, e^{2x}\}$

通解 $y = C_1 e^x + C_2 e^{2x}$

(2) $y'' + 4y' + 13y = 0$

解 二阶线性常系数ODE

特征方程 $\lambda^2 + 4\lambda + 13 = 0$

特征根 $\lambda_{1,2} = -2 \pm 3i$

共轭复根, 基解组 $\{e^{-2x} \cos 3x, e^{-2x} \sin 3x\}$

通解 $y = C_1 e^{-2x} \cos 3x + C_2 e^{-2x} \sin 3x$

$$x''(t) + p x'(t) + q x(t) = 0$$

$$\lambda^2 + p \lambda + q = 0$$

- 互异实根 $\{e^{\lambda_1 t}, e^{\lambda_2 t}\}$
- 相同实根 $\{te^{\lambda t}, e^{\lambda t}\}$
- 共轭复根 $\{e^{\alpha t} \cos \beta t, e^{\alpha t} \sin \beta t\}$

(3) $y'' + 4y' + 4y = 0$

解法1 特征方程法 二阶线性常系数ODE

特征方程 $\lambda^2 + 4\lambda + 4 = 0$

特征根 $\lambda_{1,2} = -2$

相同实根, 基解组 $\{e^{-2x}, xe^{-2x}\}$

通解 $y = C_1 e^{-2x} + C_2 x e^{-2x}$

二阶齐次ODE求解 → 寻找2个线性无关基解

$$C_1 f_1(x) + C_2 f_2(x) = 0$$

线性无关: 仅有 C_1, C_2 同时为0才成立



3.1 二阶线性ODE幂级数解法

常系数二阶线性ODE的基解组

(3) $y'' + 4y' + 4y = 0$

解法2 摄动法 设原方程常系数有微小变化

摄动方程 $y'' + (4 + \varepsilon)y' + (4 + 2\varepsilon)y = 0$

特征方程 $\lambda^2 + (4 + \varepsilon)\lambda + (4 + 2\varepsilon) = 0$

特征根 $\lambda_1 = -2, \lambda_2 = -2 - \varepsilon$ 基解组 $\{e^{-2x}, e^{-2x - \varepsilon x}\}$

基解的线性组合仍然是解

$$y_\varepsilon = \frac{e^{-2x - \varepsilon x} - e^{-2x}}{-\varepsilon} = e^{-2x} \frac{e^{-\varepsilon x} - 1}{-\varepsilon}$$

当 $\varepsilon \rightarrow 0$ 时, 摄动方程趋近于原方程

$$\begin{aligned} y &= \lim_{\varepsilon \rightarrow 0} y_\varepsilon = \lim_{\varepsilon \rightarrow 0} e^{-2x} \frac{e^{-\varepsilon x} - 1}{-\varepsilon} \\ &= \lim_{\varepsilon \rightarrow 0} e^{-2x} \frac{-xe^{-\varepsilon x}}{-1} = xe^{-2x} \end{aligned}$$

L'Hospital 法则
对 ε 求导

第1类Bessel函数 $\rightarrow$ 摄动法 $\rightarrow$ 第2类Bessel函数

摄动方程中为什么取 $4y \rightarrow (4 + 2\varepsilon)y$

摄动方程当 $\varepsilon \rightarrow 0$ 时, 不仅两个方程要形似, 还需保持原方程性态 $\rightarrow$ 二阶ODE方程性态 $\rightarrow$ 取决于特征方程判别式

原方程 $y'' + 4y' + 4y = 0$

特征方程 $s^2 + 4s + 4 = 0$

判别式 $\Delta = 4^2 - 4 \cdot 4 = 0$ 相同实根

摄动方程 $y'' + (4 + \varepsilon)y' + (4 + \alpha\varepsilon)y = 0$

特征方程 $s^2 + (4 + \varepsilon)s + (4 + \alpha\varepsilon) = 0$

判别式 $\begin{aligned} \Delta_\varepsilon &= (4 + \varepsilon)^2 - 4 \cdot (4 + \alpha\varepsilon) \\ &= 4^2 + 8\varepsilon + \varepsilon^2 - 4^2 - 4\alpha\varepsilon \\ &= 8\varepsilon + \varepsilon^2 - 4\alpha\varepsilon \end{aligned}$

欲使 $\lim_{\varepsilon \rightarrow 0} \Delta_\varepsilon = \lim_{\varepsilon \rightarrow 0} (8\varepsilon + \varepsilon^2 - 4\alpha\varepsilon) = 0 = \Delta$

需 $\alpha = 2$

$$x''(t) + p(t)x'(t) + q(t)x(t) = 0$$

$$x_2(t) = x_1(t) \int \frac{1}{x_1^2(t)} e^{-\int p(t) dt} dt$$

3.1 二阶线性ODE幂级数解法

Liouville公式

二阶常/变系数齐次ODE

代入原ODE

$$x''(t) + p(t)x'(t) + q(t)x(t) = 0$$

二阶系数为1

求解的关键是确定2个基解 $\{x_1(t), x_2(t)\}$

如何根据 基解1 $\rightarrow$ 基解2 ?

假设已知1个基解 $x_1(t)$

另一个基解 $x_2(t)$ 与 $x_1(t)$ 线性无关

$$\frac{x_2(t)}{x_1(t)} = \varphi(t) \neq C \quad x_2(t) = \varphi(t)x_1(t)$$

各阶导数 $x_2'(t) = \varphi'(t)x_1(t) + \varphi(t)x_1'(t)$

$$\begin{aligned} x_2''(t) &= \varphi''(t)x_1(t) + \varphi'(t)x_1'(t) \\ &\quad + \varphi'(t)x_1'(t) + \varphi(t)x_1''(t) \\ &= \varphi''(t)x_1(t) + 2\varphi'(t)x_1'(t) + \varphi(t)x_1''(t) \end{aligned}$$

$$x_2''(t) + p(t)x_2'(t) + q(t)x_2(t) = 0$$

$$\begin{aligned} &\varphi''(t)x_1(t) + 2\varphi'(t)x_1'(t) + \varphi(t)x_1''(t) \\ &\quad + p(t)[\varphi'(t)x_1(t) + \varphi(t)x_1'(t)] \\ &\quad + q(t)\varphi(t)x_1(t) = 0 \end{aligned}$$

$$\varphi(t)[x_1''(t) + p(t)x_1'(t) + q(t)x_1(t)] = 0$$

$$+ \varphi''(t)x_1(t) + \varphi'(t)[2x_1'(t) + p(t)x_1(t)] = 0$$

$$\varphi''(t)x_1(t) + \varphi'(t)[2x_1'(t) + p(t)x_1(t)] = 0$$

分离变量法 $\frac{\varphi''(t)}{\varphi'(t)} = -2\frac{x_1'(t)}{x_1(t)} - p(t)$

$$\varphi'(t) = \frac{1}{x_1^2(t)} \int e^{-\int p(t) dt} dt \quad \rightarrow \quad \varphi(t) = \int \frac{1}{x_1^2(t)} \int e^{-\int p(t) dt} dt$$

另一线性无关基解 $x_2(t) = x_1(t) \int \frac{1}{x_1^2(t)} e^{-\int p(t) dt} dt$



J. Liouville
1809-1882
French Math.

3.1 二阶线性ODE幂级数解法

常系数二阶线性ODE的基解组

(3) $y'' + 4y' + 4y = 0$

解法3 Liouville公式

二阶变系数ODE $y''(x) + p(x)y'(x) + q(x)y(x) = 0$

本例已知一个基解为 $y_1(x) = e^{-2x}$

$$\begin{aligned} y_2(x) &= y_1(x) \int \frac{1}{y_1^2(x)} e^{-\int p(x) dx} dx \\ &= e^{-2x} \int \frac{1}{(e^{-2x})^2} e^{-\int 4 dx} dx \quad \curvearrowright \quad p(x) = 4 \\ &= e^{-2x} \int e^{4x} e^{-4x+C} dx \\ &= e^{-2x} \int e^C dx \\ &= e^{-2x} (e^C x + C_2) \\ &= e^C x e^{-2x} + C_2 e^{-2x} \\ &= C_1 x e^{-2x} + C_2 e^{-2x} \end{aligned}$$

于是原方程另1个线性无关的基解为

$$y_2(x) = x e^{-2x}$$

$$\begin{aligned} x''(t) + p(t)x'(t) + q(t)x(t) &= 0 \\ x_2(t) &= x_1(t) \int \frac{1}{x_1^2(t)} e^{-\int p(t) dt} dt \end{aligned}$$



J. Liouville
1809-1882
French Math.

3.1 二阶线性ODE幂级数解法

常系数二阶线性ODE的基解组

例2 用Liouville公式解Euler方程

$$t^2 \frac{d^2 x}{dt^2} + t \frac{dx}{dt} + a_n x = 0 \quad (a_n < 0)$$

解 原方程变型为 $\frac{d^2 x}{dt^2} + \frac{1}{t} \frac{dx}{dt} + \frac{a_n}{t^2} x = 0$ 基解 $\{t^{\sqrt{-a_n}}, t^{-\sqrt{-a_n}}\} \quad (a_n < 0)$

$$\begin{aligned} x_2(t) &= x_1(t) \int \frac{1}{x_1^2(t)} e^{-\int p(t) dt} dt \\ &= t^{\sqrt{-a_n}} \int \frac{1}{t^{2\sqrt{-a_n}}} e^{-\int \frac{1}{t} dt} dt = t^{\sqrt{-a_n}} \int \frac{1}{t^{2\sqrt{-a_n}}} e^{-\ln t + C_1} dt \\ &= t^{\sqrt{-a_n}} \int \frac{1}{t^{2\sqrt{-a_n}}} \frac{C_1}{t} dt = t^{\sqrt{-a_n}} C_1 \int t^{-(1+2\sqrt{-a_n})} dt \\ &= t^{\sqrt{-a_n}} \left[C_1 \frac{t^{-(1+2\sqrt{-a_n})+1}}{-(1+2\sqrt{-a_n})+1} + C_2 \right] = t^{\sqrt{-a_n}} C_1 \frac{1}{-2\sqrt{-a_n}} t^{-2\sqrt{-a_n}} + C_2 t^{\sqrt{-a_n}} \\ &= C_1 t^{-\sqrt{-a_n}} + C_2 t^{\sqrt{-a_n}} \end{aligned}$$

吸收至 C_1 中

于是原方程另1个线性无关的基解为

$$x_2(t) = t^{-\sqrt{-a_n}}$$



J. Liouville
1809-1882
French Math.



L. Euler
1707-1783
Swiss Math.
Physical Scientist

1. 数学建模和基本原理介绍

2. 分离变量法

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5. Green函数法

6. 特征线法

7. Legendre多项式

3.1 二阶线性常微分方程的幂级数解法

3.2 Bessel函数

3.3 多个自变量分离变量法举例

常系数二阶线性ODE
的基解组

变系数二阶线性ODE
的幂级数解法

3.1 二阶线性ODE幂级数解法

$$y''(x) + p(x)y'(x) + q(x)y(x) = 0$$

变系数二阶线性ODE的幂级数解法

1. 基解组仍为2个线性无关的解→基解一般求解困难
2. 广义待定函数法→方程两边用幂级数展开→比较/确定系数→得级数形式解
3. 关键: 幂级数的选择/收敛性

幂级数解存在定理1

对二阶ODE $y''(x) + p(x)y'(x) + q(x)y(x) = 0$, 若 $p(x)$, $q(x)$ 在邻域 $\{x \in \mathbb{R} \mid |x - x_0| < \delta\}$ 内解析, 即可展开为Taylor级数, 则原方程在该邻域内存在幂级数解 $y(x) = \sum_{k=0}^{\infty} a_k (x - x_0)^k$ 其中系数 a_k 可由待定系数法求出

Taylor级数

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \frac{f^{<3>}(x_0)}{3!}(x - x_0)^3 + \cdots + \frac{f^{<n>}(x_0)}{n!}(x - x_0)^n + \cdots$$

推论 若 $p(x)$, $q(x)$ 在 $x=0$ 可展开, 收敛半径无穷大→幂级数解可定义为

$$y(x) = \sum_{k=0}^{\infty} a_k x^k$$

序号/幂次一致
k从0开始

3.1 二阶线性ODE幂级数解法

$$y''(x) + p(x)y'(x) + q(x)y(x) = 0$$

例1 幂级数法同样适于常系数 $y'(x) - ay(x) = 0$ 分离变量法 $y(x) = Ce^{ax} = y(0)e^{ax}$

解 幂级数法 变系数 $p(x) = -a, q(x) = 0$ 在实数域解析 $\rightarrow$ 有解

初始条件

幂级数展开 多项式求导

$$y(x) = \sum_{k=0}^{\infty} b_k x^k$$

$$y(x) =$$
$$+ b_k x^k$$
$$+ b_{k-1} x^{k-1}$$
$$+ b_{k-2} x^{k-2}$$
$$+ \dots$$
$$+ b_3 x^3$$
$$+ b_2 x^2$$
$$+ b_1 x^1$$
$$+ b_0 x^0$$

$b_0 = x(0)$ 是初始条件/是已知的

对应幂次系数相等

$$ay(x) = y'(x)$$
$$a \cdot b_{k-1} x^{k-1} = k \cdot b_k x^{k-1}$$
$$a \cdot b_{k-2} x^{k-2} = (k-1) \cdot b_{k-1} x^{k-2}$$
$$a \cdot b_{k-3} x^{k-3} = (k-2) \cdot b_{k-2} x^{k-3}$$
$$\dots$$
$$a \cdot b_2 x^2 = 3 \cdot b_3 x^2$$
$$a \cdot b_1 x^1 = 2 \cdot b_2 x^1$$
$$a \cdot b_0 x^0 = 1 \cdot b_1 x^0$$

得到递推式

$$b_k = \frac{a}{k} \cdot b_{k-1}$$
$$b_{k-1} = \frac{a}{(k-1)} \cdot b_{k-2}$$
$$b_{k-2} = \frac{a}{(k-2)} b_{k-3}$$
$$\dots$$
$$b_3 = \frac{a}{3} \cdot b_2$$
$$b_2 = \frac{a}{2} \cdot b_1$$
$$b_1 = \frac{a}{1} \cdot b_0$$

逐级回代

$$y(x) = y(0)e^{ax} = y(0) \sum_{k=0}^{\infty} \frac{(ax)^k}{k!} = b_0 \sum_{k=0}^{\infty} \frac{a^k}{k!} \cdot x^k = \sum_{k=0}^{\infty} b_k x^k$$
$$b_k = \frac{a}{k} \cdot \frac{a}{(k-1)} \cdot \frac{a}{(k-2)} \dots \frac{a}{3} \cdot \frac{a}{2} \cdot \frac{a}{1} \cdot b_0 = \frac{a^k}{k!} \cdot b_0$$
$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} = 1 + x + \frac{x^2}{2!} + \dots + \frac{x^k}{k!} + \dots$$
$$y(0) = b_0$$

3.1 二阶线性ODE幂级数解法

$$y''(x) + p(x)y'(x) + q(x)y(x) = 0$$

初始条件

例1 幂级数法同样适于常系数 $x' - ax = 0$

分离变量法 $x(t) = Ce^{at} = x(0)e^{at}$

一阶齐次ODE幂级数解法的意义

幂级数法 $x(t) = \sum_{k=0}^{\infty} b_k t^k = x(0) \sum_{k=0}^{\infty} \frac{a^k}{k!} \cdot t^k$

线性高阶非齐次ODE $y^{(n)} + a_{n-1}y^{(n-1)} + \dots + a_1\dot{y} + a_0y = u$

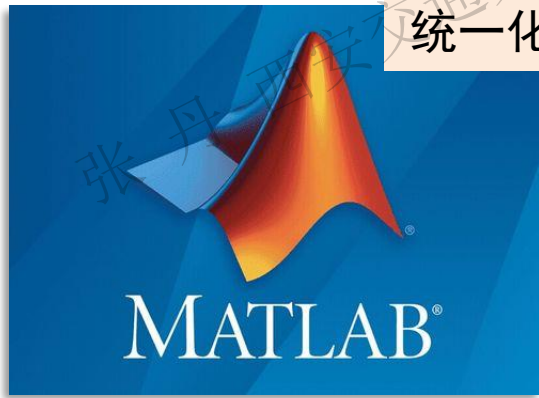
变量代换 $\begin{cases} x_1 = y \\ x_2 = \dot{y} \\ \vdots \\ x_n = y^{(n-1)} \end{cases}$

改造为线性ODE组

写成矩阵形式 $\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = x_3 \\ \vdots \\ \dot{x}_{n-1} = x_n \\ \dot{x}_n = -a_0x_1 - \dots - a_{n-1}x_n + u \end{cases} \Rightarrow \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \vdots \\ \dot{x}_{n-1} \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \mathbf{A} & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ -a_0 & -a_1 & -a_2 & \dots & -a_{n-1} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} u$

高阶ODE求解 统一化/程序化

$x = \sum_{k=0}^{\infty} b_k t^k$ 齐次部分



$y = x_1 \leftarrow x(t) = \sum_{k=0}^{\infty} \frac{A^k}{k!} x(0) t^k = x(0) \sum_{k=0}^{\infty} \frac{A^k}{k!} t^k = e^{At} x(0)$

$x(t) = \sum_{k=0}^{\infty} b_k t^k = x(0) \sum_{k=0}^{\infty} \frac{a^k}{k!} \cdot t^k$

$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} = 1 + x + \frac{x^2}{2!} + \dots + \frac{x^k}{k!} + \dots$

3.1 二阶线性ODE幂级数解法

$$y''(x) + p(x)y'(x) + q(x)y(x) = 0$$

变系数二阶线性ODE的幂级数解法

例2 用幂级数法解变系数ODE $y'' + xy' + y = 0$

解 变系数 $p(x) = x, q(x) = 1$ 在实数域解析
存在幂级数解

$$y(x) = \sum_{k=0}^{\infty} a_k x^k = a_0 + a_1 x + a_2 x^2 + \dots + a_k x^k + \dots$$

$$y'(x) = a_1 + 2a_2 x^1 + \dots + a_k k x^{k-1} + \dots = \sum_{k=1}^{\infty} a_k k x^{k-1}$$

起点已变

$$y''(x) = 2a_2 + \dots + a_k k(k-1) x^{k-2} + \dots = \sum_{k=2}^{\infty} a_k k(k-1) x^{k-2}$$

起点已变

代入原方程

$$\sum_{k=2}^{\infty} a_k k(k-1) x^{k-2} + x \sum_{k=1}^{\infty} a_k k x^{k-1} + \sum_{k=0}^{\infty} a_k x^k = 0$$

$$\sum_{k=0}^{\infty} a_{k+2} (k+2)(k+1) x^k + \sum_{k=0}^{\infty} a_k k x^k + \sum_{k=0}^{\infty} a_k x^k = 0$$

变量代换,
从0累加

首项为0, 累加
直接改自0起

序号/幂次同时调整一致, 同次幂系数相等

$$a_{k+2} (k+2)(k+1) + a_k (k+1) = 0, \quad (k \geq 0)$$

得到递推式 $a_{k+2} = -\frac{1}{k+2} a_k, \quad (k \geq 0)$

系数间隔传递, 分奇/偶讨论

$$a_{2k} = \frac{(-1)^k}{(2k)!!} a_0, \quad a_{2k+1} = \frac{(-1)^k}{(2k+1)!!} a_1, \quad (k \geq 0)$$

半阶乘

$$(2k)!! = 2k \cdot (2k-2) \cdot (2k-4) \cdot \dots \cdot 2$$

$$(2k+1)!! = (2k+1) \cdot (2k-1) \cdot (2k-3) \cdot \dots \cdot 1$$

半阶乘 双叹号不是阶乘两次

$(2k)!!$ 不大于 $(2k)$ 的偶数乘积

$(2k+1)!!$ 不大于 $(2k+1)$ 的奇数乘积

例如 $(6)!! = 2 \cdot 4 \cdot 6 = 48$

$$(7)!! = 1 \cdot 3 \cdot 5 \cdot 7 = 105$$

$$(2k+1)!! (2k)!! = (2k+1)!$$

原方程的解

$$y(x) = a_0 \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!!} x^{2k} + a_1 \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!!} x^{2k+1}$$

本例幂级数方程序号/幂次可同时调整一致

3.1 二阶线性ODE幂级数解法

$$y''(x) + p(x)y'(x) + q(x)y(x) = 0$$

变系数二阶线性ODE的幂级数解法

例3 用幂级数法解ODE $y'' + x^2y' = 0$

解 易知 a_0 是原方程的解, 找另1个线性无关解
因 $p(x) = x^2$ 在实数域解析 $\rightarrow$ 存在幂级数解

$$y(x) = \sum_{n=0}^{\infty} a_n x^n, (a_0 \neq 0)$$

$$y'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}, y''(x) = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}$$
$$\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} + x^2 \sum_{n=2}^{\infty} n a_n x^{n-1} = 0$$
$$\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} + \sum_{n=1}^{\infty} n a_n x^{n+1} = 0$$

幂次/起点难以调整一致

n	列表	
n = 1		$a_1 x^2$
n = 2	$2 \cdot 1 \cdot a_2 x^0$	$2 a_2 x^3$
n = 3	$3 \cdot 2 \cdot a_3 x^1$	$3 a_3 x^4$
n = 4	$4 \cdot 3 \cdot a_4 x^2$	$4 a_4 x^5$
n = 5	$5 \cdot 4 \cdot a_5 x^3$	$5 a_5 x^6$
n = 6	$6 \cdot 5 \cdot a_6 x^4$	
n = 7		

同次幂系数为0

x^0	$2 \cdot 1 \cdot a_2 = 0$
x^1	$3 \cdot 2 \cdot a_3 = 0$
x^2	$4 \cdot 3 \cdot a_4 + a_1 = 0$
x^3	$5 \cdot 4 \cdot a_5 + 2 a_2 = 0$
x^4	$6 \cdot 5 \cdot a_6 + 3 a_3 = 0$
x^5	$7 \cdot 6 \cdot a_7 + 4 a_4 = 0$

解得

$$a_2 = 0$$
$$a_3 = 0$$

$$a_4 = -\frac{1}{4 \cdot 3} a_1 = \frac{(-1)^1}{3^1 \cdot (3 \cdot 1 + 1) \cdot 1!} a_1$$

$$a_5 = -\frac{2}{5 \cdot 4} a_2 = 0$$

$$a_6 = -\frac{3}{6 \cdot 5} a_3 = 0$$

$$a_7 = (-1)^2 \frac{4}{7 \cdot 6} \frac{1}{4 \cdot 3} a_1 = (-1)^2 \frac{1}{3^2 \cdot (3 \cdot 2 + 1) \cdot 2!} a_1$$

只有能递推至 a_1 的系数不为0

系数隔3传递 $\dots \rightarrow a_7 \rightarrow a_4 \rightarrow a_1$

用连续下标 k 表示

$$a_{3k+1} = (-1)^k \frac{1}{3^k \cdot (3 \cdot k + 1) \cdot k!} a_1$$

原ODE解为

a_1 是 x^1 的系数

$$y(x) = a_0 + a_1 \left[x + \sum_{k=1}^{\infty} \frac{(-1)^k}{3^k \cdot (3k+1) \cdot k!} x^{3k+1} \right]$$

- 1. 本例幂级数方程序号/幂次难以同时调整一致 $\rightarrow$ 列表计算系数
- 2. 注意验证常数项是否为解

3.1 二阶线性ODE幂级数解法

$$y''(x) + p(x)y'(x) + q(x)y(x) = 0$$

变系数二阶线性ODE的幂级数解法

$$x''(t) + p(t)x'(t) + q(t)x(t) = 0$$
$$x_2(t) = x_1(t) \int \frac{1}{x_1^2(t)} e^{-\int p(t)dt} dt$$

例3 解ODE $y'' + x^2 y' = 0$

解法2 Liouville公式 易知 a_0 是原方程的解, 找另1个线性无关解

$$x_2(t) = x_1(t) \int \frac{1}{x_1^2(t)} e^{-\int p(t)dt} dt = a_0 \int \frac{1}{a_0^2} e^{-\int x^2 dx} dx$$

$$= C_1 \int e^{-\int x^2 dx} dx$$

$$= C_1 \int e^{-\frac{x^3}{3}} dx$$

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} = 1 + x + \frac{x^2}{2!} + \dots + \frac{x^k}{k!} + \dots$$

$$= C_1 \int \left[\sum_{k=0}^{\infty} \frac{1}{k!} \left(-\frac{x^3}{3} \right)^k \right] dx$$

$$= C_1 \sum_{k=0}^{\infty} \int \frac{(-1)^k}{3^k k!} x^{3k} dx$$

累加/积分交换次序

$$= C_2 + C_1 \sum_{k=0}^{\infty} \frac{(-1)^k x^{3k+1}}{3^k (3k+1) k!}$$

另1个基解

$$y(x) = a_0 + a_1 \left[x + \sum_{k=1}^{\infty} \frac{(-1)^k}{3^k \cdot (3k+1) \cdot k!} x^{3k+1} \right]$$

与解法1相同



J. Liouville
1809-1882
French Math.

变系数二阶线性ODE的幂级数解法

小结

- 幂级数法基本步骤
- 1. 判定解存在→给出幂级数解的基本形式→序号不一定从0开始
 - 2. 幂级数解求导/展开 $p(x), q(x)$ →代入原方程→根据需要调整序号起点/幂次
 - 3. 按照 x 同次幂合并同类项→得出幂级数解系数的递推公式
 - 4. 系数向 a_0, a_1 化简→得基解组/原方程通解→ a_0, a_1 是幂级数前两项系数/通解常数
 - 5. 注意验证常数项 a_0 是否为解

幂级数解存在定理2

拓展

求解Bessel方程的理论支撑

对二阶ODE $x''(t) + p(x)x'(t) + q(x)x(t) = 0$, 若 $(x-x_0)p(x), (x-x_0)^2q(x)$ 在 x_0 的邻域 $\{x \in R \mid |x-x_0| < \delta\}$ 内解析, 即 x_0 最多为 $p(x), q(x)$ 的一阶和二阶极点, 则原方程在该去心邻域 $\{x \in R \mid 0 < |x-x_0| < \delta\}$ 内存在幂级数解

$$y(x) = (x-x_0)^\rho \sum_{k=0}^\infty a_k (x-x_0)^k = \sum_{k=0}^\infty a_k (x-x_0)^{k+\rho}$$

其中常数 $\rho, a_k (a_0 \neq 0)$ 可由待定系数法求出

$$p(x) = \frac{A(x)}{x-x_0}$$

一阶极点

$$q(x) = \frac{B(x)}{(x-x_0)^2}$$

二阶极点

相当于调整了序号起点

序号/幂次可独立调整
序号不一定从0开始

3.1 二阶线性ODE幂级数解法

$$y''(x) + p(x)y'(x) + q(x)y(x) = 0$$

变系数二阶线性ODE的幂级数解法

例4 用幂级数法解Euler方程 $x^2 y'' + xy' - \alpha^2 y = 0, (\alpha > 0)$

解 方程变型 $y'' + \frac{1}{x}y' - \frac{\alpha^2}{x^2}y = 0$

因存在 $x \cdot p(x) = x \cdot \frac{1}{x} = 1$, $-x^2 \cdot q(x) = x^2 \cdot \frac{\alpha^2}{x^2} = -1$ 在实数域内解析, 原方程有幂级数解存在

设解为 $y(x) = x^\rho \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} a_n x^{n+\rho}, (a_0 \neq 0)$ $\rightarrow$ 各阶导数 $y'(x) = \sum_{n=0}^{\infty} (n+\rho) a_n x^{n+\rho-1}$
 $y''(x) = \sum_{n=0}^{\infty} (n+\rho)(n+\rho-1) a_n x^{n+\rho-2}$

代入原方程

$$y'' + \frac{1}{x}y' - \frac{\alpha^2}{x^2}y = 0$$

$$\sum_{n=0}^{\infty} (n+\rho)(n+\rho-1) a_n x^{n+\rho-2} + \frac{1}{x} \sum_{n=0}^{\infty} (n+\rho) a_n x^{n+\rho-1} - \frac{\alpha^2}{x^2} \sum_{n=0}^{\infty} a_n x^{n+\rho} = 0$$

$$\sum_{n=0}^{\infty} (n+\rho)(n+\rho-1) a_n x^{n+\rho-2} + \sum_{n=0}^{\infty} (n+\rho) a_n x^{n+\rho-2} - \alpha^2 \sum_{n=0}^{\infty} a_n x^{n+\rho-2} = 0$$

$$(n+\rho)(n+\rho-1) + (n+\rho) - \alpha^2 = 0$$

$$(n+\rho)^2 = \alpha^2$$

$$n+\rho = \pm \alpha$$

$$y(x) = \sum_{n=0}^{\infty} a_n x^{n+\rho} = \sum_{n=0}^{\infty} a_{n1} x^{+\alpha} + \sum_{n=0}^{\infty} a_{n2} x^{-\alpha}$$
$$= C_1 x^{+\alpha} + C_2 x^{-\alpha}$$

3.1 二阶线性ODE幂级数解法

$y''(x) + p(x)y'(x) + q(x)y(x) = 0$

变系数二阶线性ODE的幂级数解法

例5

用幂级数法求方程 $y'' - 2xy' - 4y = 0$ 满足初始条件 $y(0) = 0, y'(0) = 1$ 的解

解 因 $p(x) = -2x$, $q(x) = -4$ 在实数域内解析, 原方程有幂级数解存在

设解为 $y(x) = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x^1 + a_2 x^2 + \cdots + a_n x^n + \cdots$

由初始条件 $y(0) = a_0 = 0$ $a_1 = 1$

$y'(0) = a_1 + a_2 \cdot 0^2 + \cdots + a_n \cdot 0^n + \cdots = 1$

各阶导数 $y'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}$ 代入原方程

$y''(x) = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}$

$$\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} - 2x \sum_{n=1}^{\infty} n a_n x^{n-1} - 4 \sum_{n=0}^{\infty} a_n x^n = 0$$
$$\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} - 2 \sum_{n=1}^{\infty} n a_n x^n - 4 \sum_{n=0}^{\infty} a_n x^n = 0$$
$$\sum_{n=0}^{\infty} (n+2)(n+2-1) a_{n+2} x^n - \sum_{n=0}^{\infty} 2n a_n x^n - \sum_{n=0}^{\infty} 4a_n x^n = 0$$
$$(n+2)(n+1) a_{n+2} - 2(n+2) a_n = 0$$

变量代换
调整序号

换成连续下标 k 只需考虑奇数项 偶数项系数均为0 $a_{n+2} = \frac{2}{n+1} a_n$

原方程的解

$$a_{2k+1} = \frac{2}{2k} \cdot a_{2k-1} = \frac{2}{2k} \cdot \frac{2}{2(k-1)} a_{2k-3} = \frac{2}{2k} \cdot \frac{2}{2(k-1)} \cdots \frac{2}{2} a_1 = \frac{1}{k!} 1$$

$$y(x) = \sum_{k=0}^{\infty} \frac{a_1}{k!} x^{2k+1} = 1 \cdot x^1 + \frac{x^3}{2!} + \frac{x^5}{3!} + \cdots + \frac{x^{2k+1}}{k!} + \cdots = x \left[1 + \frac{1}{2!} \cdot x^2 + \frac{1}{3!} \cdot (x^2)^2 + \cdots + \frac{1}{k!} \cdot (x^2)^k + \cdots \right] = x e^{x^2}$$

1. 数学建模和基本原理介绍

2. 分离变量法

3. Bessel函数

4. 积分变换法

5. Green函数法

6. 特征线法

7. Legendre多项式

3.1 二阶线性常微分方程的幂级数解法

3.2 Bessel函数

3.3 多个自变量分离变量法举例

Gamma函数

Bessel方程和Bessel函数

Bessel函数的性质

Bessel方程的特征值问题

圆域上Laplace算子的特征值问题

3.2 Bessel函数

$$\Gamma(\alpha) = \int_0^{\infty} x^{\alpha-1} e^{-x} dx$$

Gamma函数

当 $\alpha > 0$ 时, 广义积分 $\int_0^{\infty} x^{\alpha-1} e^{-x} dx$ 收敛

→可定义以 α 为自变量的函数→ Γ 函数

→难以用初等函数表示→超越函数

自变量
是 α

$$\Gamma(\alpha) = \int_0^{\infty} x^{\alpha-1} e^{-x} dx = \int_0^{\infty} \frac{x^{\alpha}}{x e^x} dx, \quad (\alpha > 0)$$

Γ 函数的基本性质

(1) 几个基本值 $\Gamma(1) = 1, \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$

证 1

$$\Gamma(1) = \int_0^{\infty} x^{1-1} e^{-x} dx = \int_0^{\infty} e^{-x} dx = -e^{-x} \Big|_0^{\infty} = 1$$

得证

如何研究超越函数

1. 紧抓定义式→可否利用递推拓展定义域?
2. 特殊点是否有简单表达式?
3. 是否有递推式?
4. 和差积商初等计算规则
5. 微/积分计算规则

3.2 Bessel函数

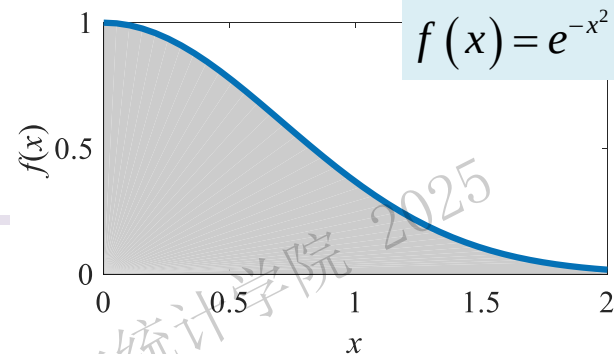
$$\Gamma(\alpha) = \int_0^{\infty} x^{\alpha-1} e^{-x} dx$$

Gamma函数

(1) 几个基本值

$$\Gamma(1) = 1$$

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$



证2 先证明预备等式 $I = \int_0^{\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$

$I = \int_0^{\infty} e^{-x^2} dx$ 不易积分 $\rightarrow$ 先积分其平方

定义域为半径 ∞ 扇域 $\{(x, y) | x \geq 0, y \geq 0, x^2 + y^2 \leq \rho^2 \rightarrow \infty\}$

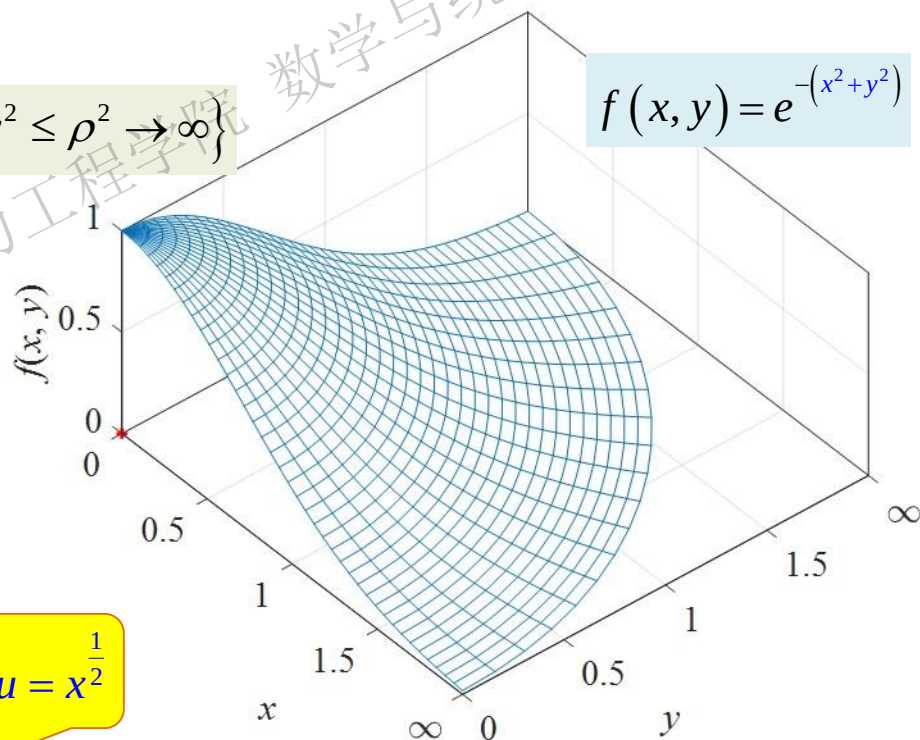
$$\begin{aligned} I^2 &= \int_0^{\infty} e^{-x^2} dx \cdot \int_0^{\infty} e^{-y^2} dy = \int_0^{\infty} e^{-x^2} dx \cdot \int_0^{\infty} e^{-y^2} dy \\ &= \int_0^{\infty} \int_0^{\infty} e^{-(x^2+y^2)} dx dy = \lim_{\rho \rightarrow \infty} \int_0^{\frac{\pi}{2}} d\theta \int_0^{\rho} e^{-\rho^2} \rho d\rho \\ &= \frac{\pi}{4} \lim_{\rho \rightarrow \infty} \left(-e^{-\rho^2} \right) \Big|_0^{\rho} = \frac{\pi}{4} \end{aligned}$$

扇域

即 $I^2 = \frac{\pi}{4} \rightarrow I = \frac{\sqrt{\pi}}{2}$

变量代换 $u = x^2$

$$\Gamma\left(\frac{1}{2}\right) = \int_0^{\infty} x^{\frac{1}{2}-1} e^{-x} dx = \int_0^{\infty} x^{-\frac{1}{2}} e^{-x} dx = 2 \int_0^{\infty} e^{-x} d\left(x^{\frac{1}{2}}\right) = 2 \int_0^{\infty} e^{-u^2} du = 2 \cdot I = 2 \cdot \frac{\sqrt{\pi}}{2} = \sqrt{\pi}$$



$$\Gamma(\alpha) = \int_0^{\infty} x^{\alpha-1} e^{-x} dx$$

3.2 Bessel函数

Gamma函数

(2) 递推公式 $\Gamma(\alpha+1) = \alpha \cdot \Gamma(\alpha), (\alpha > 0)$

分部积分

$$\begin{aligned} \text{证 } \Gamma(\alpha+1) &= \int_0^{\infty} x^{\alpha+1-1} e^{-x} dx = \int_0^{\infty} x^{\alpha} e^{-x} dx = \int_0^{\infty} x^{\alpha} d(-e^{-x}) \\ &= \underbrace{x^{\alpha}(-e^{-x})}\bigg|_0^{\infty} - \alpha \int_0^{\infty} x^{\alpha-1} (-e^{-x}) dx = 0 + \alpha \int_0^{\infty} x^{\alpha-1} e^{-x} dx = \alpha \cdot \Gamma(\alpha) \end{aligned}$$

=0

推论 若 α 为正整数 n $\Gamma(n+1) = n!, \Gamma\left(n + \frac{1}{2}\right) = \frac{(2n-1)!!}{2^n} \sqrt{\pi}$

证 $\Gamma(n+1) = n \cdot \Gamma(n)$

$$\begin{aligned} &= n \cdot (n-1) \cdot \Gamma(n-1) \\ &= n \cdot (n-1) \cdot (n-2) \cdot \Gamma(n-2) \\ &= \dots \\ &= n \cdot (n-1) \cdot (n-2) \cdot \dots \cdot \underbrace{1 \cdot \Gamma(1)}_{=1} \\ &= n! \end{aligned}$$

推论 $\Gamma(2) = \Gamma(1+1) = 1 \cdot \Gamma(1) = 1$

证

奇数

半阶乘

$$\begin{aligned} \Gamma\left(n + \frac{1}{2}\right) &= \Gamma\left(n - \frac{1}{2} + 1\right) = \left(n - \frac{1}{2}\right) \Gamma\left(n - \frac{1}{2}\right) \\ &= \frac{2n-1}{2} \Gamma\left(n - \frac{1}{2}\right) = \frac{2n-1}{2} \cdot \frac{2n-3}{2} \Gamma\left(n - \frac{3}{2}\right) \\ &= \frac{2n-1}{2} \cdot \frac{2n-3}{2} \cdot \frac{2n-5}{2} \Gamma\left(n - \frac{5}{2}\right) \\ &= \frac{2n-1}{2} \cdot \frac{2n-3}{2} \cdot \dots \cdot \frac{2n-(2n-1)}{2} \Gamma\left(\frac{1}{2}\right) \\ &= \frac{(2n-1)!!}{2^n} \Gamma\left(\frac{1}{2}\right) = \frac{(2n-1)!!}{2^n} \sqrt{\pi} \end{aligned}$$

1. Γ 函数是阶乘由整数 $\rightarrow$ 实数域内的推广
2. 两个推论为定积分的求解拓展了思路

$$\Gamma(\alpha) = \int_0^{\infty} x^{\alpha-1} e^{-x} dx$$

3.2 Bessel函数

Gamma函数

原始定义 当 $\alpha > 0$ 时, 广义积分 $\int_0^{\infty} x^{\alpha-1} e^{-x} dx$ 收敛 $\rightarrow$ 定义为 $\Gamma(\alpha)$ 函数

(3) 负实数域延拓

① 当 $\alpha > 0$ 时 $\Gamma(\alpha)$ 可按原广义积分定义计算

$$\Gamma(\alpha) = \int_0^{\infty} x^{\alpha-1} e^{-x} dx$$

$$\Gamma(\alpha+1) = \alpha \cdot \Gamma(\alpha) \quad \rightarrow \quad \Gamma(\alpha) = \frac{\Gamma(\alpha+1)}{\alpha}$$

② 当 $-1 < \alpha < 0$ 时 $0 < \alpha+1 < 1$, $\Gamma(\alpha+1)$ 可按原广义积分定义计算 $\rightarrow$ 递推式变型计算 $\Gamma(\alpha)$

例

$$\Gamma(-0.5) = \frac{\Gamma(-0.5+1)}{-0.5} = \frac{\Gamma(0.5)}{-0.5} = -2\sqrt{\pi}$$

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

③ 当 $-2 < \alpha < -1$ 时, $-1 < \alpha+1 < 0$, $\Gamma(\alpha+1)$ 已按上述递推定义 $\rightarrow$ 再次利用递推式计算 $\Gamma(\alpha)$

例

$$\Gamma(-1.5) = \frac{\Gamma(-1.5+1)}{-1.5} = \frac{\Gamma(-0.5)}{-1.5} = \frac{4\sqrt{\pi}}{3}$$

④ 如此类推 $\rightarrow$ 可将定义域推广至除0/负整数外的所有实数域

⑤ 规定0/负整数处 $\Gamma(\alpha) = \infty$ ($\alpha = 0, Z^-$)

不同定义域下计算汇总

$$\Gamma(\alpha) = \begin{cases} \int_0^{\infty} x^{\alpha-1} e^{-x} dx = \int_0^{\infty} \frac{x^{\alpha}}{x e^x} dx, & (\alpha > 0) \\ \frac{\Gamma(\alpha+1)}{\alpha}, & (\alpha < 0, \alpha \neq Z^-) \\ \infty, & (\alpha = 0, Z^-) \end{cases}$$

$$\Gamma(\alpha) = \int_0^{\infty} x^{\alpha-1} e^{-x} dx$$

3.2 Bessel函数

Gamma函数 图线

$$\Gamma(\alpha) = \begin{cases} \int_0^{\infty} x^{\alpha-1} e^{-x} dx = \int_0^{\infty} \frac{x^{\alpha}}{xe^x} dx, & (\alpha > 0) \\ \frac{\Gamma(\alpha+1)}{\alpha}, & (\alpha < 0, \alpha \neq Z^-) \\ \infty, & (\alpha = 0, Z^-) \end{cases}$$

推论 ① $\Gamma(\alpha) \neq 0$

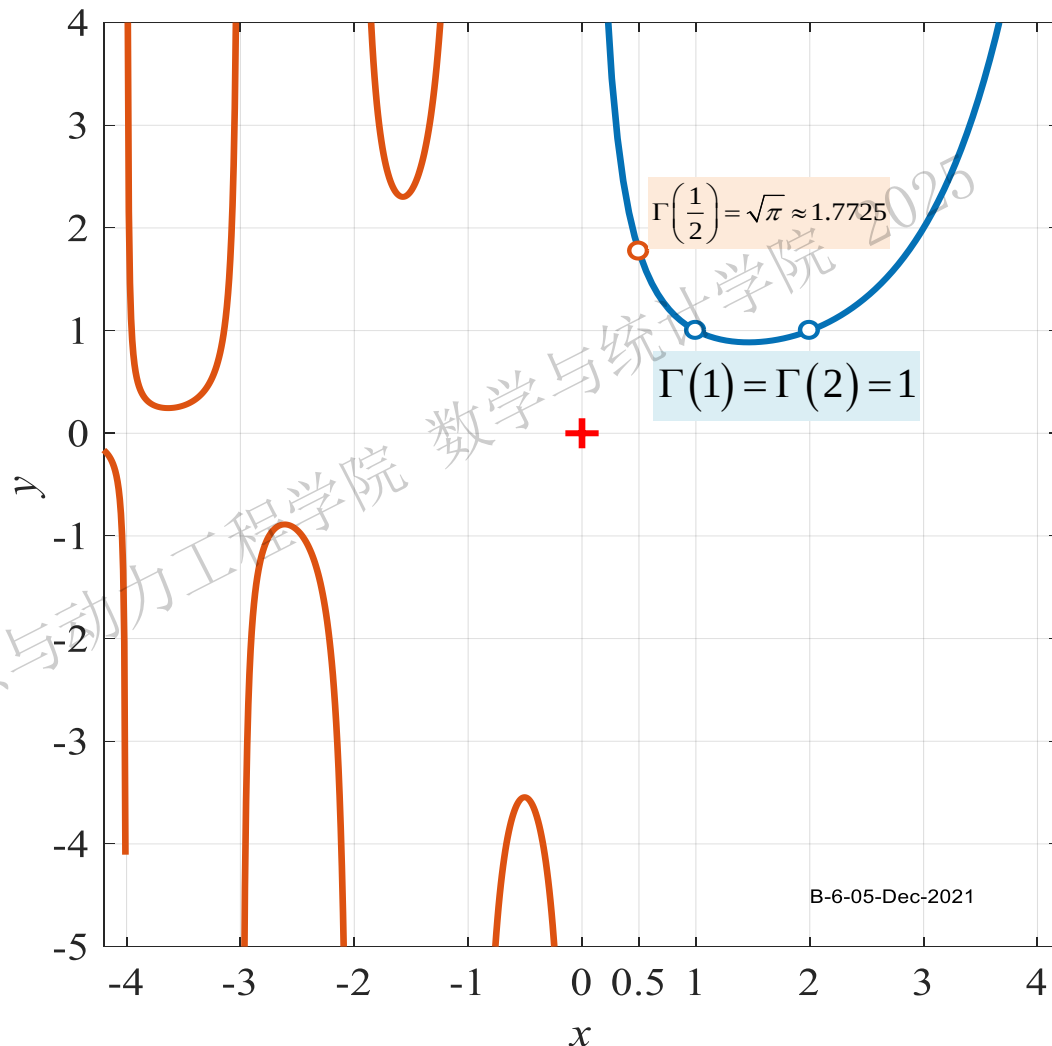
② $\Gamma(\alpha+1) = \alpha \cdot \Gamma(\alpha)$

$$\frac{1}{\alpha} = \frac{\Gamma(\alpha)}{\Gamma(\alpha+1)}$$

超越函数之
比为实数

$$2 = \frac{1}{\frac{1}{2}} = \frac{\Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{1}{2}+1\right)} = \frac{\sqrt{\pi}}{\sqrt{\pi}/2} = 2$$

Bessel函数化简发挥重要作用



B-6-05-Dec-2021

3.2 Bessel函数

$$\Gamma(\alpha) = \int_0^{\infty} x^{\alpha-1} e^{-x} dx$$



Gamma函数

小结

Γ 函数常用计算

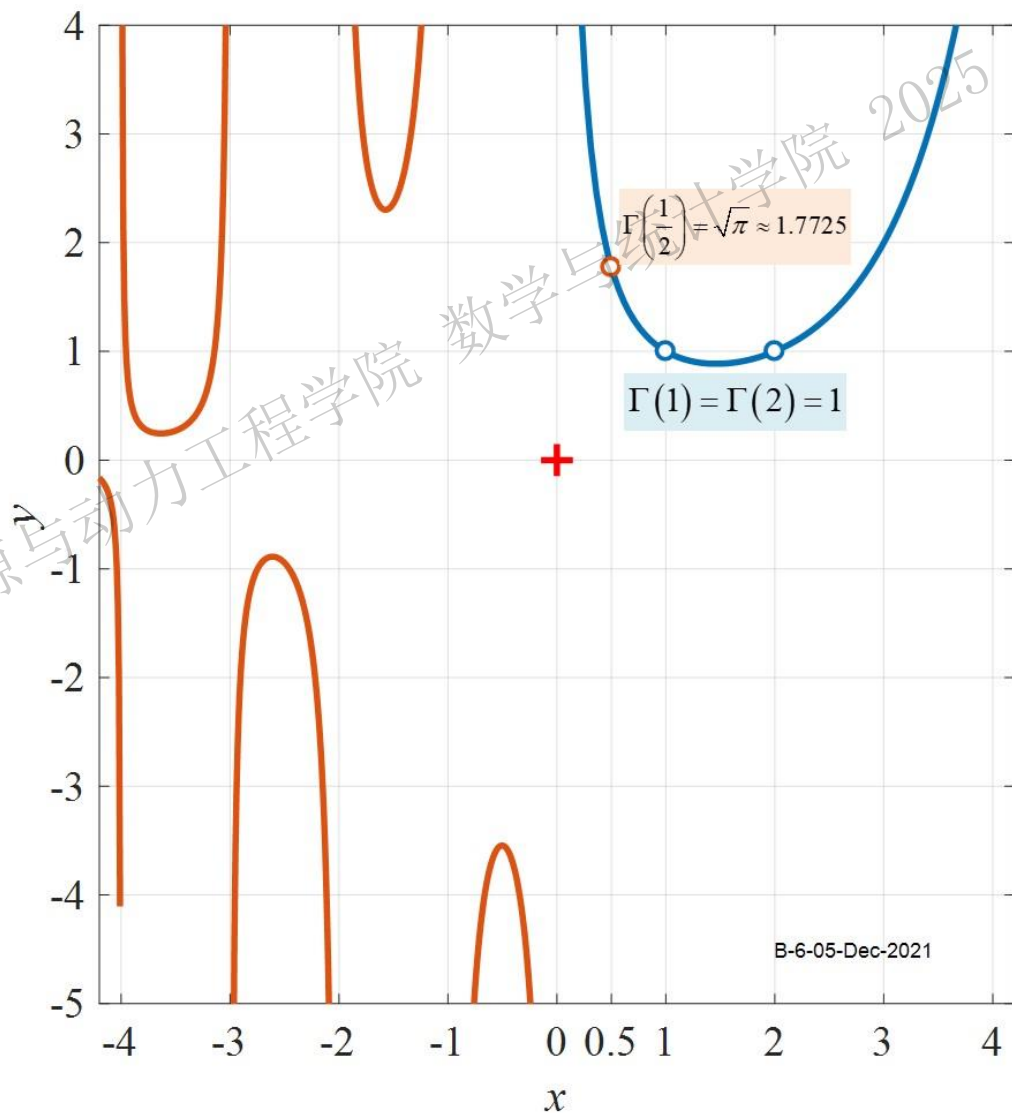
$$\Gamma(\alpha) = \begin{cases} \int_0^{\infty} x^{\alpha-1} e^{-x} dx, & (\alpha > 0) \\ \frac{\Gamma(\alpha+1)}{\alpha}, & (\alpha < 0, \alpha \neq \mathbb{Z}^-) \\ \infty, & (\alpha = 0, \mathbb{Z}^-) \end{cases}$$

$$\Gamma(\alpha) \neq 0, \Gamma(2) = \Gamma(1) = 1, \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

$$\Gamma(\alpha+1) = \alpha \cdot \Gamma(\alpha), \frac{1}{\alpha} = \frac{\Gamma(\alpha)}{\Gamma(\alpha+1)}$$

$$\Gamma(n+1) = n!$$

$$\Gamma\left(n + \frac{1}{2}\right) = \frac{(2n-1)!!}{2^n} \sqrt{\pi}$$



3.2 Bessel函数

$$\Gamma(\alpha) = \int_0^{\infty} x^{\alpha-1} e^{-x} dx$$

Gamma函数

例1 计算下列Gamma函数值

$$\Gamma(0) \quad \Gamma(0) = +\infty$$

$$\Gamma(1) \quad \Gamma(1) = 1$$

$$\Gamma(2) \quad \Gamma(2) = 1$$

$$\Gamma\left(\frac{1}{2}\right) \quad \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

$$\Gamma\left(\frac{3}{2}\right) \quad \Gamma\left(\frac{3}{2}\right) = \Gamma\left(1 + \frac{1}{2}\right) = \frac{(2-1)!!}{2^1} \sqrt{\pi} = \frac{\sqrt{\pi}}{2}$$

$$\Gamma\left(\frac{5}{2}\right) \quad \Gamma\left(\frac{5}{2}\right) = \Gamma\left(2 + \frac{1}{2}\right) = \frac{(2 \cdot 2 - 1)!!}{2^2} \sqrt{\pi} = \frac{(1 \cdot 3)}{2^2} \sqrt{\pi} = \frac{3\sqrt{\pi}}{4}$$

$$\Gamma\left(\frac{5}{2}\right) = \Gamma\left(\frac{3}{2} + 1\right) = \frac{3}{2} \Gamma\left(\frac{3}{2}\right) = \frac{3}{2} \Gamma\left(\frac{1}{2} + 1\right) = \frac{3}{2} \cdot \frac{1}{2} \Gamma\left(\frac{1}{2}\right) = \frac{3\sqrt{\pi}}{4}$$

$$\Gamma\left(-\frac{1}{2}\right) \quad \Gamma\left(-\frac{1}{2}\right) = \frac{\Gamma\left(-\frac{1}{2} + 1\right)}{-\frac{1}{2}} = -2\Gamma\left(\frac{1}{2}\right) = -2\sqrt{\pi}$$

$$\Gamma(\alpha) = \begin{cases} \int_0^{\infty} x^{\alpha-1} e^{-x} dx, & (\alpha > 0) \\ \frac{\Gamma(\alpha+1)}{\alpha}, & (\alpha < 0, \alpha \neq Z^-) \\ \infty, & (\alpha = 0, Z^-) \end{cases}$$

$$\Gamma(\alpha+1) = \alpha \cdot \Gamma(\alpha)$$

$$\Gamma(n+1) = n!, \quad \Gamma(2) = \Gamma(1) = 1$$

$$\Gamma\left(n + \frac{1}{2}\right) = \frac{(2n-1)!!}{2^n} \sqrt{\pi}, \quad \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

$$\Gamma(\alpha) = \int_0^{\infty} x^{\alpha-1} e^{-x} dx$$

3.2 Bessel函数

Gamma函数

应用之一→为广义积分的计算另辟蹊径

例2

计算积分

$$\int_0^{\infty} x^6 e^{-x^2} dx$$

解 变量代换 $t = x^2, x = \sqrt{t}$

$$\int_0^{\infty} x^6 e^{-x^2} dx$$

$$= \int_0^{\infty} (x^2)^3 e^{-x^2} dx$$

调整积分变量

$$= \int_0^{\infty} t^3 e^{-t} d(\sqrt{t}) = \frac{1}{2} \int_0^{\infty} t^3 e^{-t} \cdot t^{-\frac{1}{2}} dt$$

$$= \frac{1}{2} \int_0^{\infty} t^{\frac{5}{2}} e^{-t} dt = \frac{1}{2} \int_0^{\infty} t^{\frac{7}{2}-1} e^{-t} dt$$

Γ函数定义

$$= \frac{1}{2} \Gamma\left(\frac{7}{2}\right) = \frac{1}{2} \Gamma\left(3 + \frac{1}{2}\right)$$

$$\Gamma\left(n + \frac{1}{2}\right) = \frac{(2n-1)!!}{2^n} \sqrt{\pi}$$

$$= \frac{1}{2} \cdot \frac{(2 \cdot 3 - 1)!!}{2^3} \sqrt{\pi} = \frac{1}{2} \cdot \frac{(1 \cdot 3 \cdot 5)}{2^3} \sqrt{\pi}$$

$$= \frac{15}{16} \sqrt{\pi}$$

技巧 通过变量代换向Γ函数定义转化

例3

计算积分

$$\int_{-\infty}^{\infty} x^4 e^{-\frac{x^2}{2}} dx$$

解 变量代换 $t = \frac{x^2}{2}, x^2 = 2t, x = \sqrt{2t}$

$$\int_{-\infty}^{\infty} x^4 e^{-\frac{x^2}{2}} dx = 2 \int_0^{\infty} (x^2)^2 e^{-\frac{x^2}{2}} dx$$

偶函数

调整积分变量

$$= 2 \int_0^{\infty} (2t)^2 e^{-t} d(\sqrt{2t}) = 2 \int_0^{\infty} 4t^2 e^{-t} \sqrt{2} d\left(t^{\frac{1}{2}}\right)$$

$$= 2 \int_0^{\infty} 4t^2 e^{-t} \cdot \sqrt{2} \frac{1}{2} (t)^{-\frac{1}{2}} dt$$

Γ函数定义

$$= 4\sqrt{2} \int_0^{\infty} t^{\frac{3}{2}} e^{-t} dt = 4\sqrt{2} \int_0^{\infty} t^{\frac{5}{2}-1} e^{-t} dt$$

$$= 4\sqrt{2} \cdot \Gamma\left(\frac{5}{2}\right) = 4\sqrt{2} \Gamma\left(2 + \frac{1}{2}\right)$$

$$= 4\sqrt{2} \cdot \frac{(2 \cdot 2 - 1)!!}{2^2} \sqrt{\pi}$$

$$= 4\sqrt{2} \cdot \frac{(1 \cdot 3)}{2^2} \sqrt{\pi} = 3\sqrt{2\pi}$$

3.2 Bessel函数

$$\Gamma(\alpha) = \int_0^{\infty} x^{\alpha-1} e^{-x} dx$$

Gamma函数

例3 下列表达式错误的是

$$\int_0^{+\infty} x^{\frac{5}{2}} e^{-x^2} dx = \frac{3}{8} \Gamma\left(\frac{3}{4}\right)$$



$$x^2 = t, x = t^{\frac{1}{2}}$$



$$\int_0^{+\infty} x^{\frac{5}{2}} e^{-x^2} dx = \int_0^{+\infty} t^{\frac{1}{2} \cdot \frac{5}{2}} e^{-t} d\left(t^{\frac{1}{2}}\right)$$

$$= \frac{1}{2} \int_0^{+\infty} t^{\frac{1}{2} \cdot \frac{5}{2}} e^{-t} t^{-\frac{1}{2}} dt$$

$$= \frac{1}{2} \int_0^{+\infty} t^{\frac{3}{4}} e^{-t} dt$$

$$= \frac{1}{2} \int_0^{+\infty} t^{\frac{7}{4}-1} e^{-t} dt$$

$$= \frac{1}{2} \Gamma\left(\frac{7}{4}\right) = \frac{1}{2} \Gamma\left(1 + \frac{3}{4}\right)$$

$$= \frac{3}{8} \Gamma\left(\frac{3}{4}\right)$$

$$\Gamma(-2) = \infty$$

$$\Gamma\left(-\frac{1}{2}\right) = -\sqrt{\pi}$$

$$\Gamma\left(-\frac{1}{2}\right) = \frac{\Gamma\left(-\frac{1}{2} + 1\right)}{-\frac{1}{2}} = -2\Gamma\left(\frac{1}{2}\right) = -2\sqrt{\pi}$$

$$\Gamma\left(\frac{3}{2}\right) = \frac{\sqrt{\pi}}{2}$$

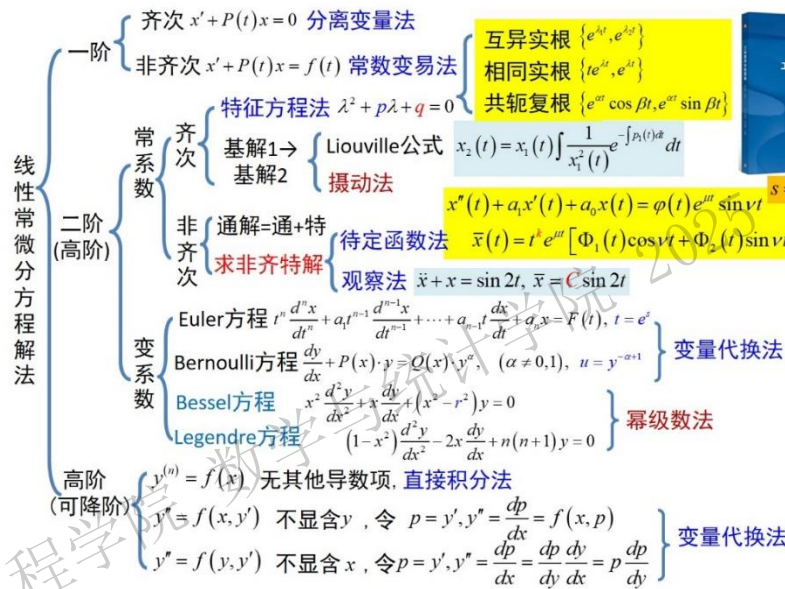
快速计算手册

边界条件齐次化常用辅助函数 $u(x,t) = v(x,t) + w(x,t)$		
边界类型	边界条件	辅助函数
[1,1]	$u(0,t) = g_1(t), u(l,t) = g_2(t)$	$w(x,t) = \frac{g_2(t) - g_1(t)}{l}x + g_1(t)$
[2,1]	$u_x(0,t) = g_1(t), u(l,t) = g_2(t)$	$w(x,t) = g_1(t)(x-l) + g_2(t)$
[1,2]	$u(0,t) = g_1(t), u_x(l,t) = g_2(t)$	$w(x,t) = g_2(t)x + g_1(t)$
[2,2]	$u_x(0,t) = g_1(t), u_x(l,t) = g_2(t)$	$w(x,t) = \frac{g_2(t) - g_1(t)}{2l}x^2 + g_1(t)x$

算子 $A = -\frac{d^2}{dx^2}$ 不同齐次边界条件下的特征值/特征函数		
齐次边界类型	特征值问题	特征值/特征函数
[1,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right), (n=1,2,3,\dots)$
[1,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \sin\left[\frac{(2n+1)\pi}{2l}x\right], (n=0,1,2,3,\dots)$
[2,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \cos\left[\frac{(2n+1)\pi}{2l}x\right], (n=0,1,2,3,\dots)$
[2,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \cos\left(\frac{n\pi}{l}x\right), (n=0,1,2,3,\dots)$
周期	$\begin{cases} \Phi''(\theta) + \lambda \Phi(\theta) = 0 \\ \Phi(0) = \Phi(2\pi), \\ \Phi'(0) = \Phi'(2\pi) \end{cases}$	$\lambda_0 = 0, \Phi_0(\theta) = 1, (n=0)$ $\lambda_n = n^2, \Phi_n(\theta) = C_1 \cos n\theta + C_2 \sin n\theta, (n=1,2,3,\dots)$

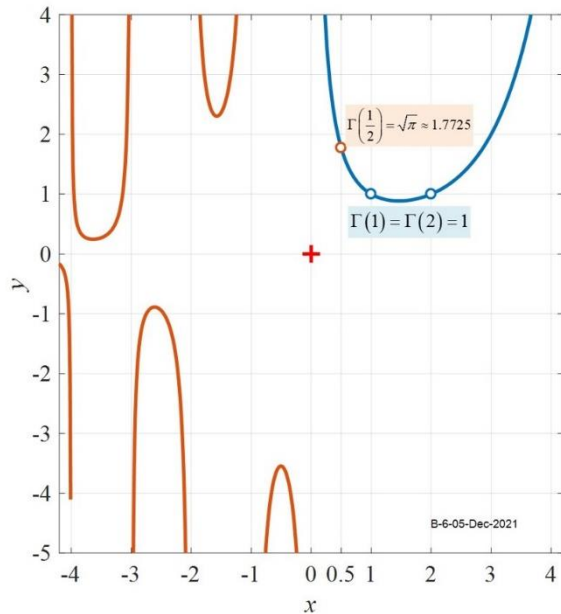
线性ODE的解法

$$x^{(n)} + P_1 x^{(n-1)} + P_2 x^{(n-2)} + \dots + P_{n-1} x^{(1)} + P_n x^{(0)} = F$$



Γ 函数常用计算

$$\Gamma(\alpha) = \begin{cases} \int_0^\infty x^{\alpha-1} e^{-x} dx, & (\alpha > 0) \\ \frac{\Gamma(\alpha+1)}{\alpha}, & (\alpha < 0, \alpha \neq Z^-) \\ \infty, & (\alpha = 0, Z^-) \end{cases}$$
$$\Gamma(\alpha) \neq 0$$
$$\Gamma(\alpha+1) = \alpha \cdot \Gamma(\alpha), \frac{1}{\alpha} = \frac{\Gamma(\alpha)}{\Gamma(\alpha+1)}$$
$$\Gamma(n+1) = n!, \Gamma(2) = \Gamma(1) = 1$$
$$\Gamma\left(n + \frac{1}{2}\right) = \frac{(2n-1)!!!}{2^n} \sqrt{\pi}, \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$



1. 数学建模和基本原理介绍

2. 分离变量法

3. Bessel函数

4. 积分变换法

5. Green函数法

6. 特征线法

7. Legendre多项式

3.1 二阶线性常微分方程的幂级数解法

3.2 Bessel函数

3.3 多个自变量分离变量法举例

Gamma函数

Bessel方程和Bessel函数

Bessel函数的性质

Bessel方程的特征值问题

圆域上Laplace算子的特征值问题

3.2 Bessel函数

→表达径向Bessel方程的解



极坐标上的波动方程

$\frac{\partial^2 u}{\partial t^2} - a^2 \Delta u = 0$ $u(\rho, \theta, t) = T(t)w(\rho, \theta)$

$T''(t)w(\rho, \theta) - a^2 T(t) \Delta w(\rho, \theta) = 0$

时空分离常数 ② 尚未确定

$\frac{T''(t)}{a^2 T(t)} = \frac{\Delta w(\rho, \theta)}{w(\rho, \theta)} = -\mu$

二维边值问题/特征值问题 $\lambda_2 = -\Delta$

初值问题

$T''(t) + \mu a^2 T(t) = 0$

$$\begin{cases} -\Delta w(\rho, \theta) = \mu w(\rho, \theta), (\rho, \theta) \in \Omega \\ w|_{\partial\Omega} = 0 \end{cases}$$

$w(\rho, \theta) = R(\rho)\Phi(\theta)$

$\Delta w = \frac{\partial^2 w}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial w}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 w}{\partial \theta^2}$

$R''(\rho)\Phi(\theta) + \frac{1}{\rho} R'(\rho)\Phi(\theta) + \frac{1}{\rho^2} R(\rho)\Phi''(\theta) = -\mu R(\rho)\Phi(\theta)$

空间分离常数

$$\frac{\Phi''(\theta)}{\Phi(\theta)} = -\frac{\mu R(\rho) + R''(\rho) + \frac{1}{\rho} R'(\rho)}{\frac{1}{\rho^2} R(\rho)} = -\lambda$$

$\lambda_0 = 0, \Phi_0(\theta) = 1, (n=0)$
 $\lambda_n = n^2, \Phi_n(\theta) = \{\cos n\theta, \sin n\theta\} (n=1, 2, 3, \dots)$

周向

径向

相当于原方程是Poisson方程

$$\begin{cases} \Phi''(\theta) + \lambda \Phi(\theta) = 0 \\ \Phi(\theta) = \Phi(\theta + 2\pi) \end{cases} \quad \begin{cases} |R(0)| < +\infty, R(\rho_0) = 0 \\ \rho^2 R''(\rho) + \rho R'(\rho) + (\mu \rho^2 - \lambda) R(\rho) = 0 \end{cases}$$

Euler方程 $\mu = 0$

$\rho^2 R''(\rho) + \rho R'(\rho) - n^2 R(\rho) = 0$

Bessel方程 $\mu > 0$ 1824

$\rho^2 R''(\rho) + \rho R'(\rho) + (\mu \rho^2 - n^2) R(\rho) = 0$

线性二阶变系数齐次ODE ① 如何求解



F.W.Bessel
1784-1846
German Math.
Astronomer

3.2 Bessel函数

Bessel方程的标准形式 预处理

变量代换

$$x = \sqrt{\mu} \rho, \quad \rho = \frac{x}{\sqrt{\mu}},$$
$$y(x) = R(\rho) = R\left(\frac{x}{\sqrt{\mu}}\right)$$

各阶导数

$$y'(x) = R'(\rho) \frac{d\rho}{dx} = \frac{1}{\sqrt{\mu}} R'(\rho)$$
$$y''(x) = \frac{1}{\sqrt{\mu}} R''(\rho) \frac{d\rho}{dx} = \frac{1}{\mu} R''(\rho)$$

Bessel方程标准形式

极坐标径向ODE

$$\begin{cases} |R(0)| < +\infty, R(\rho_0) = 0 \\ \rho^2 R''(\rho) + \rho R'(\rho) + (\mu \rho^2 - n^2) R(\rho) = 0 \end{cases}$$

时空分离常数

空间分离常数 $\lambda = n^2$

$$R'(\rho) = \sqrt{\mu} y'(x)$$
$$R''(\rho) = \mu y''(x)$$

$$\frac{x^2}{\mu} \mu y''(x) + \frac{x}{\sqrt{\mu}} \sqrt{\mu} y'(x) + \left(\mu \frac{x^2}{\mu} - n^2 \right) y(x) = 0$$
$$x^2 y''(x) + x y'(x) + (x^2 - n^2) y(x) = 0$$

定义Bessel方程阶数

$$r^2 = n^2$$

整数→实数
默认 $r \geq 0$

$$x^2 y''(x) + x y'(x) + (x^2 - r^2) y(x) = 0$$



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- 1. 阶数是周向特征值=非负整数→全面研究Bessel方程→阶数拓展→实数域
- 2. Bessel方程是线性二阶变系数齐次ODE→阶数 r 是唯一参数→决定了解的形式
- 3. 观察法→显然 $y(x)=0$ 是Bessel方程的解→关心更有意义的非零解

3.2 Bessel函数

$$x''(t) + p(x)x'(t) + q(x)x(t) = 0$$

Bessel方程/Bessel函数

Bessel方程→幂级数求解→Bessel函数

$$x^2 y'' + xy' + (x^2 - r^2)y = 0$$

阶数 $r \geq 0$

求解思路:

判定幂级数解存在→幂级数展开→2个形式解→判定线性相关性→确定基解



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1784-1846
German Math.
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1 解的存在性

方程变型 $y'' + \frac{1}{x}y' + \left(1 - \frac{r^2}{x^2}\right)y = 0$ $p(x) = \frac{1}{x}$, $q(x) = 1 - \frac{r^2}{x^2}$

而 $x \cdot p(x) = x \cdot \frac{1}{x} = 1$, $x^2 \cdot q(x) = x^2 \cdot \left(1 - \frac{r^2}{x^2}\right) = x^2 - r^2$ 在实域内解析→解存在

Bessel方程存在幂级数解

$$y(x) = x^\rho \sum_{n=0}^\infty a_n x^n = \sum_{n=0}^\infty a_n x^{n+\rho}, (a_0 \neq 0)$$

探寻非零解

1个参数 $r \rightarrow$ 2组未知系数 ρ, a_n

幂级数解存在定理2

求解Bessel方程的理论支撑

对二阶ODE $x''(t) + p(x)x'(t) + q(x)x(t) = 0$, 若 $(x-x_0)p(x)$, $(x-x_0)^2 q(x)$ 在 x_0 的邻域 $\{x \in \mathbb{R} \mid |x-x_0| < \delta\}$ 内解析, 即 x_0 最多为 $p(x)$, $q(x)$ 的一阶和二阶极点, 则原方程在该去心邻域 $\{x \in \mathbb{R} \mid 0 < |x-x_0| < \delta\}$ 内存在幂级数解

$$y(x) = (x-x_0)^\rho \sum_{k=0}^\infty a_k (x-x_0)^k$$

其中常数 $\rho, a_k (a_0 \neq 0)$ 可由待定系数法求出

$$x^2 y'' + xy' + (x^2 - r^2) y = 0$$

3.2 Bessel函数

Bessel方程/Bessel函数

2 寻找递推式

幂级数解 $y(x) = \sum_{n=0}^{\infty} a_n x^{n+\rho}, (a_0 \neq 0)$

各阶导数 $y'(x) = \sum_{n=0}^{\infty} (n+\rho) a_n x^{n+\rho-1}, y''(x) = \sum_{n=0}^{\infty} (n+\rho)(n+\rho-1) a_n x^{n+\rho-2}$

代入
原
方
程

$$x^2 \sum_{n=0}^{\infty} (n+\rho)(n+\rho-1) a_n x^{n+\rho-2} + x \sum_{n=0}^{\infty} (n+\rho) a_n x^{n+\rho-1} + (x^2 - r^2) \sum_{n=0}^{\infty} a_n x^{n+\rho} = 0$$

$$\sum_{n=0}^{\infty} (n+\rho)(n+\rho-1) a_n x^{n+\rho} + \sum_{n=0}^{\infty} (n+\rho) a_n x^{n+\rho} - r^2 \sum_{n=0}^{\infty} a_n x^{n+\rho} + \sum_{n=0}^{\infty} a_n x^{n+\rho+2} = 0$$

约去 x^ρ

$$\sum_{n=0}^{\infty} [(n+\rho)^2 - r^2] a_n x^{n+\rho} + \sum_{n=0}^{\infty} a_n x^{n+\rho+2} = 0$$

展开前两项

$$\sum_{n=0}^{\infty} [(n+\rho)^2 - r^2] a_n x^{n+\rho} + \sum_{n=0}^{\infty} a_n x^{n+\rho+2} = 0$$

$$(\rho^2 - r^2) a_0 + [(1+\rho)^2 - r^2] a_1 x + \sum_{n=2}^{\infty} [(n+\rho)^2 - r^2] a_n x^{n+\rho} + \sum_{n=2}^{\infty} a_{n-2} x^{n+\rho} = 0$$

调整起点→比较同幂次系数

比较系数→原始递推关系

$$\begin{cases} (\rho^2 - r^2) a_0 = 0 \\ [(1+\rho)^2 - r^2] a_1 = 0 \\ [(n+\rho)^2 - r^2] a_n + a_{n-2} = 0, (n \geq 2) \end{cases}$$

1. Bessel方程的阶数 r 决定→幂级数解中 x 的幂次 ρ
2. 后续根据幂次 ρ 的取值/系数序号奇偶→化简递推

$$x^2 y'' + xy' + (x^2 - r^2)y = 0$$

$$y(x) = \sum_{n=0}^{\infty} a_n x^{n+\rho}, \quad (a_0 \neq 0)$$

3.2 Bessel函数

Bessel方程/Bessel函数

2 递推式简化

$$y(x) = \sum_{n=0}^{\infty} a_n x^{n+\rho}, \quad (a_0 \neq 0)$$

$$\begin{cases} (\rho^2 - r^2)a_0 = 0 & \textcircled{1} \\ [(1+\rho)^2 - r^2]a_1 = 0 & \textcircled{2} \\ [(n+\rho)^2 - r^2]a_n + a_{n-2} = 0, \quad (n \geq 2) & \textcircled{3} \end{cases}$$

$$\textcircled{1} \quad \rho = \pm r, \quad (r \geq 0)$$

$$\begin{aligned} \textcircled{2} \quad & [(1+\rho)^2 - r^2]a_1 = 0 \\ & (1+2\rho+\rho^2-r^2)a_1 = 0 \\ & (1+2\rho)a_1 = 0 \end{aligned}$$

$$\begin{aligned} \textcircled{3} \quad & [(n+\rho)^2 - r^2]a_n + a_{n-2} = 0, \quad (n \geq 2) \\ & (n^2 + 2n\rho + \rho^2 - r^2)a_n + a_{n-2} = 0 \\ & (n^2 + 2n\rho)a_n + a_{n-2} = 0 \\ & n(n+2\rho)a_n + a_{n-2} = 0 \end{aligned}$$

递推式的简化

$$\begin{cases} \textcircled{1} \quad \rho = \pm r \\ \textcircled{2} \quad (1+2\rho)a_1 = 0 \\ \textcircled{3} \quad n(n+2\rho)a_n + a_{n-2} = 0, \quad (n \geq 2) \end{cases}$$

1. 已经确定 ρ
2. 无论 r 是否为整数; 无论 $\rho = +r$ 还是 $\rho = -r$; 递推式均成立
3. 但 ρ 的正负/取值 $\rightarrow$ 递推式 $\textcircled{2} \textcircled{3}$ 的运用 $\rightarrow$ 需对 ρ 分类讨论

$$\rho = \begin{cases} r > 0 \\ -r < 0 \\ r = 0 \end{cases} \begin{cases} r = N^+ \\ r \neq N^+ \end{cases}, \quad (r \geq 0)$$

3.2 Bessel函数

Bessel方程/Bessel函数

3 阶数 r 分类讨论

$$x^2 y'' + xy' + (x^2 - r^2)y = 0$$

$$\rho = \begin{cases} r > 0 \\ -r < 0 \\ r = 0 \end{cases} \begin{cases} r = N^+ \\ r \neq N^+ \end{cases}, \quad (r \geq 0)$$

$$y(x) = \sum_{n=0}^{\infty} a_n x^{n+\rho}, \quad (a_0 \neq 0)$$

1 当 $\rho = r > 0$ 时

- ① $\rho = \pm r \rightarrow$ 此时只可能 $\rho = +r$
- ② $(1+2\rho)a_1 = 0 \rightarrow \rho = r > 0 \rightarrow (1+2\rho) \neq 0 \rightarrow$ 只可能 $a_1 = 0$
- ③ $n(n+2\rho)a_n + a_{n-2} = 0, (n \geq 2) \rightarrow n(n+2\rho) \neq 0, (n \geq 2)$

可除至分母

$$a_n = -\frac{a_{n-2}}{n(n+2\rho)} = -\frac{a_{n-2}}{n(n+2r)}, \quad (n \geq 2)$$

说明系数跨位传递→
需对序号分奇偶讨论

1.1 当下标为奇数时→最终均可递推至 $a_1 \rightarrow$ 奇数项系数均为0

n 不连续→改用连续的 k 表示 $a_{2k-1} = a_{2k-3} = \cdots = a_1 = 0, (k \geq 1)$

3.2 Bessel函数

Bessel方程/Bessel函数

3 阶数 r 分类讨论

1 当 $\rho = r > 0$ 时

1.2 下标为偶数时 $a_0 \neq 0$

$$\rho = \begin{cases} r > 0 \\ -r < 0 \\ r = 0 \end{cases} \begin{cases} r = N^+ \\ r \neq N^+ \end{cases}, \quad (r \geq 0)$$

$$x^2 y'' + xy' + (x^2 - r^2) y = 0$$

$$y(x) = \sum_{n=0}^{\infty} a_n x^{n+\rho}, \quad (a_0 \neq 0)$$

- 1 $\rho = \pm r$
- 2 $(1 + 2\rho) a_1 = 0$
- 3 $n(n + 2\rho) a_n + a_{n-2} = 0, \quad (n \geq 2)$

$$a_n = -\frac{a_{n-2}}{n(n + 2\rho)} = -\frac{a_{n-2}}{n(n + 2r)}, \quad (n \geq 2)$$

$$a_2 = -\frac{a_0}{2(2 + 2r)} = -\frac{a_0}{2^2 \cdot 1! (1 + r)}$$

$$a_4 = -\frac{a_2}{4(4 + 2r)} = -\frac{a_2}{2^3 (2 + r)} = (-1)^2 \frac{1}{2^3 (2 + r)} \cdot \frac{a_0}{2^2 \cdot 1! (1 + r)} = (-1)^2 \frac{a_0}{2^4 \cdot 2! (2 + r) (1 + r)}$$

$$a_6 = -\frac{a_4}{6(6 + 2r)} = -\frac{a_4}{2 \cdot 3 \cdot 2 \cdot (3 + r)} = (-1)^3 \frac{1}{2^2 \cdot 3 \cdot (3 + r)} \cdot \frac{a_0}{2^4 \cdot 2! (2 + r) (1 + r)}$$

$$= (-1)^3 \frac{a_0}{2^6 \cdot 3! (3 + r) (2 + r) (1 + r)} \quad (n \geq 2)$$

$$a_{2k} \rightarrow a_{2k-2} \rightarrow a_{2k-4} \rightarrow \dots \rightarrow a_4 \rightarrow a_2 \rightarrow a_0$$

偶数项系数不为0 → 均可递推至 a_0 →
但系数表达式冗长连乘 → 如何化简?

$$a_{2k} = (-1)^k \frac{a_0}{2^{2k} \cdot k! (k + r) (k - 1 + r) \dots (1 + r)}, \quad (k \geq 1)$$

如何化简连乘

3.2 Bessel函数

Bessel方程/Bessel函数

3 阶数 r 分类讨论

1 当 $\rho=r>0$ 时

1.2 下标为偶数 分母连乘的化简

$$a_{2k}=(-1)^k \frac{a_0}{2^{2k} \cdot k! \cdot (k+r)(k-1+r) \cdots (1+r)}, (k \geq 1)$$

$$=(-1)^k \frac{1}{2^{2k} \cdot k!} \cdot \frac{1}{k+r} \cdot \frac{1}{k-1+r} \cdots \frac{1}{3+r} \cdot \frac{1}{2+r} \cdot \frac{1}{1+r} a_0$$

$$=(-1)^k \frac{1}{2^{2k} \cdot k!} \cdot \frac{\Gamma(k+r)}{\Gamma(k+r+1)} \cdot \frac{\Gamma(k-1+r)}{\Gamma(k+r)} \cdots \frac{\Gamma(3+r)}{\Gamma(4+r)} \cdot \frac{\Gamma(2+r)}{\Gamma(3+r)} \cdot \frac{\Gamma(1+r)}{\Gamma(2+r)} a_0$$

$$=(-1)^k \frac{1}{2^{2k} \cdot k!} \cdot \frac{\Gamma(1+r)}{\Gamma(k+r+1)} a_0$$

$a_0 = \frac{1}{2^r \cdot \Gamma(1+r)}$ 系数 a_0 为任意常数不妨定义为

$$=(-1)^k \frac{\Gamma(1+r)}{2^{2k} \cdot k! \cdot \Gamma(k+r+1)} \frac{1}{2^r \cdot \Gamma(1+r)}$$

$$=\frac{(-1)^k}{2^r \cdot 2^{2k} \cdot k! \cdot \Gamma(k+r+1)}$$

a_0 消失了

偶数项系数表达式

$$a_{2k} = \frac{(-1)^k}{2^r \cdot 2^{2k} \cdot k! \cdot \Gamma(k+r+1)}$$

$$x^2 y'' + xy' + (x^2 - r^2) y = 0$$

$$\rho = \begin{cases} r > 0 \\ -r < 0 \\ r = 0 \end{cases} \begin{cases} r = N^+ \\ r \neq N^+ \end{cases}, (r \geq 0)$$

- ① $\rho = \pm r$

② $(1+2\rho)a_1 = 0$

③ $n(n+2\rho)a_n + a_{n-2} = 0, (n \geq 2)$
- $$y(x) = \sum_{n=0}^{\infty} a_n x^{n+\rho}, (a_0 \neq 0)$$

$$\frac{1}{\alpha} = \frac{\Gamma(\alpha)}{\Gamma(\alpha+1)}$$

$$\Gamma(\alpha+1) = \alpha \cdot \Gamma(\alpha)$$

Bessel方程给定
→阶数 r 为定值

3.2 Bessel函数

Bessel方程/Bessel函数

3 阶数 r 分类讨论

1 当 $\rho=r>0$ 时 小结

$$\rho = \begin{cases} r > 0 \\ -r < 0 \\ r = 0 \end{cases} \begin{cases} r = N^+ \\ r \neq N^+ \end{cases}, \quad (r \geq 0)$$

$$x^2 y'' + xy' + (x^2 - r^2) y = 0$$

$$y(x) = \sum_{n=0}^{\infty} a_n x^{n+\rho}, \quad (a_0 \neq 0)$$

- 1 $\rho = \pm r$

2 $(1+2\rho)a_1 = 0$

3 $n(n+2\rho)a_n + a_{n-2} = 0, \quad (n \geq 2)$

奇数项 $a_{2k-1} = \dots = a_1 = 0, \quad (k \geq 1)$

$$a_{2k} = \frac{(-1)^k}{2^r \cdot 2^{2k} \cdot k! \cdot \Gamma(k+r+1)}$$

偶数项

代入幂级数解

$$y_1(x) = \sum_{n=0}^{\infty} a_n x^{n+\rho} = \sum_{n=0}^{\infty} a_n x^{n+r}$$

$$= \sum_{k=1}^{\infty} a_{2k-1} x^{2k-1+r} + \sum_{k=0}^{\infty} a_{2k} x^{2k+r}$$

$$= \sum_{k=0}^{\infty} \frac{(-1)^k}{2^r \cdot 2^{2k} \cdot k! \cdot \Gamma(k+r+1)} x^{2k+r}$$

$$= \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \cdot \Gamma(k+r+1)} \cdot \frac{x^{2k}}{2^{2k}} \cdot \frac{x^r}{2^r}$$

递推式从 $k=1$ 开始, 但此处要包含 a_0

只需累加偶数项 $\rightarrow$ 换为连续序号 k

$$= \left(\frac{x}{2}\right)^r \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \cdot \Gamma(k+r+1)} \left(\frac{x}{2}\right)^{2k} = J_r(x)$$

与 k 无关的移出 Σ

$$J_r(x) = \left(\frac{x}{2}\right)^r \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \cdot \Gamma(k+r+1)} \left(\frac{x}{2}\right)^{2k}$$



F.W.Bessel
1784-1846
German Math.
Astronomer

r 阶 Bessel 函数

仅仅是 Bessel 方程的 1 个基解? $\rightarrow$ 另一个基解在哪里?

3.2 Bessel函数

Bessel方程/Bessel函数

3 阶数 r 分类讨论

2 当 $\rho = -r < 0$ 时 $a_0 \neq 0$

$$\rho = \begin{cases} r > 0 \\ -r < 0 \\ r = 0 \end{cases} \begin{cases} r = N^+ \\ r \neq N^+ \end{cases}, \quad (r \geq 0)$$

$$x^2 y'' + xy' + (x^2 - r^2) y = 0$$

$$y(x) = \sum_{n=0}^{\infty} a_n x^{n+\rho}, \quad (a_0 \neq 0)$$

化简递推式

- ① $\rho = \pm r \rightarrow \rho = -r$
- ② $(1+2\rho)a_1 = 0 \rightarrow$ 不妨仍令 $a_1 = 0 \rightarrow$ 奇数下标全为0 $a_{2k-1} = \dots = a_1 = 0, (k \geq 1)$
- ③ $n(n+2\rho)a_n + a_{n-2} = 0, (n \geq 2)$ 聚焦 $\rightarrow n$ 为偶数 $n = 2k$ 时递推式 ③ 的运用

按 n 为偶数化简

$\rho = -r$

$$n(n+2\rho)a_n + a_{n-2} = 0, \quad (n \geq 2)$$
$$n(n-2r)a_n + a_{n-2} = 0$$

$n = 2k$

$$2k(2k-2r)a_{2k} + a_{2k-2} = 0, \quad (k \geq 1)$$
$$4k(k-r)a_{2k} + a_{2k-2} = 0$$

非负整数

若 r 为正整数 $\rightarrow (k-r)$ 有可能 $= 0 \rightarrow$ 不能直接除至分母
 $\rightarrow$ 需根据 r 是否为正整数分亚类讨论

$$\rho = \begin{cases} r > 0 \\ -r < 0 \\ r = 0 \end{cases} \begin{cases} r = N^+ \\ r \neq N^+ \end{cases}, \quad (r \geq 0)$$

3.2 Bessel函数

Bessel方程/Bessel函数

3阶数 r 分类讨论

2 当 $\rho = -r < 0$ 时 $a_0 \neq 0$

2.1 当 r 不为正整数时 $k-r \neq 0$

递推式 3 $4k(k-r)a_{2k} + a_{2k-2} = 0$ 可除至分母

递推可变型为 $a_{2k} = -\frac{1}{4k(k-r)}a_{2k-2}$ 偶数项递推至 a_0 $a_{2k} \rightarrow a_{2k-2} \rightarrow \dots \rightarrow a_4 \rightarrow a_2 \rightarrow a_0$

同理可得 $a_{2k} = \frac{(-1)^k}{2^{-r} \cdot 2^{2k} \cdot k! \Gamma(k-r+1)}$ $\rho = -r < 0$ 对比 $\rho = +r > 0$ $a_{2k} = \frac{(-1)^k}{2^r \cdot 2^{2k} \cdot k! \Gamma(k+r+1)}$ 相当于 $r \rightarrow -r$

得到 r 不为整数时Bessel方程另一个幂级数解

-r 阶Bessel函数

$$J_{-r}(x) = \left(\frac{x}{2}\right)^{-r} \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k-r+1)} \left(\frac{x}{2}\right)^{2k}$$

$$x^2 y'' + xy' + (x^2 - r^2) y = 0$$

$$y(x) = \sum_{n=0}^{\infty} a_n x^{n+\rho}, (a_0 \neq 0)$$

- 1 $\rho = \pm r$
- 2 $(1+2\rho)a_1 = 0$
- 3 $n(n+2\rho)a_n + a_{n-2} = 0, (n \geq 2)$

$$4k(k-r)a_{2k} + a_{2k-2} = 0$$

$$x^2 y'' + xy' + (x^2 - r^2)y = 0$$

$$y(x) = \sum_{n=0}^{\infty} a_n x^{n+\rho}, \quad (a_0 \neq 0)$$

$$\rho = \begin{cases} r > 0 \\ -r < 0 \\ r = 0 \end{cases} \begin{cases} r \neq N^+ \\ r = N^+ \end{cases} \quad (r \geq 0)$$

- ① $\rho = \pm r$
- ② $(1 + 2\rho)a_1 = 0$
- ③ $n(n + 2\rho)a_n + a_{n-2} = 0, \quad (n \geq 2)$
 $4k(k - r)a_{2k} + a_{2k-2} = 0$

3.2 Bessel函数

Bessel方程/Bessel函数

3阶数 r 分类讨论

2 当 $\rho = -r < 0$ 时 $a_0 \neq 0$

2.2 当 r 为正整数时 $r = l$

任意确定正整数

连续变化的整数 k 总有 $= l$ 的可能

当 $k - r = k - l = 0$ 时根据递推式

$$4k(k - r)a_{2k} + a_{2k-2} = 0, \quad (k \geq 1)$$

$$4k(k - l)a_{2l} + a_{2l-2} = 0$$

=0

故有 $a_{2l-2} = 0$

令 $k = l - 1$, 再次使用递推式

$$4[(l - 1) - 1][(l - 1) - l]a_{2l-2} + a_{2l-4} = 0$$

只可能 $a_{2l-4} = 0$

如此反复运用递推式可得

$$a_{2l-2} = a_{2l-4} = \cdots = a_4 = a_2 = a_0 = 0$$

$$a_{2(l-1)} = a_{2(l-2)} = a_{2(l-3)} = \cdots = a_{2 \cdot 2} = a_{2 \cdot 1} = a_{2 \cdot 0} = 0$$

与假设 $a_0 \neq 0$ 矛盾

问题出在哪里呢?

3.2 Bessel函数

Bessel方程/Bessel函数

3 阶数 r 分类讨论

$$x^2 y'' + xy' + (x^2 - r^2) y = 0$$

$$\rho = \begin{cases} r > 0 \\ -r < 0 \\ r = 0 \end{cases} \begin{cases} r \neq N^+ \\ r = N^+ \end{cases} \quad (r \geq 0)$$

$$y(x) = \sum_{n=0}^{\infty} a_n x^{n+\rho}, \quad (a_0 \neq 0)$$

- ① $\rho = \pm r$
- ② $(1 + 2\rho) a_1 = 0$
- ③ $n(n + 2\rho) a_n + a_{n-2} = 0, \quad (n \geq 2)$

2 当 $\rho = -r < 0$ 时 $a_0 \neq 0$

2.2 当 r 为正整数时 $r = l$ 任意确定正整数 原幂级数解

$$\begin{aligned} y(x) &= x^\rho \sum_{n=0}^{\infty} a_n x^n = x^{-r} \sum_{n=0}^{\infty} a_n x^n = \sum_{k=0}^{\infty} a_{2k} x^{-l+2k}, \quad (a_0 \neq 0) \\ &= \left[a_0 x^{-l} + a_2 x^{-l+2} + a_4 x^{-l+4} + \dots + a_{2(l-1)} x^{-l+2(l-1)} \right] + a_{2l} x^{-l+2l} + a_{2(l+1)} x^{-l+2(l+1)} + \dots + a_{2k} x^{-l+2k} + \dots \\ &= \sum_{k=l}^{\infty} a_{2k} x^{-l+2k} \end{aligned}$$

$$a_{2l-2} = a_{2l-4} = \dots = a_0 = 0$$

系数依然满足递推式3

$$4k(k-l) a_{2l} + a_{2l-2} = 0, \quad (k > l)$$

所有系数只可递推至 a_{2l}

$$a_{2k} \rightarrow a_{2k-2} \rightarrow a_{2k-4} \rightarrow \dots \rightarrow a_{2(l+2)} \rightarrow a_{2(l+1)} \rightarrow a_{2l}$$

说明原幂级数解形式不妥→不应有幂次低于 $2l$ 的项

原形式解序号/幂次对应关系不合理→不妨把前 l 项 ($k = 0 \sim l-1$) 取掉→新假设解为

$$y(x) = x^\rho \sum_{k=l}^{\infty} a_{2k} x^{2k} = x^{-l} \sum_{k=l}^{\infty} a_{2k} x^{2k} = \sum_{k=l}^{\infty} a_{2k} x^{-l+2k}, \quad (a_0 \neq 0)$$

从 l 开始累加

3.2 Bessel函数

Bessel方程/Bessel函数

3阶数 r 分类讨论

2 当 $\rho = -r < 0$ 时 $a_0 \neq 0$

$$\rho = \begin{cases} r > 0 \\ -r < 0 \\ r = 0 \end{cases} \begin{cases} r \neq N^+ \\ r = N^+ \end{cases} \quad (r \geq 0)$$

$$x^2 y'' + xy' + (x^2 - r^2) y = 0$$

$$y(x) = \sum_{n=0}^{\infty} a_n x^{n+\rho}, \quad (a_0 \neq 0)$$
$$\begin{cases} \rho = \pm r \\ (1+2\rho)a_1 = 0 \\ n(n+2\rho)a_n + a_{n-2} = 0, \quad (n \geq 2) \end{cases}$$

2.2 当 r 为正整数时 $r = l$ 任意确定正整数

新假设解为 $y(x) = x^{-l} \sum_{k=l}^{\infty} a_{2k} x^{2k}$

相当于除去了原级数的前 l 项 → 新解代入 → 系数递推式是前述递推式在 $k > l$ 的特殊区段

$$4k(k-l)a_{2k} + a_{2k-2} = 0, \quad (k > l)$$

$$a_{2k} = -\frac{a_{2k-2}}{4k(k-l)}$$

其系数是 $\rho = -r < 0, (r \neq N^+)$ 时系数表达式在 $k > l$ 时的特殊情况

$$\rho = -r < 0, \quad (r \neq N^+)$$

$$a_{2k} = \frac{(-1)^k}{2^{-r} \cdot 2^{2k} \cdot k! \cdot \Gamma(k-r+1)}$$

$$\rho = -r = -l, \quad (r = N^+, k \geq l)$$

$$a_{2k} = \frac{(-1)^k}{2^{-l} \cdot 2^{2k} \cdot k! \cdot \Gamma(k-l+1)}$$

代入新幂级数解 $y(x) = \sum_{k=l}^{\infty} \frac{(-1)^k}{2^{-l} \cdot 2^{2k} \cdot k! \cdot \Gamma(k-l+1)} x^{2k-l}$

$$= \sum_{k=0}^{\infty} \frac{(-1)^k}{2^{-l} \cdot 2^{2k} \cdot k! \cdot \Gamma(k-l+1)} x^{2k-l}$$

起点可改回 0

惊喜 当 $k < l$ 时, $k-l+1$ 是负整数 $\frac{1}{\Gamma(k-l+1)} \equiv 0$

k 起点改回 0 → 形式与 $J_r(x)$ 统一 → 得另 1 个非 0 解

$$J_{-r}(x) = \left(\frac{x}{2}\right)^{-r} \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \cdot \Gamma(k-r+1)} \left(\frac{x}{2}\right)^{2k}$$

3.2 Bessel函数

Bessel方程/Bessel函数

4 解的线性相关性

$$\rho = \begin{cases} r > 0 \\ -r < 0 \\ r = 0 \end{cases} \begin{cases} r \neq N^+ \\ r = N^+ \end{cases}, \quad (r \geq 0)$$

$$x^2 y'' + xy' + (x^2 - r^2) y = 0$$

$$y(x) = \sum_{n=0}^{\infty} a_n x^{n+\rho}, \quad (a_0 \neq 0)$$

至此, 已得到 $r > 0$ 时Bessel方程的两个解

$$J_r(x) = \left(\frac{x}{2}\right)^r \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+r+1)} \left(\frac{x}{2}\right)^{2k}$$

$$J_{-r}(x) = \left(\frac{x}{2}\right)^{-r} \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k-r+1)} \left(\frac{x}{2}\right)^{2k}$$

当 $r > 0$ 不为正整数

幂级数部分

$$\begin{aligned} S_k(0) &= \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k \pm r + 1)} \left(\frac{x}{2}\right)^{2k} \\ &= \frac{1}{\Gamma(\pm r + 1)} + \sum_{k=1}^{\infty} \frac{(-1)^k}{k! \Gamma(k \pm r + 1)} \left(\frac{0}{2}\right)^{2k} \\ &= \frac{1}{\Gamma(\pm r + 1)} \neq 0 \quad \text{有限值} \end{aligned}$$

幂函数部分

$$x=0, \left(\frac{x}{2}\right)^r = 0, \left(\frac{x}{2}\right)^{-r} \rightarrow \infty \Rightarrow \frac{J_{-r}(0)}{J_r(0)} \neq \text{Const.}$$

说明当 $r > 0$ 不为正整数时, $\{J_{-r}(x), J_r(x)\}$ 线性无关 $\rightarrow$ 是Bessel方程的2个基解

3.2 Bessel函数

Bessel方程/Bessel函数

4 解的线性相关性

$$x^2 y'' + xy' + (x^2 - r^2) y = 0$$

$$\rho = \begin{cases} r > 0 \\ -r < 0 \\ r = 0 \end{cases} \begin{cases} r \neq N^+ \\ r = N^+ \end{cases}, \quad (r \geq 0)$$

$$y(x) = \sum_{n=0}^{\infty} a_n x^{n+\rho}, \quad (a_0 \neq 0)$$

$$J_r(x) = \left(\frac{x}{2}\right)^r \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+r+1)} \left(\frac{x}{2}\right)^{2k}$$

$$J_{-r}(x) = \left(\frac{x}{2}\right)^{-r} \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k-r+1)} \left(\frac{x}{2}\right)^{2k}$$

当 $r > 0$ 且为正整数时

$$\begin{aligned} J_{-n}(x) &= \left(\frac{x}{2}\right)^{-n} \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k-n+1)} \left(\frac{x}{2}\right)^{2k} \quad \begin{matrix} \text{令 } k = n+j \\ \text{序号是 } j \end{matrix} \\ &= \left(\frac{x}{2}\right)^{-n} \sum_{j=0}^{\infty} \frac{(-1)^{n+j}}{(n+j)! \Gamma(j+1)} \left(\frac{x}{2}\right)^{2n+2j} \\ &= (-1)^n \left(\frac{x}{2}\right)^n \sum_{j=0}^{\infty} \frac{(-1)^j}{j! (j+n) \cdot (j+n-1) \cdots (j+2) \cdot (j+1) \cdot \Gamma(j+1)} \left(\frac{x}{2}\right)^{2j} \\ &= (-1)^n \left(\frac{x}{2}\right)^n \sum_{j=0}^{\infty} \frac{(-1)^j}{j! \Gamma(n+j+1)} \left(\frac{x}{2}\right)^{2j} \quad \begin{matrix} \text{利用 } \Gamma(\alpha+1) = \alpha \cdot \Gamma(\alpha) \end{matrix} \\ &= (-1)^n J_n(x) \end{aligned}$$

说明 $J_{-n}(x), J_n(x)$ 线性相关 $\rightarrow$ 还需另寻1个基解

当 $r=0$ 时 $J_{-0}(x) = J_0(x)$, 仅有一解

3.2 Bessel函数

Bessel方程/Bessel函数

4 解的线性相关性

当 $r > 0$ 为正整数时 $J_{-n}(x), J_n(x)$ 线性相关
当 $r = 0$ 时 $J_{-0}(x) = J_0(x)$, 仅有一解

另寻线性无关基解

$$\rho = \begin{cases} r > 0 \\ -r < 0 \end{cases} \begin{cases} r \neq N^+ \\ r = N^+ \end{cases} \quad (r \geq 0)$$

$$x^2 y'' + xy' + (x^2 - r^2)y = 0$$
$$J_r(x) = \left(\frac{x}{2}\right)^r \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+r+1)} \left(\frac{x}{2}\right)^{2k}$$
$$J_{-r}(x) = \left(\frac{x}{2}\right)^{-r} \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k-r+1)} \left(\frac{x}{2}\right)^{2k}$$

摄动法 对于任意非负整数 $n \geq 0$, 假设阶数 $n < r < n+1$

定义 $N_r(x) = \frac{J_r(x) \cos(r\pi) - J_{-r}(x)}{\sin(r\pi)} \rightarrow$ 是Bessel方程解

$$N_n(x) = \lim_{r \rightarrow n^+} N_r(x)$$
$$= \lim_{r \rightarrow n^+} \frac{J_r(x) \cos(r\pi) - J_{-r}(x)}{\sin(r\pi)}$$

视为 x, r 二元函数

L'Hospital 法则

$$= \lim_{r \rightarrow n^+} \frac{\frac{\partial [J_r(x)]}{\partial r} \cos(r\pi) - \pi \sin(r\pi) J_r(x) - \frac{\partial [J_{-r}(x)]}{\partial r}}{\pi \cos(r\pi)}$$
$$= \frac{1}{\pi} \left[\lim_{r \rightarrow n^+} \frac{\partial [J_r(x)]}{\partial r} - (-1)^n \lim_{r \rightarrow n^+} \frac{\partial [J_{-r}(x)]}{\partial r} \right]$$

数学与统计学院

C. Neumann
1832-1925
German Math.



- 1. 该极限存在 $\rightarrow$ 第二类Bessel函数
 - 2. $J_n(0) = 0, 1$ 有限值 $\lim_{x \rightarrow 0^+} N_n(x) = -\infty$
- $J_n(x), N_n(x)$ 线性无关 $\rightarrow$ 整数阶Bessel方程基解 $\{J_n(x), N_n(x)\}$

3.2 Bessel函数

$$x^2 y'' + xy' + (x^2 - r^2) y = 0$$

Bessel方程/Bessel函数 **小结**

$$y(x) = \sum_{n=0}^{\infty} a_n x^{n+\rho}, \quad (a_0 \neq 0)$$

r阶Bessel方程		r阶Bessel函数	
$x^2 y'' + xy' + (x^2 - r^2) y = 0$			
对任意 $r \geq 0$ 无论是否为 整数	$y(x) = C_1 J_r(x) + C_2 N_r(x)$	$J_r(x) = \left(\frac{x}{2}\right)^r \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+r+1)} \left(\frac{x}{2}\right)^{2k}$ $N_n(x) = \lim_{r \rightarrow n^+} \frac{J_r(x) \cos(r\pi) - J_{-r}(x)}{\sin(r\pi)}$	
对任意 $r > 0$ 且不为整数 时, 通解为	$y(x) = C_1 J_r(x) + C_2 J_{-r}(x)$	$J_r(x) = \left(\frac{x}{2}\right)^r \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+r+1)} \left(\frac{x}{2}\right)^{2k}$ $J_{-r}(x) = \left(\frac{x}{2}\right)^{-r} \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k-r+1)} \left(\frac{x}{2}\right)^{2k}$	

3.2 Bessel函数

Bessel方程/Bessel函数

例1 写出以下方程的通解

1 $x^2 y'' + xy' + (x^2 - 2)y = 0$

$y(x) = C_1 J_{\sqrt{2}}(x) + C_2 J_{-\sqrt{2}}(x)$ $y(x) = C_1 J_{\sqrt{2}}(x) + C_2 N_{\sqrt{2}}(x)$

2 $x^2 y'' + xy' + (4x^2 - 2)y = 0$

解 令 $t = 2x$ 原方程中 $y' = y'(x)$

$$y'(x) = y'(t) \frac{dt}{dx} = 2y'(t)$$

$$y''(x) = y''(t) \frac{dt}{dx} = 4y''(t)$$

$$\frac{t^2}{4} \cdot 4y''(t) + \frac{t}{2} \cdot 2y'(t) + (t^2 - 2)y = 0$$

$$t^2 y''(t) + ty'(t) + (t^2 - (\sqrt{2})^2)y = 0$$

得 $y(t) = C_1 J_{\sqrt{2}}(t) + C_2 J_{-\sqrt{2}}(t)$

$$y(x) = C_1 J_{\sqrt{2}}(2x) + C_2 J_{-\sqrt{2}}(2x)$$

$$y(x) = C_1 J_{\sqrt{2}}(2x) + C_2 N_{\sqrt{2}}(2x)$$

还原
变量

3 $x^2 y'' + xy' + (x^2 - 16)y = 0$

$$y(x) = C_1 J_4(x) + C_2 N_4(x)$$

4 $xy'' + y' + xy = 0$

$$y(x) = C_1 J_0(x) + C_2 N_0(x)$$

$$x^2 y'' + xy' + (x^2 - r^2)y = 0$$

$$J_r(x) = \left(\frac{x}{2}\right)^r \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+r+1)} \left(\frac{x}{2}\right)^{2k}$$

$$J_{-r}(x) = \left(\frac{x}{2}\right)^{-r} \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k-r+1)} \left(\frac{x}{2}\right)^{2k}$$

$$N_r(x) = \frac{J_r(x) \cos(r\pi) - J_{-r}(x)}{\sin(r\pi)}$$

$$x^2 y'' + xy' + (x^2 - r^2)y = 0$$

3.2 Bessel函数

Bessel方程/Bessel函数

例2 求ODE的通解 $x^2 y'' + 2xy' + [x^2 - r(r+1)]y = 0$

解 函数代换, 令 $y(x) = x^{-\frac{1}{2}}u(x)$ 各阶导数 $y'(x) = -\frac{1}{2}x^{-\frac{3}{2}}u(x) + x^{-\frac{1}{2}}u'(x)$
 $y''(x) = \frac{3}{4}x^{-\frac{5}{2}}u(x) - \frac{1}{2}x^{-\frac{3}{2}}u'(x) - \frac{1}{2}x^{-\frac{3}{2}}u'(x) + x^{-\frac{1}{2}}u''(x) = \frac{3}{4}x^{-\frac{5}{2}}u(x) - x^{-\frac{3}{2}}u'(x) + x^{-\frac{1}{2}}u''(x)$

代入原方程

$$x^2 \left[\frac{3}{4}x^{-\frac{5}{2}}u(x) - x^{-\frac{3}{2}}u'(x) + x^{-\frac{1}{2}}u''(x) \right] + 2x \left[-\frac{1}{2}x^{-\frac{3}{2}}u(x) + x^{-\frac{1}{2}}u'(x) \right] + [x^2 - r(r+1)]x^{-\frac{1}{2}}u(x) = 0$$

$$\left[\frac{3}{4}x^{-\frac{1}{2}}u(x) - x^{\frac{1}{2}}u'(x) + x^{\frac{3}{2}}u''(x) \right] + \left[-x^{-\frac{1}{2}}u(x) + 2x^{\frac{1}{2}}u'(x) \right] + [x^2 - r(r+1)]x^{-\frac{1}{2}}u(x) = 0$$

两边同乘以 $x^{\frac{1}{2}}$

$$\left[\frac{3}{4}u(x) - xu'(x) + x^2u''(x) \right] + [-u(x) + 2xu'(x)] + [x^2 - r(r+1)]u(x) = 0$$

$$x^2u''(x) + xu'(x) - \frac{1}{4}u(x) + [x^2 - r(r+1)]u(x) = 0$$

$$x^2u''(x) + xu'(x) + \left[x^2 - r(r+1) - \frac{1}{4} \right]u(x) = 0$$

配平方

$$x^2u''(x) + xu'(x) + \left[x^2 - \left(r + \frac{1}{2} \right)^2 \right]u(x) = 0$$

Bessel方程

原方程的通解可表示为

$$y(x) = x^{-\frac{1}{2}}u(x) = x^{-\frac{1}{2}} \left[C_1 J_{r+\frac{1}{2}}(x) + C_2 N_{r+\frac{1}{2}}(x) \right]$$

可能是整数

$$r + \frac{1}{2} \geq 0$$

1. 数学建模和基本原理介绍

2. 分离变量法

3. Bessel函数

4. 积分变换法

5. Green函数法

6. 特征线法

7. Legendre多项式

3.1 二阶线性常微分方程的幂级数解法

3.2 Bessel函数

3.3 多个自变量分离变量法举例

Gamma函数

Bessel方程和Bessel函数

Bessel函数的性质

Bessel方程的特征值问题

圆域上Laplace算子的特征值问题

3.2 Bessel函数

第一类Bessel函数

收敛半径

Bessel函数幂级数部分 $S_k(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k \pm r + 1)} \left(\frac{x}{2}\right)^{2k}$

$$J_r(x) = \left(\frac{x}{2}\right)^r \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+r+1)} \left(\frac{x}{2}\right)^{2k}$$

$$\sin x = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} x^{2k+1}$$

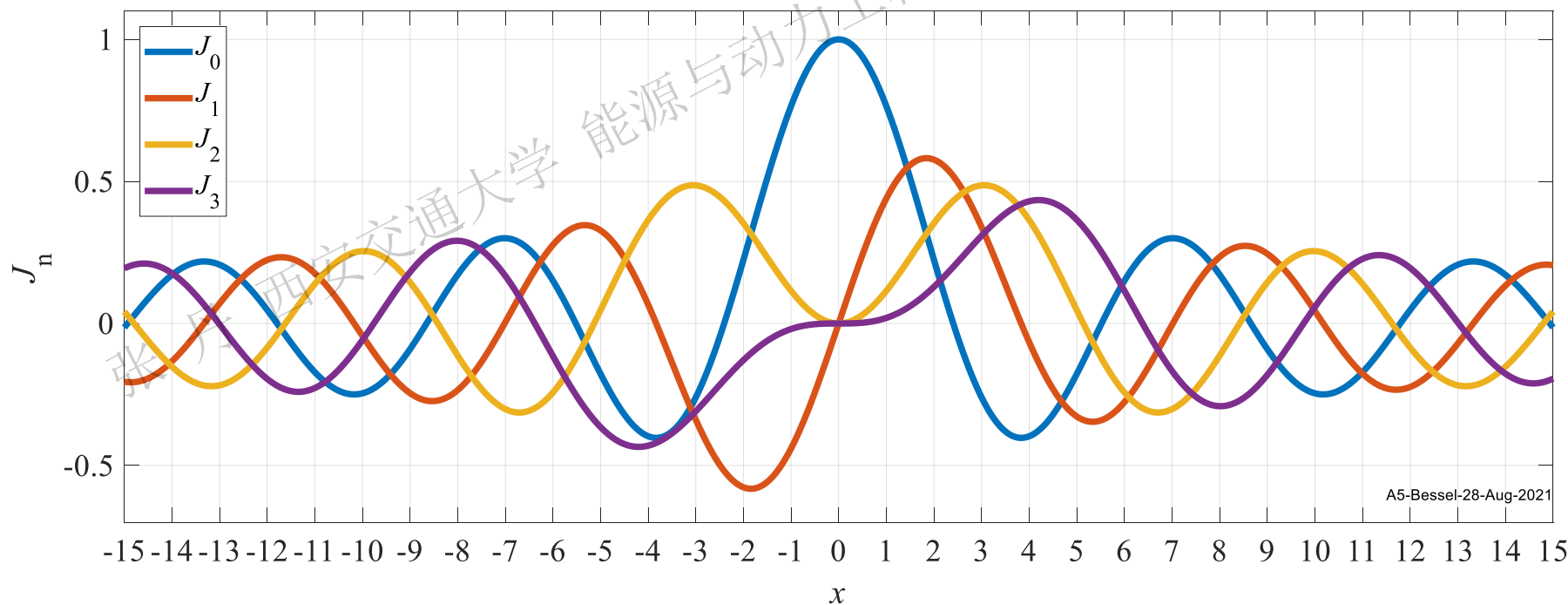
$$\cos x = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} x^{2k}$$

D'Alembert 判别法

$$\lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| = \lim_{n \rightarrow \infty} \frac{(n+1)! \Gamma(n+2+r) 2^{2n+2}}{n! \Gamma(n+1+r) 2^{2n}} = \lim_{n \rightarrow \infty} 4(n+1)(n+1+r) = \infty$$

Bessel函数 $J_r(x), J_{-r}(x)$ 收敛半径无穷大 $\rightarrow$ 幂级数在实数域 $\mathbf{R}$ 上收敛 $\rightarrow J_r(x)$ 解析

类似 $\sin(x), \cos(x) \rightarrow$ 后面重点讨论 r 为正整数 n 时 $\rightarrow J_n(x)$ 在实数域上的性质



3.2 Bessel函数

第一类Bessel函数

基本值

① $J_n(0) = \begin{cases} 1, & (n=0) \\ 0, & (n \geq 1) \end{cases}$

② $J'_n(0) = \begin{cases} 0, & (n=0, n > 1) \\ \frac{1}{2}, & (n=1) \end{cases}$

后续证明

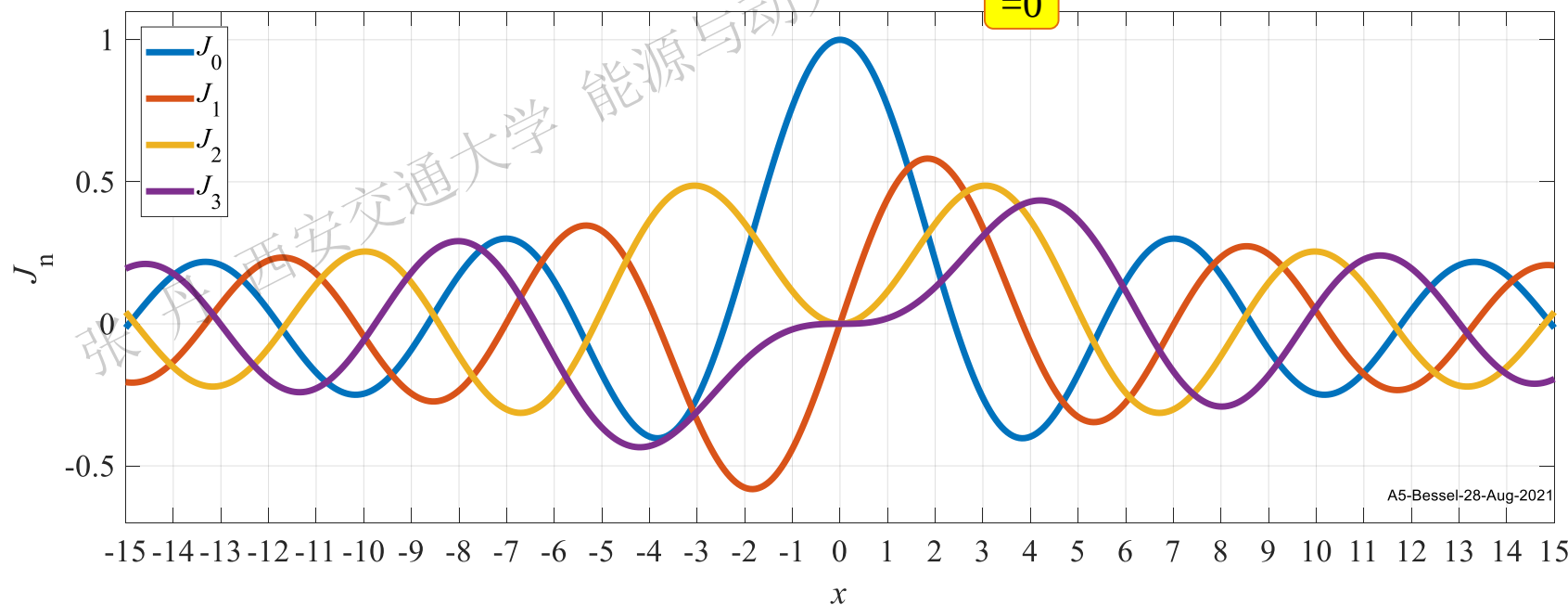
③ $|J_n(x)| < +\infty$ 有界性

证 ① $J_n(x) = \left(\frac{x}{2}\right)^n \left[\frac{1}{\Gamma(n+1)} + \sum_{k=1}^{\infty} \frac{(-1)^k}{k! \Gamma(k+n+1)} \left(\frac{x}{2}\right)^{2k} \right]$

$J_n(0)$ 的不同由幂函数造成

$$J_n(0) = \left(\frac{0}{2}\right)^n \left[\frac{1}{\Gamma(n+1)} + \sum_{k=1}^{\infty} \frac{(-1)^k}{k! \Gamma(k+n+1)} \left(\frac{0}{2}\right)^{2k} \right] = \begin{cases} 1, & (n=0) \\ 0, & (n > 0) \end{cases}$$

=0



3.2 Bessel函数

第一类Bessel函数

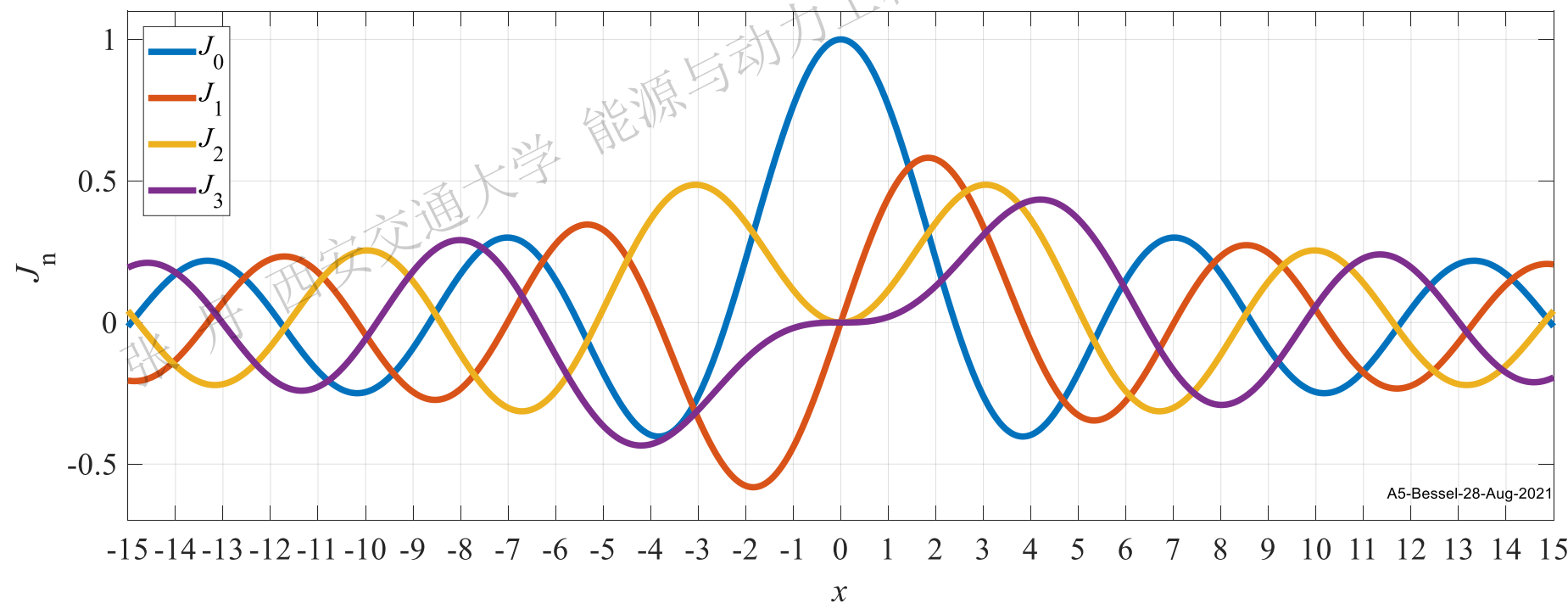
奇偶性

$$J_n(x) = (-1)^n J_n(-x) \begin{cases} n \text{ 为奇数时, } J_n(x) \text{ 为奇函数} \\ n \text{ 为偶数时, } J_n(x) \text{ 为偶函数} \end{cases}$$

$$J_n(x) = \left(\frac{x}{2}\right)^n \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+n+1)} \left(\frac{x}{2}\right)^{2k}$$

证 由定义式

$$J_n(-x) = \left(\frac{-x}{2}\right)^n \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+n+1)} \left(\frac{-x}{2}\right)^{2k} = (-1)^n \left(\frac{x}{2}\right)^n \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+n+1)} \left(\frac{x}{2}\right)^{2k} = (-1)^n J_n(x)$$



3.2 Bessel函数

第一类Bessel函数

递推性

证① $\frac{d}{dx} [x^n J_n(x)]$

① $\frac{d}{dx} [x^n J_n(x)] = x^n J_{n-1}(x)$

② $\frac{d}{dx} [x^{-n} J_n(x)] = -x^{-n} J_{n+1}(x)$

③ $J_{n-1}(x) + J_{n+1}(x) = \frac{2n}{x} J_n(x)$

④ $J_{n-1}(x) - J_{n+1}(x) = 2J'_n(x)$

分子/分母
同乘以 2^n

$$= \frac{d}{dx} \left[x^n \left(\frac{x}{2} \right)^n \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+n+1)} \left(\frac{x}{2} \right)^{2k} \right] = \frac{d}{dx} \left[2^n \frac{x^{2n}}{2^n \cdot 2^n} \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+n+1)} \left(\frac{x}{2} \right)^{2k} \right]$$

$$= 2^n \cdot \frac{d}{dx} \left[\sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+n+1)} \left(\frac{x}{2} \right)^{2k+2n} \right]$$

与幂次合并
→方便求导

$$= 2^n \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+n+1)} \frac{1}{2} \left(\frac{x}{2} \right)^{2k+2n-1}$$

$$\frac{\alpha}{\Gamma(\alpha+1)} = \frac{1}{\Gamma(\alpha)}$$

$$\Gamma(\alpha+1) = \alpha \cdot \Gamma(\alpha)$$

$$= 2^n \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+n-1+1)} \left(\frac{x}{2} \right)^{2k} \cdot \left(\frac{x}{2} \right)^n \cdot \left(\frac{x}{2} \right)^{n-1}$$

$$= 2^n \frac{x^n}{2^n} \cdot \left[\left(\frac{x}{2} \right)^{n-1} \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+n-1+1)} \left(\frac{x}{2} \right)^{2k} \right]$$

$$= x^n J_{n-1}(x)$$

同理可证

$$\frac{d}{dx} [x^{-n} J_n(x)] = -x^{-n} J_{n+1}(x)$$

3.2 Bessel函数

第一类Bessel函数

递推性

$$J_n(x) = \left(\frac{x}{2}\right)^n \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+n+1)} \left(\frac{x}{2}\right)^{2k}$$

$$\textcircled{1} \quad \frac{d}{dx} [x^n J_n(x)] = x^n J_{n-1}(x)$$

$$\textcircled{2} \quad \frac{d}{dx} [x^{-n} J_n(x)] = -x^{-n} J_{n+1}(x)$$

$$\textcircled{3} \quad J_{n-1}(x) + J_{n+1}(x) = \frac{2n}{x} J_n(x)$$

$$\textcircled{4} \quad J_{n-1}(x) - J_{n+1}(x) = 2J'_n(x)$$

证 式 $\textcircled{1}$ $\textcircled{2}$ 左端按乘积求导法则展开

$$x^n J'_n(x) + n x^{n-1} J_n(x) = x^n J_{n-1}(x)$$

$$x^{-n} J'_n(x) - n x^{-n-1} J_n(x) = -x^{-n} J_{n+1}(x)$$

约去 x^{n-1}, x^{-n-1} ↓

$$\left. \begin{aligned} xJ'_n(x) + nJ_n(x) &= xJ_{n-1}(x) \\ xJ'_n(x) - nJ_n(x) &= -xJ_{n+1}(x) \end{aligned} \right\} \begin{aligned} &\text{相减} \quad J_{n-1}(x) + J_{n+1}(x) = \frac{2n}{x} J_n(x) \quad \textcircled{3} \\ &\text{相加} \quad J_{n-1}(x) - J_{n+1}(x) = 2J'_n(x) \quad \textcircled{4} \end{aligned}$$

后续计算→高阶Bessel函数尽量
用低阶Bessel表示→变换下标

$$J_{n+2}(x) = \frac{2(n+1)}{x} J_{n+1}(x) - J_n(x) = J_n(x) - 2J'_{n+1}(x)$$

$$J_n(x) = \frac{(n+1)}{x} J_{n+1}(x) + J'_{n+1}(x)$$

1. 任意阶Bessel函数最终可用 $J_0(x), J_1(x)$ 表示
2. 递推式对非整数阶Bessel函数亦成立

3.2 Bessel函数

第一类Bessel函数

递推性 基本值

利用导数递推式
可证明基本值

$$J'_n(0) = \begin{cases} 0, & (n=0, n>1) \\ \frac{1}{2}, & (n=1) \end{cases}$$

$$J_{n-1}(x) + J_{n+1}(x) = \frac{2n}{x} J_n(x)$$

$$J_{n-1}(x) - J_{n+1}(x) = 2J'_n(x)$$

$$J_n(x) = \left(\frac{x}{2}\right)^n \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+n+1)} \left(\frac{x}{2}\right)^{2k}$$

证 ① 当 $n=0$ 时

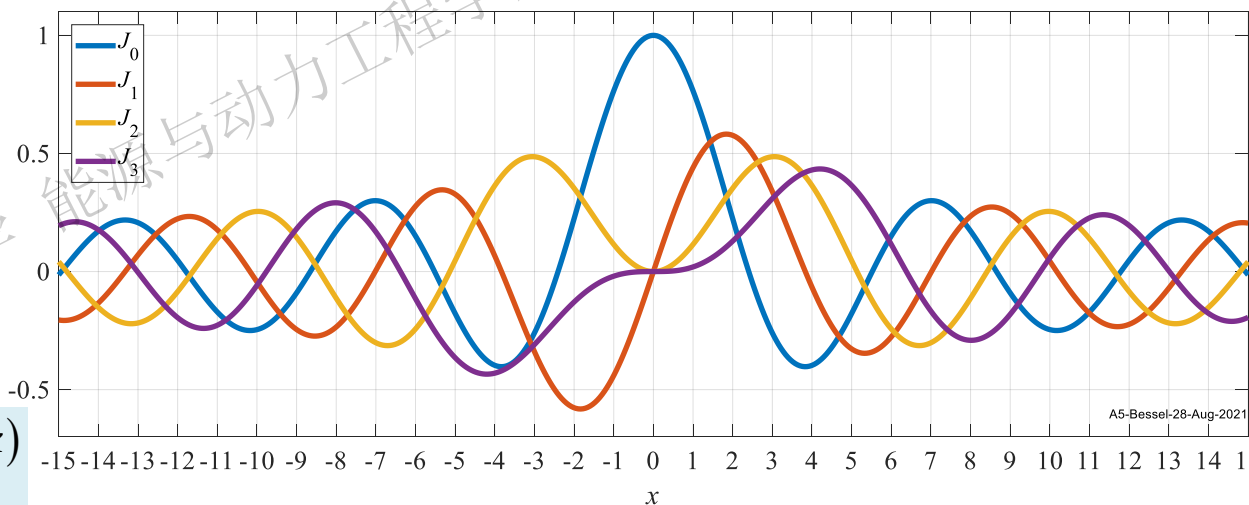
$$J'_0(0) = \frac{d}{dx} [1 \cdot J_0(0)]$$

$$= \frac{d}{dx} [x^{-0} J_0(0)] = -\frac{J_1(0)}{x^0}$$

$$= -\frac{J_1(0)}{1}$$

$$= 0$$

$$\frac{d}{dx} [x^{-n} J_n(x)] = -x^{-n} J_{n+1}(x)$$



② 当 $n=1$ 时

$$J'_1(0) = \frac{1}{2} [J_0(0) - J_2(0)]$$

$$= \frac{1}{2} (1 - 0)$$

$$= \frac{1}{2}$$

$$J_{n-1}(x) - J_{n+1}(x) = 2J'_n(x)$$

③ 当 $n>1$ 时 $J'_n(0) = \frac{1}{2} [J_{n-1}(0) - J_{n+1}(0)] = \frac{1}{2} [0 - 0] = 0$

3.2 Bessel函数

第一类Bessel函数 递推性

Bessel函数积分公式

$$J_n(x) = \left(\frac{x}{2}\right)^n \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+n+1)} \left(\frac{x}{2}\right)^{2k}$$

$$\textcircled{1} \quad \frac{d}{dx} [x^n J_n(x)] = x^n J_{n-1}(x)$$

$$\textcircled{2} \quad \frac{d}{dx} [x^{-n} J_n(x)] = -x^{-n} J_{n+1}(x)$$

$$\textcircled{3} \quad J_{n-1}(x) + J_{n+1}(x) = \frac{2n}{x} J_n(x)$$

$$\textcircled{4} \quad J_{n-1}(x) - J_{n+1}(x) = 2J'_n(x)$$

变形 ① 式 → 升阶积分公式 → 积分值用相邻高阶Bessel函数表示

$$\frac{d}{dx} [x^n J_n(x)] = x^n J_{n-1}(x) \quad \rightarrow \quad \int x^n J_{n-1}(x) dx = x^n J_n(x) + C$$

变形 ② 式 → 降阶积分公式 → 积分值用相邻低阶Bessel函数表示

$$\frac{d}{dx} [x^{-n} J_n(x)] = -x^{-n} J_{n+1}(x) \quad \rightarrow \quad \int x^{-n} J_{n+1}(x) dx = -x^{-n} J_n(x) + C$$
$$\int \frac{J_{n+1}(x)}{x^n} dx = -\frac{J_n(x)}{x^n} + C$$

本质是含有Bessel函数的积分 → 后续Bessel展开/极坐标PDE的求解发挥重要作用

3.2 Bessel函数

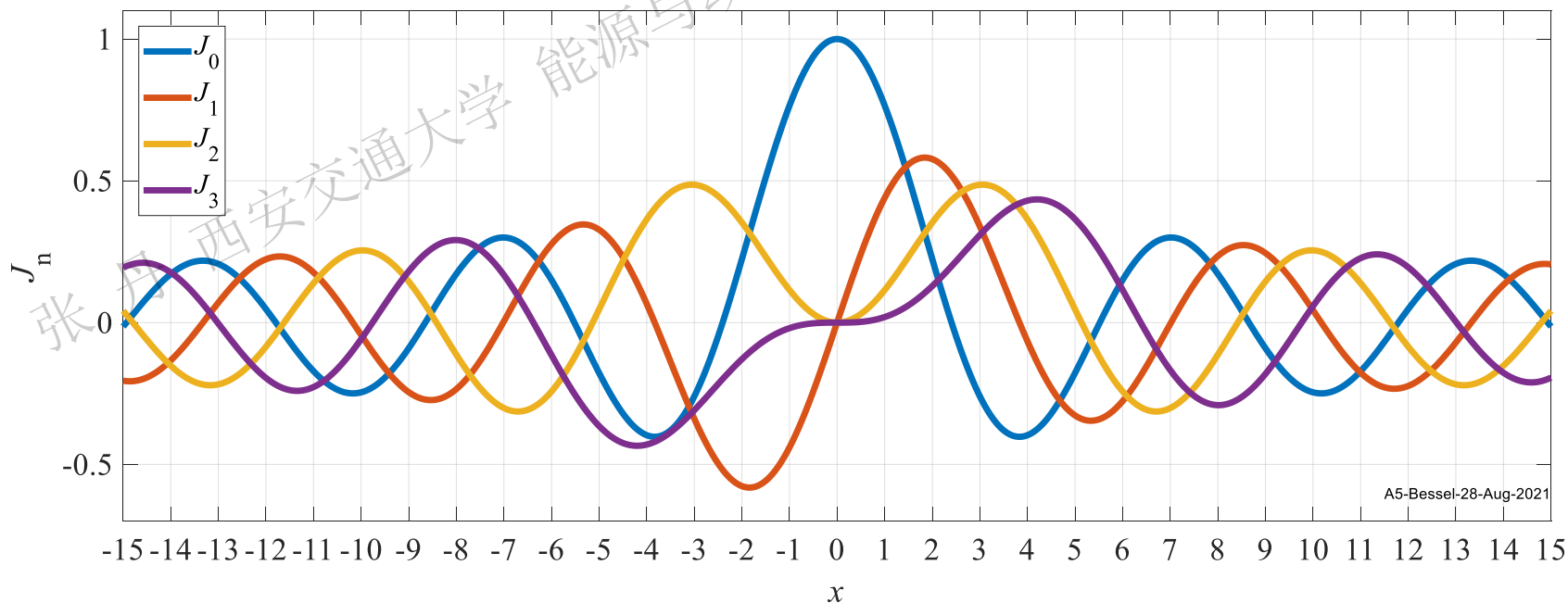
第一类Bessel函数

零点 方程 $J_n(x)=0$ 的根

1. 零点表示符号

- Bessel函数 $J_n(x)$ 有**无数**个**实零点**, 实零点在 x 轴上关于 原点 $x=0$ 对称→关注**正根**
- 原点 $x=0$ 不是 $J_0(x)$ 的零点; 但是 $J_n(x)$ ($n > 0$) 的 **n 重零点**. 如 $x=0$ 即是 $J_3(x)$ 的零点也是 $J_5(x)$ 的零点
- 但除 $x=0$ 外, $J_n(x)$ 的其他零点均为单零点. 如除 $x=0$ 外, $J_3(x)$ 的其他零点一定不是 $J_5(x)$ 的零点

$$J_n[\mu_m^{(n)}] = 0, J_q[\mu_m^{(n)}] \neq 0 \quad (m \geq 1, n \neq q)$$



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Bessel函数
阶数

$\mu_m^{(n)}$

原点起正根
序号

$$J_n(x) = \left(\frac{x}{2}\right)^n \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+n+1)} \left(\frac{x}{2}\right)^{2k}$$

3.2 Bessel函数

第一类Bessel函数

零点

方程 $J_n(x) = 0$ 的根

$\mu_m^{(n)}$

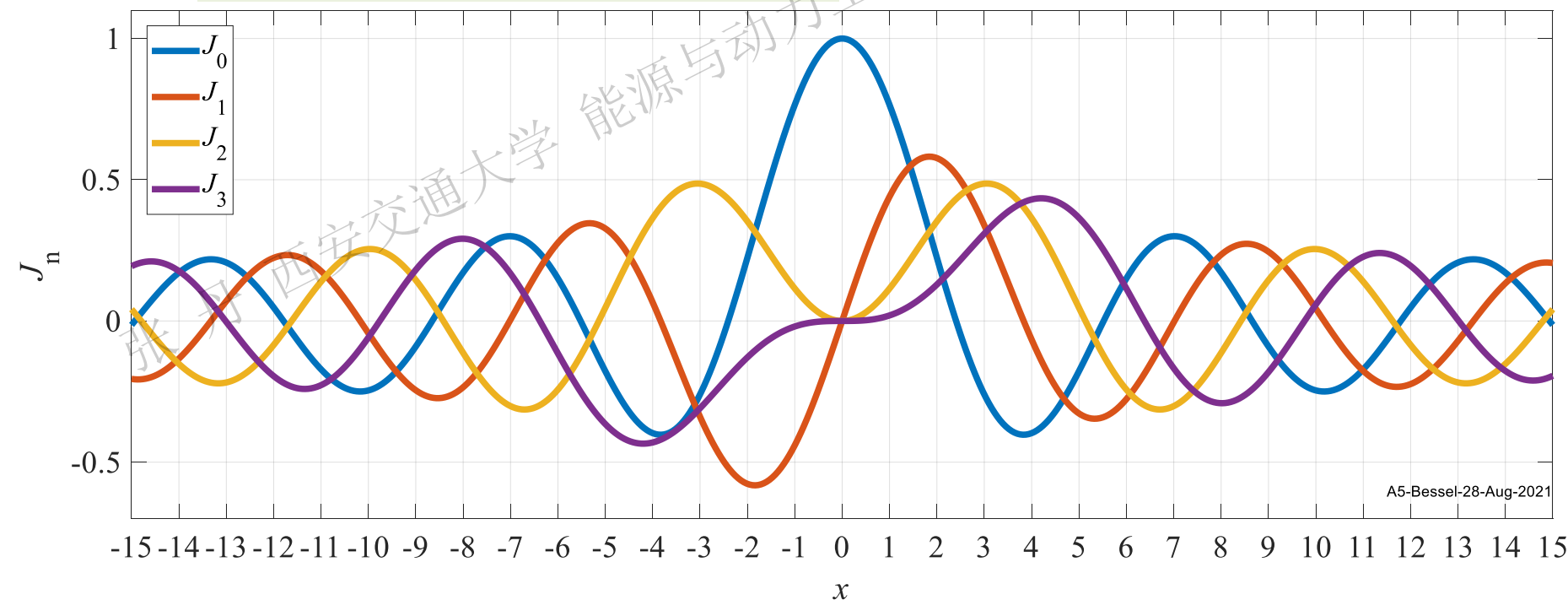
Bessel函数阶数

原点起正根序号

$$J_n(x) = \left(\frac{x}{2}\right)^n \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+n+1)} \left(\frac{x}{2}\right)^{2k}$$

5. 相邻阶数的零点交错分布. 如 2 个 $J_1(x)$ 的零点之间只有 1 个 $J_2(x)$ 的零点
6. 随 x 的增大 $\rightarrow$ 任意阶 Bessel 函数的零点间距越来越接近 π , $\lim_{m \rightarrow \infty} (\mu_{m+1}^{(n)} - \mu_m^{(n)}) = \pi$
7. 随 x 的增大 $\rightarrow$ 任意阶 Bessel 函数 $\rightarrow$ 渐进表达式 $\rightarrow$ 趋近于三角函数 $\rightarrow$ 减幅振荡/有界

$$\lim_{x \rightarrow \infty} J_n(x) = \sqrt{\frac{2}{\pi x}} \cos\left(x - \frac{n\pi}{2} - \frac{\pi}{4}\right) + O\left(x^{-\frac{3}{2}}\right) \rightarrow |J_n(x)| < +\infty \text{ 有界性}$$



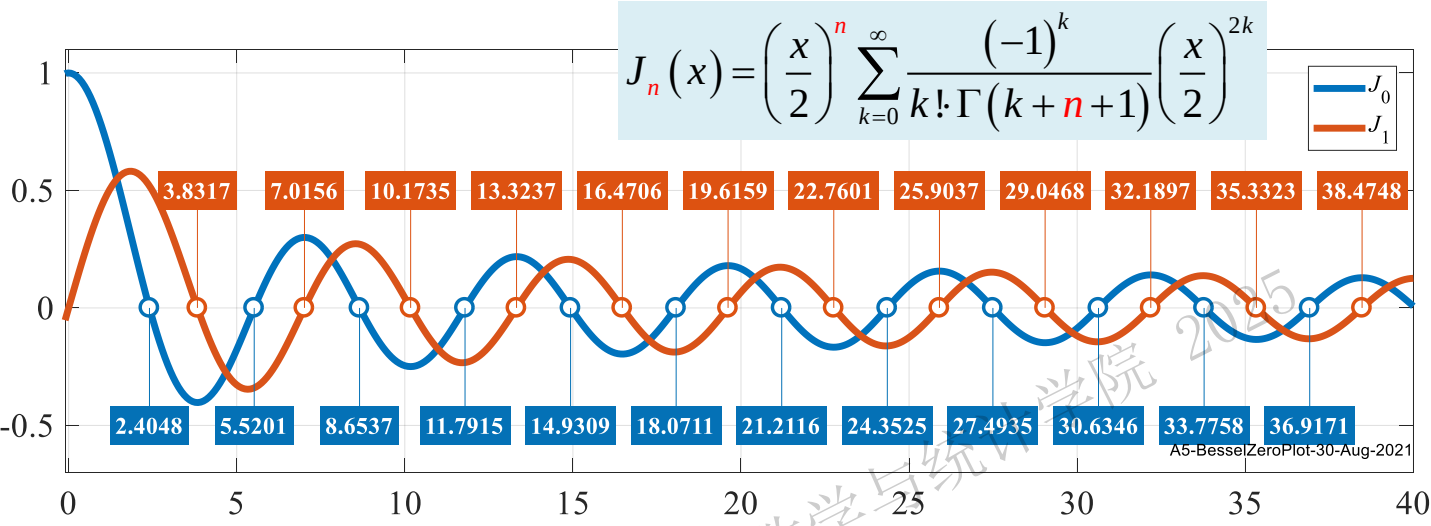
3.2 Bessel函数

第一类Bessel函数

零点

符号/无解析式

$\mu_m^{(n)}$



正零点→时空分离常数/Bessel展开→但Bessel函数正零点无解析式→数值计算→高精度近似值

各阶Bessel函数的正零点

$J_n(x)$ $\mu_m^{(n)}$	0	1	2	3	4	5	
1	2.404826	3.831706	5.135622	6.380162	7.588342	8.771484	
2	5.520078	7.015587	8.417244	9.761023	11.06471	12.33860	
3	8.653728	10.17347	11.61984	13.01520	14.37254	15.70017	
4	11.79153	13.32369	14.79595	16.22347	17.61597	18.98013	
5	14.93092	16.47063	17.95982	19.40942	20.82693	22.21780	
6	18.07106	19.61586	21.11700	22.58273	24.01902	25.43034	
7	21.21164	22.76008	24.27011	25.74817	27.19909	28.62662	
8	24.35247	25.90367	27.42057	28.90835	30.37101	31.81172	
9	27.49348	29.04683	30.56920	32.06485	33.53714	34.98878	
10	30.63461	32.18968	33.71652	35.21867	36.69900	38.15987	
11	33.77582	35.33231	36.86286	38.37047	39.85763	41.32638	
.....							



3.2 Bessel函数

第一类Bessel函数

小结

$$J_n(x) = \left(\frac{x}{2}\right)^n \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+n+1)} \left(\frac{x}{2}\right)^{2k}$$

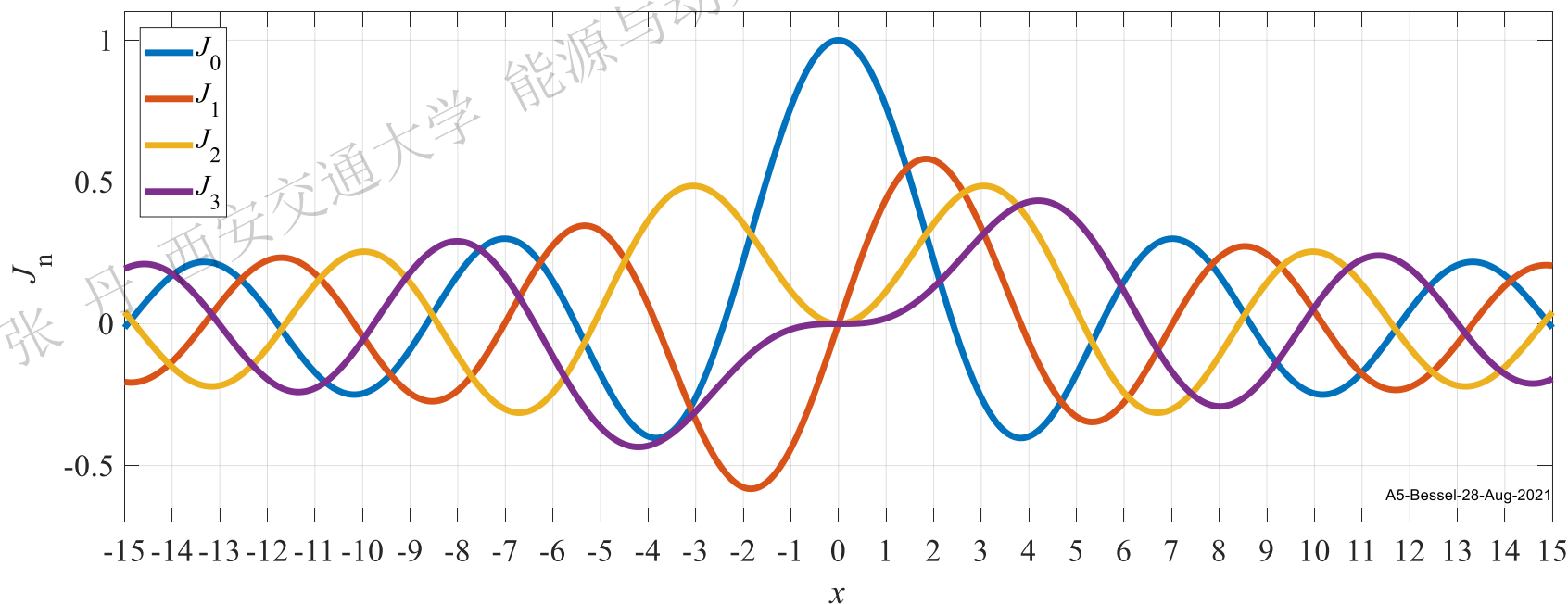
$$J_n(0) = \begin{cases} 1, & (n=0) \\ 0, & (n \geq 1) \end{cases} \quad J'_n(0) = \begin{cases} 0, & (n=0, n > 1) \\ \frac{1}{2}, & (n=1) \end{cases}$$

$$J_n(-x) = (-1)^n J_n(x) \quad \frac{d}{dx} [x^n J_n(x)] = x^n J_{n-1}(x) \quad \frac{d}{dx} [x^{-n} J_n(x)] = -x^{-n} J_{n+1}(x)$$

$$\int x^n J_{n-1}(x) dx = x^n J_n(x) + C, \quad \int \frac{J_{n+1}(x)}{x^n} dx = -\frac{J_n(x)}{x^n} + C$$

$$J_{n-1}(x) + J_{n+1}(x) = \frac{2n}{x} J_n(x) \quad J_{n-1}(x) - J_{n+1}(x) = 2J'_n(x)$$

$$\lim_{m \rightarrow \infty} (\mu_{m+1}^{(n)} - \mu_m^{(n)}) = \pi \quad \lim_{x \rightarrow \infty} J_n(x) = \sqrt{\frac{2}{\pi x}} \cos\left(x - \frac{n\pi}{2} - \frac{\pi}{4}\right) + O\left(x^{-\frac{3}{2}}\right)$$



3.2 Bessel函数

$$J_n(x) = \left(\frac{x}{2}\right)^n \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+n+1)} \left(\frac{x}{2}\right)^{2k}$$

第二类Bessel函数 常识性的结论

$$J_r(x) = \left(\frac{x}{2}\right)^r \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+r+1)} \left(\frac{x}{2}\right)^{2k} \quad N_n(x) = \lim_{r \rightarrow n^+} \frac{J_r(x) \cos(r\pi) - J_{-r}(x)}{\sin(r\pi)}$$

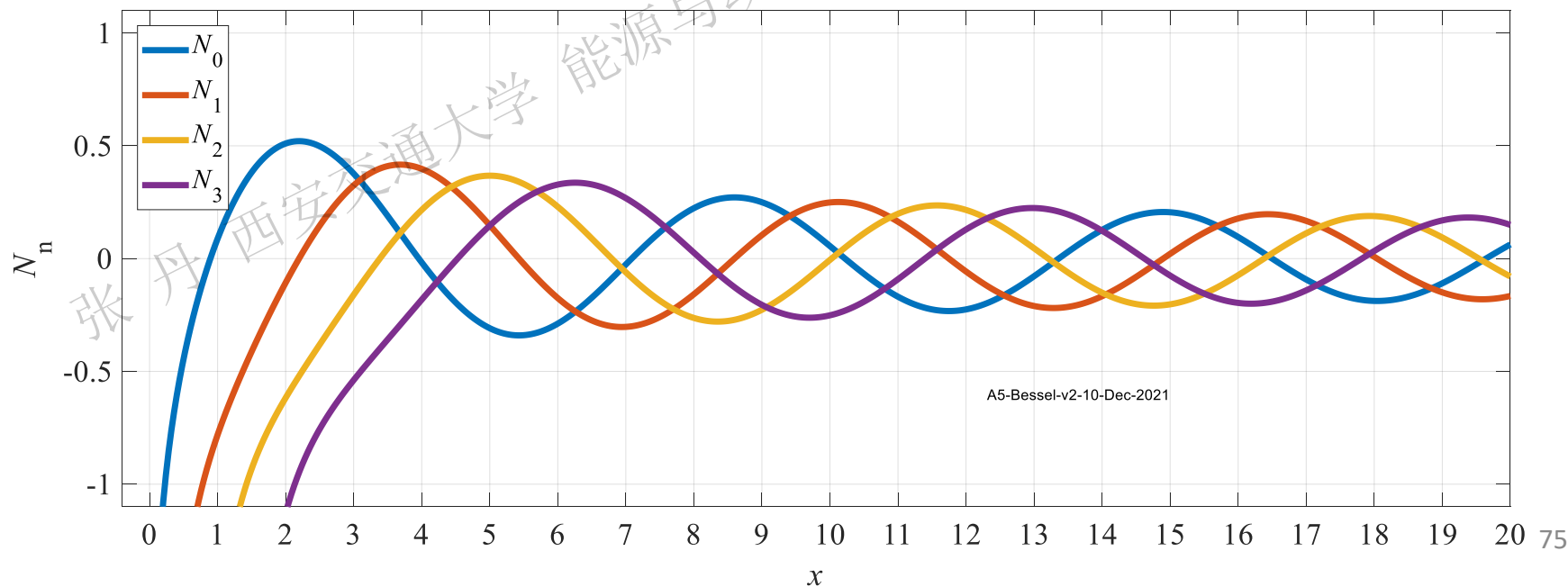
$$N_n(0) = -\infty, \quad N_n(-x) = (-1)^n N_n(x)$$

$$\frac{d}{dx} [x^n N_n(x)] = x^n N_{n-1}(x), \quad \frac{d}{dx} [x^{-n} N_n(x)] = -x^{-n} N_{n+1}(x)$$

$$N_{n-1}(x) + N_{n+1}(x) = \frac{2n}{x} N_n(x), \quad N_{n-1}(x) - N_{n+1}(x) = 2N'_n(x)$$

$$N_n(x) = \sqrt{\frac{2}{x\pi}} \sin\left(x - \frac{1}{4}\pi - \frac{n}{2}\pi\right) + O\left(\frac{1}{x^{3/2}}\right), \quad \lim_{x \rightarrow \infty} J_n(x) = \lim_{x \rightarrow \infty} N_n(x) = 0$$

减幅振荡



3.2 Bessel函数

Bessel函数比较

$$J_n(0) = \begin{cases} 1, & (n=0) \\ 0, & (n \geq 1) \end{cases} \quad J'_n(0) = \begin{cases} 0, & (n=0, n > 1) \\ \frac{1}{2}, & (n=1) \end{cases}$$

$$J_n(-x) = (-1)^n J_n(x) \quad \frac{d}{dx} [x^n J_n(x)] = x^n J_{n-1}(x)$$

$$\frac{d}{dx} [x^{-n} J_n(x)] = -x^{-n} J_{n+1}(x)$$

$$\int x^n J_{n-1}(x) dx = x^n J_n(x) + C, \quad \int \frac{J_{n+1}(x)}{x^n} dx = -\frac{J_n(x)}{x^n} + C$$

$$J_{n-1}(x) + J_{n+1}(x) = \frac{2n}{x} J_n(x) \quad J_{n-1}(x) - J_{n+1}(x) = 2J'_n(x)$$

$$\lim_{m \rightarrow \infty} (\mu_{m+1}^{(n)} - \mu_m^{(n)}) = \pi \quad \lim_{x \rightarrow \infty} J_n(x) = \sqrt{\frac{2}{\pi x}} \cos\left(x - \frac{n\pi}{2} - \frac{\pi}{4}\right) + O\left(x^{-\frac{3}{2}}\right)$$

$$J_r(x) = \left(\frac{x}{2}\right)^r \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+r+1)} \left(\frac{x}{2}\right)^{2k}, \quad N_n(x) = \lim_{r \rightarrow n^+} \frac{J_r(x) \cos(r\pi) - J_{-r}(x)}{\sin(r\pi)}$$

$$N_n(0) = -\infty, \quad N_n(-x) = (-1)^n N_n(x)$$

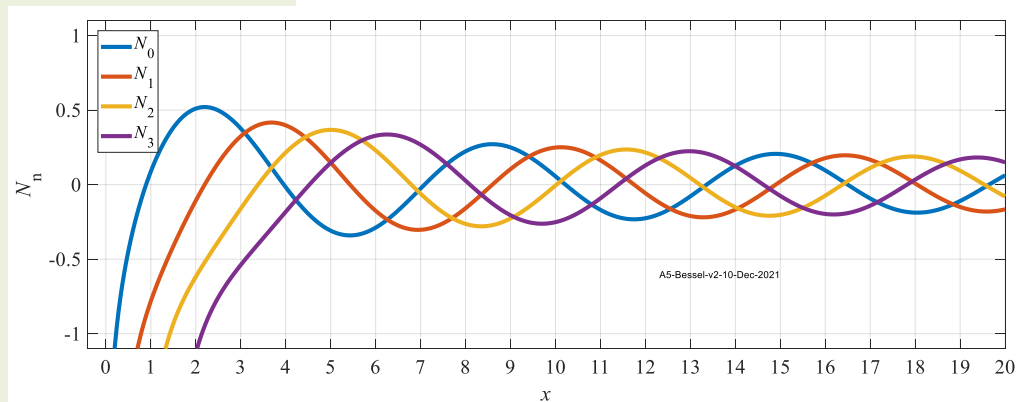
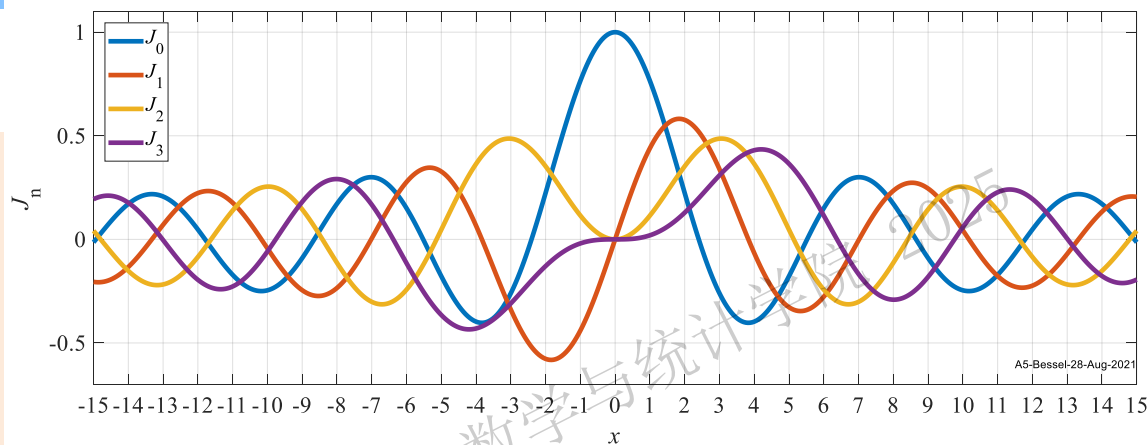
$$\frac{d}{dx} [x^n N_n(x)] = x^n N_{n-1}(x), \quad \frac{d}{dx} [x^{-n} N_n(x)] = -x^{-n} N_{n+1}(x)$$

$$N_{n-1}(x) + N_{n+1}(x) = \frac{2n}{x} N_n(x), \quad N_{n-1}(x) - N_{n+1}(x) = 2N'_n(x)$$

$$N_n(x) = \sqrt{\frac{2}{x\pi}} \sin\left(x - \frac{1}{4}\pi - \frac{n}{2}\pi\right) + O\left(\frac{1}{x^{3/2}}\right),$$

$$\lim_{x \rightarrow \infty} J_n(x) = \lim_{x \rightarrow \infty} N_n(x) = 0$$

$$J_n(x) = \left(\frac{x}{2}\right)^n \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+n+1)} \left(\frac{x}{2}\right)^{2k}$$



3.2 Bessel函数

$$J_n(x) = \left(\frac{x}{2}\right)^n \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+n+1)} \left(\frac{x}{2}\right)^{2k}$$

Bessel函数比较

例

下列说法**正确**的是

$N_n(x)$ 是有界函数

$N_n(x)$ 是奇函数

$N_n(x)$ 的零点无解析式

$N_n(x)$ 是非振荡衰减函数

对**整数阶**Bessel函数, 下列说法**错误**的是

$J_n(x)$ 与 $J_{n+1}(x)$ 的零点相间分布

$J_n(x)$ 与 $J_{-n}(x)$ 线性相关

$J_n(x) = 0$ 的根全部为单重根

$J'_n(x) = 0$ 有无穷多个零点

3.2 Bessel函数

$$J_n(x) = \left(\frac{x}{2}\right)^n \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+n+1)} \left(\frac{x}{2}\right)^{2k}$$

Bessel函数比较

例 当 $x \rightarrow 0$ 时, 以下表达式不等于0的是

$$J'_0(x)$$

$$\frac{1}{N_1(x)}$$

$$J'_1(x)$$

$$\frac{1}{\Gamma(x)}$$

张丹 西安交通大学 能源与动力工程学院 数学与统计学院 2025

快速计算手册

边界条件齐次化常用辅助函数 $u(x,t)=v(x,t)+w(x,t)$		
边界类型	边界条件	辅助函数
[1,1]	$u(0,t)=g_1(t), u(l,t)=g_2(t)$	$w(x,t)=\frac{g_2(t)-g_1(t)}{l}x+g_1(t)$
[2,1]	$u_x(0,t)=g_1(t), u(l,t)=g_2(t)$	$w(x,t)=g_1(t)(x-l)+g_2(t)$
[1,2]	$u(0,t)=g_1(t), u_x(l,t)=g_2(t)$	$w(x,t)=g_2(t)x+g_1(t)$
[2,2]	$u_x(0,t)=g_1(t), u_x(l,t)=g_2(t)$	$w(x,t)=\frac{g_2(t)-g_1(t)}{2l}x^2+g_1(t)x$

线性常微分方程解法

一阶

齐次 $x'+P(t)x=0$ 分离变量法

非齐次 $x'+P(t)x=f(t)$ 常数变易法

特征方程法 $\lambda^2+p\lambda+q=0$

常系数

齐次

非齐次

二阶 (高阶)

变系数

高阶 (可降阶)

多元实根 $\{e^{\alpha t}, e^{\beta t}\}$

相同实根 $\{te^{\alpha t}, e^{\alpha t}\}$

共轭复根 $\{e^{\alpha t} \cos \beta t, e^{\alpha t} \sin \beta t\}$

Liouville公式 基解1→基解2

摄动法

通解=通+特

求非齐特解

待定函数法

观察法

Euler方程 $t^2 \frac{d^2 x}{dt^2} + a_1 t \frac{dx}{dt} + a_2 x = F(t), t=e^t$

Bernoulli方程 $\frac{dy}{dx} + P(x)y = Q(x)y^\alpha, (\alpha \neq 0,1), u=y^{1-\alpha}$

Bessel方程 $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (x^2 - \nu^2)y = 0$

Legendre方程 $(1-x^2) \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} + n(n+1)y = 0$

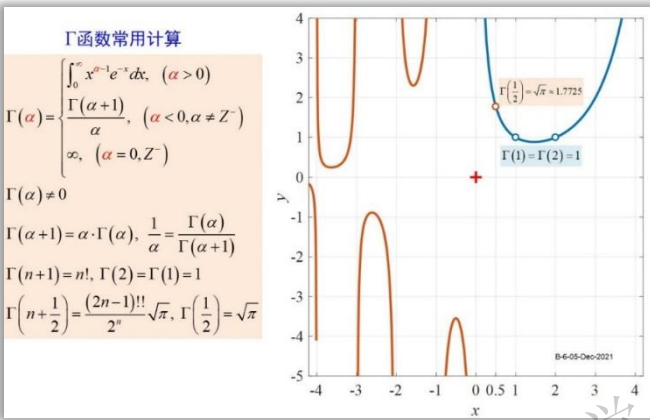
变量代换法

幂级数法

直接积分法

变量代换法

算子 $A = -\frac{d^2}{dx^2}$ 不同齐次边界条件下的特征值/特征函数		
齐次边界类型	特征值问题	特征值/特征函数
[1,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right), (n=1,2,3,\dots)$
[1,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \sin\left[\frac{(2n+1)\pi}{2l}x\right], (n=0,1,2,3,\dots)$
[2,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \cos\left[\frac{(2n+1)\pi}{2l}x\right], (n=0,1,2,3,\dots)$
[2,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \cos\left(\frac{n\pi}{l}x\right), (n=0,1,2,3,\dots)$
周期	$\begin{cases} \Phi''(\theta) + \lambda \Phi(\theta) = 0 \\ \Phi(0) = \Phi(2\pi), \\ \Phi'(0) = \Phi'(2\pi) \end{cases}$	$\lambda_0 = 0, \Phi_0(\theta) = 1, (n=0)$ $\lambda_n = n^2, \Phi_n(\theta) = C_1 \cos n\theta + C_2 \sin n\theta, (n=1,2,3,\dots)$



第一类Bessel函数 小结

$J_n(0) = \begin{cases} 1, (n=0) \\ 0, (n \geq 1) \end{cases}, J'_n(0) = \begin{cases} 0, (n=0, n>1) \\ \frac{1}{2}, (n=1) \end{cases}$

$J_{-n}(-x) = (-1)^n J_n(x), \frac{d}{dx}[x^n J_n(x)] = x^n J_{n-1}(x), \frac{d}{dx}[x^{-n} J_n(x)] = -x^{-n} J_{n+1}(x)$

$\int x^n J_{n-1}(x) dx = x^n J_n(x) + C, \int \frac{J_{n+1}(x)}{x^n} dx = -\frac{J_n(x)}{x^n} + C$

$J_{n-1}(x) + J_{n+1}(x) = \frac{2n}{x} J_n(x), J_{n-1}(x) - J_{n+1}(x) = 2J'_n(x)$

$\lim_{n \rightarrow \infty} (\mu_{n+1}^{(n)} - \mu_n^{(n)}) = \pi, \lim_{x \rightarrow \infty} J_n(x) = \sqrt{\frac{2}{\pi x}} \cos\left(x - \frac{n\pi}{2} - \frac{\pi}{4}\right) + O\left(x^{-\frac{3}{2}}\right)$

r 阶Bessel方程 $x^2 y'' + xy' + (x^2 - r^2)y = 0$		r 阶Bessel函数
对任意 $r > 0$ 无论是否为整数	$y(x) = C_1 J_r(x) + C_2 N_r(x)$	$J_r(x) = \left(\frac{x}{2}\right)^r \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+r+1)} \left(\frac{x}{2}\right)^{2k}$ $N_n(x) = \lim_{r \rightarrow n^+} \frac{J_r(x) \cos(r\pi) - J_{-r}(x)}{\sin(r\pi)}$
对任意 $r > 0$ 且不为整数时, 通解为	$y(x) = C_1 J_r(x) + C_2 J_{-r}(x)$	$J_r(x) = \left(\frac{x}{2}\right)^r \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+r+1)} \left(\frac{x}{2}\right)^{2k}$ $J_{-r}(x) = \left(\frac{x}{2}\right)^{-r} \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k-r+1)} \left(\frac{x}{2}\right)^{2k}$

3.2 Bessel函数

第一类Bessel函数

$$J_n(x) = \left(\frac{x}{2}\right)^n \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+n+1)} \left(\frac{x}{2}\right)^{2k}$$

例1

求不定积分 $\int x^3 J_0(x) dx$

解 $\int x^3 J_0(x) dx$

升阶积分

$$= \int x^2 [x \cdot J_0(x)] dx$$

$$= \int x^2 \cdot d[x \cdot J_1(x)]$$

$$= x^3 \cdot J_1(x) - \int x \cdot J_1(x) \cdot 2x \cdot dx$$

$$= x^3 \cdot J_1(x) - 2 \int x^2 \cdot J_1(x) dx$$

$$= x^3 \cdot J_1(x) - 2x^2 \cdot J_2(x) + C$$

$$\int x^{n+1} J_n(x) dx = x^{n+1} J_{n+1}(x) + C$$

$$\int x^{n+1} J_n(x) dx = x^{n+1} J_{n+1}(x) + C$$

技巧 $\int x^\alpha J_n(x) dx$

1. 积分 x 幂次与 $J_n(x)$ 阶次不一致 $\rightarrow$ 调整幂次 $\rightarrow$ 升降阶积分

2. 降阶积分中常需要 $1 = x^0$

求不定积分 $\int x J_2(x) dx$

解 $\int x J_2(x) dx$

降阶积分

$$= \int x^2 \left[\frac{J_2(x)}{x} \right] dx$$

$$= - \int x^2 d \left[\frac{J_1(x)}{x} \right]$$

$$= - \left[x J_1(x) - \int \frac{J_1(x)}{x} 2x dx \right]$$

$$= - \left[x J_1(x) - 2 \int \frac{J_1(x)}{x^0} dx \right]$$

$$= - \left[x J_1(x) + 2 \frac{J_0(x)}{x^0} \right] + C$$

$$= -x J_1(x) - 2 J_0(x) + C$$

$$\int \frac{J_n(x)}{x^{n-1}} dx = - \frac{J_{n-1}(x)}{x^{n-1}} + C$$

$$\int \frac{J_n(x)}{x^{n-1}} dx = - \frac{J_{n-1}(x)}{x^{n-1}} + C$$

3.2 Bessel函数

第一类Bessel函数

例2 求定积分

1

$$\int_0^1 x^2 J_1(2x) dx$$

解 变量代换 $t = 2x, x = \frac{t}{2}$

调整积分限

$$\int_0^1 x^2 J_1(2x) dx = \int_0^2 \left(\frac{t}{2}\right)^2 J_1(t) d\left(\frac{t}{2}\right)$$

$$= \frac{1}{8} \int_0^2 t^2 J_1(t) dt$$

升阶积分

$$= \frac{1}{8} t^2 J_2(t) \Big|_0^2$$

$$= \frac{1}{8} [2^2 J_2(2) - 0^2 J_2(0)]$$

=0

$$= \frac{1}{2} J_2(2)$$

$$J_n(x) = \left(\frac{x}{2}\right)^n \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+n+1)} \left(\frac{x}{2}\right)^{2k}$$

$$J_n(0) = \begin{cases} 1, & (n=0) \\ 0, & (n \geq 1) \end{cases} \quad J'_n(0) = \begin{cases} 0, & (n=0, n>1) \\ \frac{1}{2}, & (n=1) \end{cases}$$

$$J_n(-x) = (-1)^n J_n(x)$$

$$\frac{d}{dx} [x^n J_n(x)] = x^n J_{n-1}(x) \quad \frac{d}{dx} [x^{-n} J_n(x)] = -x^{-n} J_{n+1}(x)$$

$$J_{n-1}(x) + J_{n+1}(x) = \frac{2n}{x} J_n(x) \quad J_{n-1}(x) - J_{n+1}(x) = 2J'_n(x)$$

$$\int x^{n+1} J_n(x) dx = x^{n+1} J_{n+1}(x) + C, \quad \int \frac{J_n(x)}{x^{n-1}} dx = -\frac{J_{n-1}(x)}{x^{n-1}} + C$$

3.2 Bessel函数

第一类Bessel函数

例2 求定积分 2

$$\int_0^a x^3 J_0\left(\frac{\mu_m^{(0)}}{a} x\right) dx$$

解 变量代换 $t = \frac{\mu_m^{(0)}}{a} x, x = \frac{a}{\mu_m^{(0)}} t$
调整积分限

$$\begin{aligned} \int_0^a x^3 J_0\left(\frac{\mu_m^{(0)}}{a} x\right) dx &= \int_0^{\mu_m^{(0)}} \left(\frac{a}{\mu_m^{(0)}} t\right)^3 J_0(t) d\left(\frac{a}{\mu_m^{(0)}} t\right) \\ &= \left(\frac{a}{\mu_m^{(0)}}\right)^4 \int_0^{\mu_m^{(0)}} t^3 J_0(t) dt = \left(\frac{a}{\mu_m^{(0)}}\right)^4 \int_0^{\mu_m^{(0)}} t^2 \cdot t \cdot J_0(t) dt \\ &= \left(\frac{a}{\mu_m^{(0)}}\right)^4 \int_0^{\mu_m^{(0)}} t^2 \cdot d[t \cdot J_1(t)] \quad \text{升阶积分} \\ &= \left(\frac{a}{\mu_m^{(0)}}\right)^4 \left[t^3 \cdot J_1(t) \Big|_0^{\mu_m^{(0)}} - \int_0^{\mu_m^{(0)}} t \cdot J_1(t) 2t \cdot dt \right] \\ &= \left(\frac{a}{\mu_m^{(0)}}\right)^4 \left[t^3 \cdot J_1(t) \Big|_0^{\mu_m^{(0)}} - 2 \int_0^{\mu_m^{(0)}} t^2 J_1(t) dt \right] \quad \text{升阶积分} \end{aligned}$$

$$J_n(x) = \left(\frac{x}{2}\right)^n \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+n+1)} \left(\frac{x}{2}\right)^{2k}$$

$$J_n(0) = \begin{cases} 1, & (n=0) \\ 0, & (n \geq 1) \end{cases} \quad J'_n(0) = \begin{cases} 0, & (n=0, n > 1) \\ \frac{1}{2}, & (n=1) \end{cases}$$

$$J_n(-x) = (-1)^n J_n(x)$$

$$\frac{d}{dx} [x^n J_n(x)] = x^n J_{n-1}(x) \quad \frac{d}{dx} [x^{-n} J_n(x)] = -x^{-n} J_{n+1}(x)$$

$$J_{n-1}(x) + J_{n+1}(x) = \frac{2n}{x} J_n(x) \quad J_{n-1}(x) - J_{n+1}(x) = 2J'_n(x)$$

$$\int x^{n+1} J_n(x) dx = x^{n+1} J_{n+1}(x) + C, \quad \int \frac{J_n(x)}{x^{n-1}} dx = -\frac{J_{n-1}(x)}{x^{n-1}} + C$$

$$\begin{aligned} &= \left(\frac{a}{\mu_m^{(0)}}\right)^4 \left[t^3 \cdot J_1(t) \Big|_0^{\mu_m^{(0)}} - 2t^2 J_2(t) \Big|_0^{\mu_m^{(0)}} \right] \\ &= \left(\frac{a}{\mu_m^{(0)}}\right)^4 \left[\left(\mu_m^{(0)}\right)^3 \cdot J_1\left(\mu_m^{(0)}\right) - 2\left(\mu_m^{(0)}\right)^2 J_2\left(\mu_m^{(0)}\right) \right] \\ &= a^4 \frac{\mu_m^{(0)} \cdot J_1\left(\mu_m^{(0)}\right) - 2J_2\left(\mu_m^{(0)}\right)}{\left(\mu_m^{(0)}\right)^2} \end{aligned}$$

3.2 Bessel函数

第一类Bessel函数

例3

证明 $J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \sin x$

$$J_n(x) = \left(\frac{x}{2}\right)^n \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+n+1)} \left(\frac{x}{2}\right)^{2k}$$

证 $J_{\frac{1}{2}}(x) = \left(\frac{x}{2}\right)^{\frac{1}{2}} \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma\left(k + \frac{1}{2} + 1\right)} \left(\frac{x}{2}\right)^{2k}$

按定义式展开

$$= \left(\frac{x}{2}\right)^{\frac{1}{2}} \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \frac{(2k+1)!!}{2^{k+1}} \sqrt{\pi} \cdot 2^{2k}} \cdot x^{2k}$$

$$\Gamma\left(n + \frac{1}{2}\right) = \frac{(2n-1)!!}{2^n} \sqrt{\pi}$$

变型分母

$$= \left(\frac{x}{2}\right)^{\frac{1}{2}} \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \frac{(2k+1)!!}{2^k} \cdot \frac{\sqrt{\pi}}{2} \cdot 2^k} \cdot x^{2k}$$

将 $\frac{\sqrt{\pi}}{2}$ 移出 Σ

$$= \frac{1}{x} \sqrt{\frac{2x}{\pi}} \sum_{k=0}^{\infty} \frac{(-1)^k}{2^k k! (2k+1)!!} x^{2k+1}$$

每项乘以/除以 x

$$= \sqrt{\frac{2}{\pi x}} \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!! (2k+1)!!} x^{2k+1}$$

半阶乘合并为阶乘

$$= \sqrt{\frac{2}{\pi x}} \left[\sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} x^{2k+1} \right] = \sqrt{\frac{2}{\pi x}} \sin x$$

恰为 $\sin(x)$ 的 Taylor 展开

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} x^{2k+1}$$



N.H.Abel
1802-1829
Norway Math.

利用级数定义式的经典证明

3.2 Bessel函数

第一类Bessel函数

例4

证明 $J_2(x) = J_0''(x) - \frac{1}{x} J_0'(x)$

证 $J_0'(x) = \frac{d}{dx} \left[\frac{J_0(x)}{1} \right] = \frac{d}{dx} \left[\frac{J_0(x)}{x^0} \right]$
 $= -\frac{J_1(x)}{x^0} = -J_1(x)$



即

$$-J_0'(x) = J_1(x)$$

两侧除以 x

两侧求导

$$-\frac{J_0'(x)}{x} = \frac{J_1(x)}{x} = \frac{1}{2} [J_0(x) + J_2(x)]$$

$$J_0''(x) = -J_1'(x) = -\frac{1}{2} [J_0(x) - J_2(x)]$$

$$J_2(x) = -\frac{1}{2} [J_0(x) - J_2(x)] + \frac{1}{2} [J_0(x) + J_2(x)]$$

$$= J_0''(x) - \frac{1}{x} J_0'(x)$$

得证

$$J_n(x) = \left(\frac{x}{2}\right)^n \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+n+1)} \left(\frac{x}{2}\right)^{2k}$$

$$J_n(0) = \begin{cases} 1, & (n=0) \\ 0, & (n \geq 1) \end{cases} \quad J_n'(0) = \begin{cases} 0, & (n=0, n>1) \\ \frac{1}{2}, & (n=1) \end{cases}$$

$$J_n(-x) = (-1)^n J_n(x)$$

$$\frac{d}{dx} [x^n J_n(x)] = x^n J_{n-1}(x) \quad \frac{d}{dx} [x^{-n} J_n(x)] = -x^{-n} J_{n+1}(x)$$

$$J_{n-1}(x) + J_{n+1}(x) = \frac{2n}{x} J_n(x) \quad J_{n-1}(x) - J_{n+1}(x) = 2J_n'(x)$$

3.2 Bessel函数

第一类Bessel函数

例5 证明 $\alpha \neq 0$ 时

$$\frac{d}{dx}[J_0(\alpha x)] = \alpha J'_0(\alpha x) = -\alpha J_1(\alpha x)$$

证

$$\begin{aligned} J'_0(\alpha x) &= \frac{d}{dx}[J_0(\alpha x)] \\ &= \frac{d}{dx}\left[\frac{J_0(\alpha x)}{1}\right] \\ &= \alpha \frac{d}{dx}\left[\frac{J_0(\alpha x)}{(\alpha x)^0}\right] \\ &= -\alpha \frac{J_1(\alpha x)}{(\alpha x)^0} = -\alpha J_1(\alpha x) \end{aligned}$$

$$J_n(x) = \left(\frac{x}{2}\right)^n \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+n+1)} \left(\frac{x}{2}\right)^{2k}$$

$$J_n(0) = \begin{cases} 1, & (n=0) \\ 0, & (n \geq 1) \end{cases} \quad J'_n(0) = \begin{cases} 0, & (n=0, n>1) \\ \frac{1}{2}, & (n=1) \end{cases}$$

$$J_n(-x) = (-1)^n J_n(x)$$

$$\frac{d}{dx}[x^n J_n(x)] = x^n J_{n-1}(x) \quad \boxed{\frac{d}{dx}[x^{-n} J_n(x)] = -x^{-n} J_{n+1}(x)}$$

$$J_{n-1}(x) + J_{n+1}(x) = \frac{2n}{x} J_n(x) \quad J_{n-1}(x) - J_{n+1}(x) = 2J'_n(x)$$

$$\int x^{n+1} J_n(x) dx = x^{n+1} J_{n+1}(x) + C, \quad \int \frac{J_n(x)}{x^{n-1}} dx = -\frac{J_{n-1}(x)}{x^{n-1}} + C$$

3.2 Bessel函数

第一类Bessel函数

例5 证明

$$\frac{d}{dx} [xJ_0(x)J_1(x)] = x[J_0^2(x) - J_1^2(x)]$$

证

上例已证明

$$\begin{aligned} \frac{d}{dx} [J_0(\alpha x)] \\ = -\alpha J_1(\alpha x) \end{aligned}$$

$$\begin{aligned} &= J_0'(x)[xJ_1(x)] + J_0(x)[xJ_1(x)]' \\ &= -J_1(x)[xJ_1(x)] + J_0(x)[xJ_0'(x)] \\ &= x[-J_1^2(x) + J_0^2(x)] \end{aligned}$$

$$J_n(x) = \left(\frac{x}{2}\right)^n \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+n+1)} \left(\frac{x}{2}\right)^{2k}$$

$$J_n(0) = \begin{cases} 1, & (n=0) \\ 0, & (n \geq 1) \end{cases} \quad J_n'(0) = \begin{cases} 0, & (n=0, n>1) \\ \frac{1}{2}, & (n=1) \end{cases}$$

$$J_n(-x) = (-1)^n J_n(x)$$

$$\frac{d}{dx} [x^n J_n(x)] = x^n J_{n-1}(x) \quad \frac{d}{dx} [x^{-n} J_n(x)] = -x^{-n} J_{n+1}(x)$$

$$J_{n-1}(x) + J_{n+1}(x) = \frac{2n}{x} J_n(x) \quad J_{n-1}(x) - J_{n+1}(x) = 2J_n'(x)$$

$$\int x^{n+1} J_n(x) dx = x^{n+1} J_{n+1}(x) + C, \quad \int \frac{J_n(x)}{x^{n-1}} dx = -\frac{J_{n-1}(x)}{x^{n-1}} + C$$

3.2 Bessel函数

几个常用的推论

$$\textcircled{1} \quad \frac{d}{dx} [J_0(\alpha x)] = -\alpha J_1(\alpha x)$$

$$\textcircled{2} \quad \int x \cdot J_0(x) dx = x \cdot J_1(x) + C$$

$$\textcircled{3} \quad \int x^3 \cdot J_0(x) dx$$

$$= \int x^2 \cdot x \cdot J_0(x) dx$$

$$= \int x^2 \cdot d[x \cdot J_1(x)]$$

$$= x^3 J_1(x) - 2 \int x^2 \cdot J_1(x) dx$$

$$= x^3 J_1(x) - 2x^2 \cdot J_2(x) + C$$

$$= x^3 J_1(x) - 2x^2 \cdot \left[\frac{2}{x} J_1(x) - J_0(x) \right] + C$$

$$= x(x^2 - 4) J_1(x) + 2x^2 J_0(x) + C$$

$$J_{n+1}(x) = \frac{2n}{x} J_n(x) - J_{n-1}(x)$$

后续Bessel展开中反复使用

$$\frac{d}{dx} [J_0(\alpha x)] = -\alpha J_1(\alpha x)$$

$$\int x \cdot J_0(x) dx = x \cdot J_1(x) + C$$

$$\int x^3 \cdot J_0(x) dx = x^3 J_1(x) - 2x^2 \cdot J_2(x) + C$$

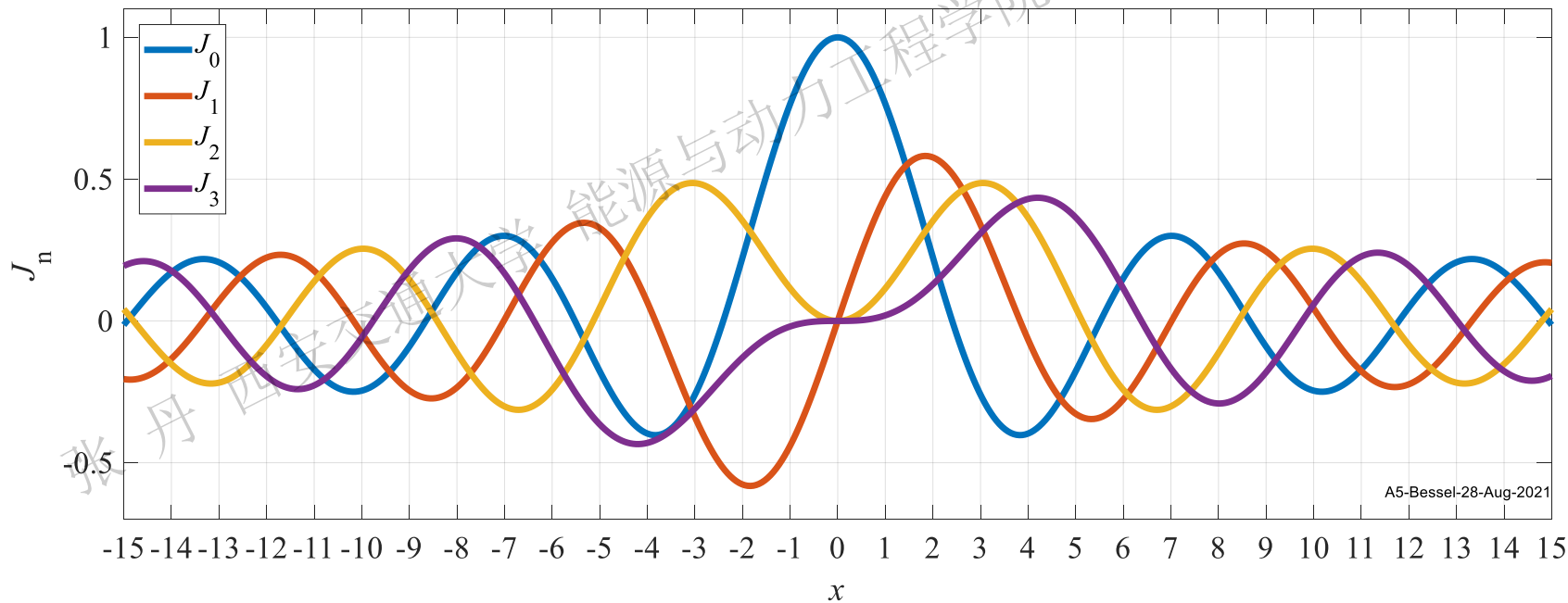
$$= x(x^2 - 4) J_1(x) + 2x^2 J_0(x) + C$$

3.2 Bessel函数

第一类Bessel函数

例6 求方程 $J'_0(x) = 0$ 的根

解 根据 $J'_0(x) = -J_1(x) = 0$ 易知方程的根为 $0, \pm \mu_m^{(1)}, (m=1, 2, 3, \dots)$



1. 数学建模和基本原理介绍

2. 分离变量法

3. Bessel函数

4. 积分变换法

5. Green函数法

6. 特征线法

7. Legendre多项式

3.1 二阶线性常微分方程的幂级数解法

3.2 Bessel函数

3.3 多个自变量分离变量法举例

Gamma函数

Bessel方程和Bessel函数

Bessel函数的性质

Bessel方程的特征值问题

圆域上Laplace算子的特征值问题

3.2 Bessel函数

极坐标下 $-\Delta w(\rho, \theta) = \mu w(\rho, \theta)$ 展开中几个特征值的关系

周/径向特征值问题

$$\frac{\partial^2 u}{\partial t^2} - a^2 \Delta u = 0 \quad u(\rho, \theta, t) = T(t)w(\rho, \theta)$$

$$T''(t)w(\rho, \theta) - a^2 T(t) \Delta w(\rho, \theta) = 0$$

时空分离
常数

$$\frac{T''(t)}{a^2 T(t)} = \frac{\Delta w(\rho, \theta)}{w(\rho, \theta)} = -\mu$$

二维边值问题/圆域- Δ 算子特征值问题

初值问题

$$T''(t) + \mu a^2 T(t) = 0$$

$$\begin{cases} \Delta w(\rho, \theta) + \mu w(\rho, \theta) = 0, (\rho, \theta) \in \Omega \\ w|_{\partial\Omega} = 0 \end{cases}$$

$$w(\rho, \theta) = R(\rho)\Phi(\theta)$$

$$\Delta w = \frac{\partial^2 w}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial w}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 w}{\partial \theta^2}$$

$$\begin{aligned} \lambda_0 &= 0, \quad \Phi_0(\theta) = 1, \\ (n=0) \\ \lambda_n &= n^2, \\ \Phi_n(\theta) &= C_1 \cos n\theta \\ &\quad + C_2 \sin n\theta, \\ (n=1, 2, 3, \dots) \end{aligned}$$

$$R''(\rho)\Phi(\theta) + \frac{1}{\rho} R'(\rho)\Phi(\theta) + \frac{1}{\rho^2} R(\rho)\Phi''(\theta) = -\mu R(\rho)\Phi(\theta)$$

$$\frac{\Phi''(\theta)}{\Phi(\theta)} = -\frac{\mu R(\rho) + R''(\rho) + \frac{1}{\rho} R'(\rho)}{\frac{1}{\rho^2} R(\rho)} = -\lambda$$

空间分离
常数

周向

径向/Bessel方程特征值问题

$$\begin{cases} \Phi''(\theta) + \lambda \Phi(\theta) = 0 \\ \Phi(\theta) = \Phi(\theta + 2\pi) \end{cases} \quad \begin{cases} |R(0)| < +\infty, R(\rho_0) = 0 \\ \rho^2 R''(\rho) + \rho R'(\rho) + (\mu \rho^2 - \lambda) R(\rho) = 0 \end{cases}$$

- 1. 圆域内- Δ 算子特征值 $\mu \rightarrow$ 周向/径向联合确定 $\rightarrow$ 如何关联的呢?
- 2. 周向: Δ 算子特征值问题/周期性边界 $\rightarrow$ 可直接求出特征值/特征函数
- 3. 径向: ODE能否写成 $AX(x) = \lambda X(x)$ 的形式? $\rightarrow$ 给定边界是否有特征值/特征函数?

3.2 Bessel函数

周/径向特征值问题

$$\begin{cases} \Delta w(\rho, \theta) + \mu w(\rho, \theta) = 0, & (\rho, \theta) \in \Omega \\ w|_{\partial\Omega} = 0 \end{cases}$$

分离变量 $\rightarrow$

二维边值问题/圆域- Δ 算子特征值问题

径向

$$-\frac{\mu R(\rho) + R''(\rho) + \frac{1}{\rho} R'(\rho)}{\frac{1}{\rho^2} R(\rho)} = -\lambda = \frac{\Phi''(\theta)}{\Phi(\theta)}$$
$$-\left[R''(\rho) + \frac{1}{\rho} R'(\rho) - \frac{\lambda}{\rho^2} R(\rho) \right] = \mu R(\rho)$$

定义运算

$$A_\rho = -\left[\frac{\partial^2}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial}{\partial \rho} - \frac{\lambda}{\rho^2} \right]$$

$\rightarrow A_\rho [R(\rho)] = \mu R(\rho)$ 特征值定义

周向 $\rightarrow \lambda = n^2$ 周向特征值

展开成为Bessel方程

$$\begin{cases} \rho^2 R''(\rho) + \rho R'(\rho) + (\mu \rho^2 - n^2) R(\rho) = 0 \\ \text{圆心自然+外缘1齐 or 圆心自然+外缘2齐} \end{cases}$$

Bessel方程特征值
圆域- Δ 的特征值
时空分离常数

- 1. 径向ODE 符合特征值定义 $\rightarrow$ 解径向ODE = 求Bessel方程特征值问题 $\rightarrow$ 本质: 寻找给定边界 $\rightarrow$ Bessel方程的非零解
- 2. 径向/周向 $\rightarrow$ 两个维度的特征值 $\rightarrow$ 通过Bessel方程零点 $\mu_m^{(n)}$ 关联

3.2 Bessel函数

Bessel方程特征值问题

$$\rho^2 R''(\rho) + \rho R'(\rho) + (\mu \rho^2 - n^2) R(\rho) = 0$$

圆心自然+外缘1齐 or 圆心自然+外缘2齐

第一类齐次边界

特征值符号变了!!
(0 < ρ < ρ₀)

$$\begin{cases} \rho^2 R''(\rho) + \rho R'(\rho) + (\lambda \rho^2 - n^2) R(\rho) = 0 \\ R(\rho_0) = 0, \quad |R(0)| < +\infty \end{cases}$$

特征值/特征函数

Bessel方程特征值改用λ表示

$$\lambda_m = \left(\frac{\mu_m^{(n)}}{\rho_0}\right)^2, \quad R_m(\rho) = J_n\left(\frac{\mu_m^{(n)}}{\rho_0} \rho\right), \quad n \geq 0, \quad m \geq 1$$

两组序号

正交性→加权正交

$$\begin{aligned} &\int_0^{\rho_0} \rho R_m(\rho) R_k(\rho) d\rho \\ &= \int_0^{\rho_0} \rho J_n\left(\frac{\mu_m^{(n)}}{\rho_0} \rho\right) \cdot J_n\left(\frac{\mu_k^{(n)}}{\rho_0} \rho\right) d\rho \\ &= \begin{cases} \frac{\rho_0^2}{2} \left[J_n'(\mu_m^{(n)})\right]^2, & (m = k) \\ 0, & (m \neq k) \end{cases} \end{aligned}$$

第二类齐次边界

$$\begin{cases} \rho^2 R''(\rho) + \rho R'(\rho) + (\lambda \rho^2 - n^2) R(\rho) = 0 \\ R_\rho(\rho_0) = 0, \quad |R(0)| < +\infty \end{cases}$$

特征值/特征函数

$$\begin{cases} n = 0, & \begin{cases} \lambda_0 = 0, R_0(\rho) = 1, m = 0 \\ \lambda_m = \left(\frac{\alpha_m}{\rho_0}\right)^2, R_m(\rho) = J_0\left(\frac{\alpha_m}{\rho_0} \rho\right), m \geq 1 \end{cases} \\ n \geq 1, & \lambda_m = \left(\frac{\alpha_m}{\rho_0}\right)^2, R_m(\rho) = J_n\left(\frac{\alpha_m}{\rho_0} \rho\right), m \geq 1 \end{cases}$$

其中 α_m (m ≥ 1) 为 J'_n(x) 的正零点

正交性→加权正交

$$\int_0^{\rho_0} \rho R_m(\rho) R_k(\rho) d\rho = \begin{cases} \frac{\rho_0^2}{2} \left[1 - \frac{n^2}{\alpha_m^2}\right] J_n^2(\alpha_m), & (m = k) \\ 0, & (m \neq k) \end{cases}$$

3.2 Bessel函数

Bessel方程特征值问题

第一类边界

$$\begin{cases} \rho^2 R''(\rho) + \rho R'(\rho) + (\lambda \rho^2 - n^2) R(\rho) = 0, & (0 < \rho < \rho_0) \\ R(\rho_0) = 0, & |R(0)| < +\infty \end{cases}$$

周向外缘齐次

自然边界条件

定理1 特征值/特征函数

$$\lambda_m = \left(\frac{\mu_m^{(n)}}{\rho_0} \right)^2, \quad R_m(\rho) = J_n \left(\frac{\mu_m^{(n)}}{\rho_0} \rho \right), \quad n \geq 0, \quad m \geq 1$$

正交性

$$\int_0^{\rho_0} \rho R_m(\rho) R_k(\rho) d\rho = \int_0^{\rho_0} \rho J_n \left(\frac{\mu_m^{(n)}}{\rho_0} \rho \right) \cdot J_n \left(\frac{\mu_k^{(n)}}{\rho_0} \rho \right) d\rho = \begin{cases} \frac{\rho_0^2}{2} \left[J_n' \left(\mu_m^{(n)} \right) \right]^2, & (m = k) \\ 0, & (m \neq k) \end{cases}$$

权函数加权正交

平方模

完备性 设 $f(\rho)$ 在 $[0, \rho_0]$ 连续, 且有分段连续的一阶导数, 则 $f(\rho)$ 在该区间上可按特征函数系展开成Fourier-Bessel级数

同阶Bessel函数正交/完备

$$f(\rho) = \sum_{m=1}^{\infty} A_m R_m(\rho) = \sum_{m=1}^{\infty} A_m J_n \left(\frac{\mu_m^{(n)}}{\rho_0} \rho \right)$$
$$A_m = \frac{1}{\frac{1}{2} \left[\rho_0 J_n' \left(\mu_m^{(n)} \right) \right]^2} \cdot \int_0^{\rho_0} \rho f(\rho) J_n \left(\frac{\mu_m^{(n)}}{\rho_0} \rho \right) d\rho$$

$n \geq 0$ 周向特征值/Bessel函数阶数
 $m \geq 1$ Bessel函数正零点序号/级数项序号
 $\mu_m^{(n)}$ n 阶Bessel方程第 m 个正零点

3.2 Bessel函数

Bessel方程特征值问题

第一类边界

$$\begin{cases} \rho^2 R''(\rho) + \rho R'(\rho) + (\lambda \rho^2 - n^2) R(\rho) = 0, & (0 < \rho < \rho_0) \\ R(\rho_0) = 0, \quad |R(0)| < +\infty \end{cases}$$

思路 特征值→正交性→平方模

证明 1 特征值 $\lambda > 0$

框定特征值的范围

$$\rho^2 R''(\rho) + \rho R'(\rho) + (\lambda \rho^2 - n^2) R(\rho) = 0$$

两边同乘以 $\frac{1}{\rho}$

$$\rho R''(\rho) + R'(\rho) + \left(\lambda \rho - \frac{n^2}{\rho} \right) R(\rho) = 0$$

求导合并

$$[\rho R'(\rho)]' + \left(\lambda \rho - \frac{n^2}{\rho} \right) R(\rho) = 0$$

重要等式
后续证明起点

两边同乘以 $R(\rho)$

$$R(\rho) [\rho R'(\rho)]' + \lambda \rho R^2(\rho) - \frac{n^2}{\rho} R^2(\rho) = 0$$

$$\int_0^{\rho_0} R(\rho) [\rho R'(\rho)]' d\rho + \lambda \int_0^{\rho_0} \rho R^2(\rho) d\rho - n^2 \int_0^{\rho_0} \frac{R^2(\rho)}{\rho} d\rho = 0$$

两边积分 $[0, \rho_0]$

$$R(\rho) \rho R'(\rho) \Big|_0^{\rho_0} - \int_0^{\rho_0} \rho R'(\rho) \cdot R'(\rho) d\rho + \lambda \int_0^{\rho_0} \rho R^2(\rho) d\rho - n^2 \int_0^{\rho_0} \frac{R^2(\rho)}{\rho} d\rho = 0$$

=0

$$\lambda = \frac{\int_0^{\rho_0} \left\{ \rho [R'(\rho)]^2 + n^2 \frac{R^2(\rho)}{\rho} \right\} d\rho}{\int_0^{\rho_0} \rho R^2(\rho) d\rho} \geq 0$$

3.2 Bessel函数

Bessel方程特征值问题

第一类边界

$$\begin{cases} \rho^2 R''(\rho) + \rho R'(\rho) + (\lambda \rho^2 - n^2) R(\rho) = 0, & (0 < \rho < \rho_0) \\ R(\rho_0) = 0, & |R(0)| < +\infty \end{cases}$$

证明 1 特征值 $\lambda > 0$

$$\lambda = \frac{\int_0^{\rho_0} \left\{ \rho [R'(\rho)]^2 + n^2 \frac{R^2(\rho)}{\rho} \right\} d\rho}{\int_0^{\rho_0} \rho R^2(\rho) d\rho} \geq 0$$

λ 可否=0 呢?

若 $\lambda = 0$, 有 $\rho [R'(\rho)]^2 + n^2 \frac{R^2(\rho)}{\rho} = 0$

需同时满足 $\begin{cases} \textcircled{1} n = 0 \\ \textcircled{2} R'(\rho) = 0 \end{cases}$ 即 $R(\rho) = C$

由边界条件 $R(\rho_0) = 0$ 只可能有 $R(\rho) = C = 0$

$\lambda = 0$ 不是特征值

故需满足 $\lambda > 0$

1. Bessel方程第1类齐次边界下特征值 λ 只能为正数
2. 实际计算时, n 可以=0

3.2 Bessel函数

Bessel方程特征值问题

第一类边界

$$\begin{cases} \rho^2 R''(\rho) + \rho R'(\rho) + (\lambda \rho^2 - n^2) R(\rho) = 0, & (0 < \rho < \rho_0) \\ R(\rho_0) = 0, & |R(0)| < +\infty \end{cases}$$

证明

2 求出特征值

变量代换

各阶
导数

$$x = \sqrt{\lambda} \rho, \quad \rho = \frac{x}{\sqrt{\lambda}} \quad y(x) = R(\rho) = R\left(\frac{x}{\sqrt{\lambda}}\right)$$
$$y'(x) = R'(\rho) \frac{d\rho}{dx} = \frac{1}{\sqrt{\lambda}} R'(\rho)$$
$$y''(x) = \frac{1}{\sqrt{\lambda}} R''(\rho) \frac{d\rho}{dx} = \frac{1}{\lambda} R''(\rho)$$

代入原方程

$$\rho^2 R''(\rho) + \rho R'(\rho) + (\lambda \rho^2 - n^2) R(\rho) = 0$$
$$\frac{x^2}{\lambda} \lambda y''(x) + \frac{x}{\sqrt{\lambda}} \sqrt{\lambda} y'(x) + \left(\lambda \frac{x^2}{\lambda} - n^2 \right) y(x) = 0$$
$$x^2 y''(x) + x y'(x) + (x^2 - n^2) y(x) = 0$$

标准 n 阶Bessel方程

n 为整数 $\rightarrow$ 通解 $y(x) = C_1 J_n(x) + C_2 N_n(x)$

还原变量 $R(\rho) = C_1 J_n(\sqrt{\lambda} \rho) + C_2 N_n(\sqrt{\lambda} \rho)$

由自然边界 $|R(0)| < +\infty$ 知 $C_2 = 0$

由外缘1齐 $R(\rho_0) = 0 = C_1 J_n(\sqrt{\lambda} \rho_0)$

为取得非零解, $\sqrt{\lambda} \rho_0$ 应取 J_n 的正零点

$$\sqrt{\lambda} \rho_0 = \mu_m^{(n)}, \quad (m \geq 1)$$

对应无数个零点 $\rightarrow$ 无数个特征值/特征函数

$$\lambda_m = \left(\frac{\mu_m^{(n)}}{\rho_0} \right)^2, \quad R_m(\rho) = J_n \left(\frac{\mu_m^{(n)}}{\rho_0} \rho \right), \quad (m \geq 1)$$

序号为 m

$\lambda \neq 0 \rightarrow m$
从1开始

3.2 Bessel函数

Bessel方程特征值问题

第一类边界

$$\begin{cases} \rho^2 R''(\rho) + \rho R'(\rho) + (\lambda \rho^2 - n^2) R(\rho) = 0, & (0 < \rho < \rho_0) \\ R(\rho_0) = 0, & |R(0)| < +\infty \end{cases}$$

$$\lambda_m = \left(\frac{\mu_m^{(n)}}{\rho_0} \right)^2, \quad R_m(\rho) = J_n \left(\frac{\mu_m^{(n)}}{\rho_0} \rho \right), \quad (m \geq 1)$$

证明 3 正交性 特征方程可推出

$$\rho^2 R''(\rho) + \rho R'(\rho) + (\lambda \rho^2 - n^2) R(\rho) = 0$$

$$\rho R''(\rho) + R'(\rho) + \left(\lambda \rho - \frac{n^2}{\rho} \right) R(\rho) = 0$$

$$[\rho R'(\rho)]' + \left(\lambda \rho - \frac{n^2}{\rho} \right) R(\rho) = 0$$

$$\begin{aligned} R_\beta(\rho) \cdot \textcircled{1} - R_\alpha(\rho) \cdot \textcircled{2} &= \\ R_\beta(\rho) [\rho R'_\alpha(\rho)]' - R_\alpha(\rho) [\rho R'_\beta(\rho)]' &+ (\lambda_\alpha - \lambda_\beta) \rho R_\alpha(\rho) R_\beta(\rho) = 0 \end{aligned}$$

特征函数系 $\{R_m(\rho) | m \geq 1\}$ 中任取两个函数

$R_\alpha(\rho), R_\beta(\rho), (\alpha \neq \beta)$ 均满足上述等式

$$[\rho R'_\alpha(\rho)]' + \left(\lambda_\alpha \rho - \frac{n^2}{\rho} \right) R_\alpha(\rho) = 0 \quad \textcircled{1}$$

$$[\rho R'_\beta(\rho)]' + \left(\lambda_\beta \rho - \frac{n^2}{\rho} \right) R_\beta(\rho) = 0 \quad \textcircled{2}$$

3.2 Bessel函数

Bessel方程特征值问题

第一类边界

$$\begin{cases} \rho^2 R''(\rho) + \rho R'(\rho) + (\lambda \rho^2 - n^2) R(\rho) = 0, & (0 < \rho < \rho_0) \\ R(\rho_0) = 0, & |R(0)| < +\infty \end{cases}$$

$$\lambda_m = \left(\frac{\mu_m^{(n)}}{\rho_0}\right)^2, \quad R_m(\rho) = J_n\left(\frac{\mu_m^{(n)}}{\rho_0} \rho\right), \quad (m \geq 1)$$

证明

3 正交性

$$R_\beta(\rho) [\rho R'_\alpha(\rho)]' - R_\alpha(\rho) [\rho R'_\beta(\rho)]' + (\lambda_\alpha - \lambda_\beta) \rho R_\alpha(\rho) R_\beta(\rho) = 0$$

移项 $(\lambda_\alpha - \lambda_\beta) \rho R_\alpha(\rho) R_\beta(\rho) = R_\alpha(\rho) [\rho R'_\beta(\rho)]' - R_\beta(\rho) [\rho R'_\alpha(\rho)]'$

两端积分

$$(\lambda_\alpha - \lambda_\beta) \int_0^{\rho_0} \rho R_\alpha(\rho) R_\beta(\rho) d\rho = \int_0^{\rho_0} R_\alpha(\rho) [\rho R'_\beta(\rho)]' d\rho - \int_0^{\rho_0} R_\beta(\rho) [\rho R'_\alpha(\rho)]' d\rho$$

右端积分 $\int_0^{\rho_0} R_\alpha(\rho) d[\rho R'_\beta(\rho)] - \int_0^{\rho_0} R_\beta(\rho) d[\rho R'_\alpha(\rho)]$

$= R_\alpha(\rho) \rho R'_\beta(\rho) \Big|_0^{\rho_0} - \int_0^{\rho_0} \rho R'_\beta(\rho) R'_\alpha(\rho) d\rho - R_\beta(\rho) \rho R'_\alpha(\rho) \Big|_0^{\rho_0} + \int_0^{\rho_0} \rho R'_\alpha(\rho) R'_\beta(\rho) d\rho$

$= \rho_0 R_\alpha(\rho_0) R'_\beta(\rho_0) - \rho_0 R_\beta(\rho_0) R'_\alpha(\rho_0)$

$= \rho_0 J_n\left(\frac{\mu_\alpha^{(n)}}{\rho_0} \rho_0\right) R'_\beta(\rho_0) - \rho_0 J_n\left(\frac{\mu_\beta^{(n)}}{\rho_0} \rho_0\right) R'_\alpha(\rho_0) = 0$

由 $\lambda_\alpha \neq \lambda_\beta$ 可知 $\int_0^{\rho_0} \rho R_\alpha(\rho) R_\beta(\rho) d\rho = 0$, 即特征函数 $R_m(\rho)$ 是带权函数 ρ 正交的

3.2 Bessel函数

Bessel方程特征值问题

第一类边界

$$\begin{cases} \rho^2 R''(\rho) + \rho R'(\rho) + (\lambda \rho^2 - n^2) R(\rho) = 0, & (0 < \rho < \rho_0) \\ R(\rho_0) = 0, \quad |R(0)| < +\infty \end{cases}$$

$$\lambda_m = \left(\frac{\mu_m^{(n)}}{\rho_0}\right)^2, \quad R_m(\rho) = J_n\left(\frac{\mu_m^{(n)}}{\rho_0} \rho\right), \quad (m \geq 1)$$

证明 4 求平方模 借鉴了摄动法思想

记 $\alpha = \sqrt{\lambda_m} = \frac{\mu_m^{(n)}}{\rho_0}$, $R_m(\rho) = J_n\left(\frac{\mu_m^{(n)}}{\rho_0} \rho\right) = J_n(\alpha \rho)$, $(m \geq 1)$ 另取 β , 并使 $|\beta - \alpha| = \left|\beta - \frac{\mu_m^{(n)}}{\rho_0}\right| \ll 1$

$R_\alpha(\rho) = J_n\left(\frac{\mu_m^{(n)}}{\rho_0} \rho\right) = J_n(\alpha \rho)$

是解

$$\begin{cases} \rho^2 R_\alpha''(\rho) + \rho R_\alpha'(\rho) + (\alpha^2 \rho^2 - n^2) R_\alpha(\rho) = 0, & (0 < \rho < \rho_0) \\ R_\alpha(\rho_0) = 0, \quad |R_\alpha(0)| < +\infty \end{cases}$$

是特征函数

$$[\rho R'_\alpha(\rho)]' + \left(\alpha^2 \rho - \frac{n^2}{\rho}\right) R_\alpha(\rho) = 0$$
$$[\rho R'_\beta(\rho)]' + \left(\beta^2 \rho - \frac{n^2}{\rho}\right) R_\beta(\rho) = 0$$

方程可改写为

不是特征函数

$$R_\beta(\rho) = J_n(\beta \rho) \neq 0$$

是方程解/但不满足外缘1齐

$$\begin{cases} \rho^2 R''(\rho) + \rho R'(\rho) + (\beta^2 \rho^2 - n^2) R(\rho) = 0, & (0 < \rho < \rho_0) \\ R(\rho_0) \neq 0, \quad |R(0)| < +\infty \end{cases}$$

3.2 Bessel函数

Bessel方程特征值问题

第一类边界

$$\begin{cases} \rho^2 R''(\rho) + \rho R'(\rho) + (\lambda \rho^2 - n^2) R(\rho) = 0, & (0 < \rho < \rho_0) \\ R(\rho_0) = 0, \quad |R(0)| < +\infty \end{cases}$$

$$\lambda_m = \left(\frac{\mu_m^{(n)}}{\rho_0} \right)^2, \quad R_m(\rho) = J_n \left(\frac{\mu_m^{(n)}}{\rho_0} \rho \right), \quad (m \geq 1)$$

$$R_\alpha(\rho) = J_n \left(\frac{\mu_m^{(n)}}{\rho_0} \rho \right) = J_n(\alpha \rho), \quad R_\beta(\rho) = J_n \left(\frac{\mu_m^{(n)}}{\rho_0} \rho \right) = J_n(\beta \rho)$$

证明

4 求平方模

$$[\rho R'_\alpha(\rho)]' + \left(\alpha^2 \rho - \frac{n^2}{\rho} \right) R_\alpha(\rho) = 0$$
$$[\rho R'_\beta(\rho)]' + \left(\beta^2 \rho - \frac{n^2}{\rho} \right) R_\beta(\rho) = 0$$

$$R_\beta(\rho)$$
$$R_\alpha(\rho)$$

$$R_\beta(\rho) [\rho R'_\alpha(\rho)]' + \left(\alpha^2 \rho - \frac{n^2}{\rho} \right) R_\alpha(\rho) R_\beta(\rho) = 0$$
$$R_\alpha(\rho) [\rho R'_\beta(\rho)]' + \left(\beta^2 \rho - \frac{n^2}{\rho} \right) R_\alpha(\rho) R_\beta(\rho) = 0$$

两端乘以

两式相减

两端在 $[0, \rho_0]$ 积分

$$R_\beta(\rho) [\rho R'_\alpha(\rho)]' - R_\alpha(\rho) [\rho R'_\beta(\rho)]' + (\alpha^2 - \beta^2) \rho R_\alpha(\rho) R_\beta(\rho) = 0$$

$$(\alpha^2 - \beta^2) \int_0^{\rho_0} \rho R_\alpha(\rho) R_\beta(\rho) d\rho = \int_0^{\rho_0} R_\alpha(\rho) [\rho R'_\beta(\rho)]' d\rho - \int_0^{\rho_0} R_\beta(\rho) [\rho R'_\alpha(\rho)]' d\rho$$

$$= \int_0^{\rho_0} R_\alpha(\rho) d[\rho R'_\beta(\rho)] - \int_0^{\rho_0} R_\beta(\rho) d[\rho R'_\alpha(\rho)]$$

$$= \rho R_\alpha(\rho) R'_\beta(\rho) \Big|_0^{\rho_0} - \int_0^{\rho_0} \rho R'_\beta(\rho) R'_\alpha(\rho) d\rho$$

$$= \rho R_\alpha(\rho) R'_\beta(\rho) \Big|_0^{\rho_0} - \int_0^{\rho_0} \rho R'_\alpha(\rho) R'_\beta(\rho) d\rho$$

$$= \rho_0 R_\alpha(\rho_0) R'_\beta(\rho_0) - \rho_0 R_\beta(\rho_0) R'_\alpha(\rho_0)$$

$$= -\rho_0 R'_\alpha(\rho_0) R_\beta(\rho_0)$$

$$\int_0^{\rho_0} \rho R_\alpha(\rho) R_\beta(\rho) d\rho = \frac{-\rho_0 R'_\alpha(\rho_0) R_\beta(\rho_0)}{\alpha^2 - \beta^2}$$

3.2 Bessel函数

Bessel方程特征值问题

第一类边界

$$\begin{cases} \rho^2 R''(\rho) + \rho R'(\rho) + (\lambda \rho^2 - n^2) R(\rho) = 0, & (0 < \rho < \rho_0) \\ R(\rho_0) = 0, & |R(0)| < +\infty \end{cases}$$

$$\lambda_m = \left(\frac{\mu_m^{(n)}}{\rho_0} \right)^2, \quad R_m(\rho) = J_n \left(\frac{\mu_m^{(n)}}{\rho_0} \rho \right), \quad (m \geq 1)$$

证明

4 求平方模

$\beta \rightarrow \alpha$ 的极限
即为平方模

$$R_\alpha(\rho) = J_n \left(\frac{\mu_m^{(n)}}{\rho_0} \rho \right) = J_n(\alpha \rho), \quad R_\beta(\rho) = J_n \left(\frac{\mu_m^{(n)}}{\rho_0} \rho \right) = J_n(\beta \rho)$$

$$\int_0^{\rho_0} \rho R_\alpha(\rho) R_\beta(\rho) d\rho = \frac{-\rho_0 R'_\alpha(\rho_0) R_\beta(\rho_0)}{\alpha^2 - \beta^2} = \frac{-\rho_0 \alpha J'_n(\alpha \rho_0) J_n(\beta \rho_0)}{\alpha^2 - \beta^2}$$

$$\int_0^{\rho_0} \rho R_\alpha^2(\rho) d\rho = \lim_{\beta \rightarrow \alpha} \int_0^{\rho_0} \rho R_\alpha(\rho) R_\beta(\rho) d\rho$$

β 是变量

$$= \lim_{\beta \rightarrow \alpha} \frac{\rho_0 \alpha J'_n(\alpha \rho_0) J_n(\beta \rho_0)}{\beta^2 - \alpha^2}$$

$$= \lim_{\beta \rightarrow \alpha} \frac{\rho_0^2 \alpha J'_n(\alpha \rho_0) J'_n(\beta \rho_0)}{2\beta}$$

L'Hospital 法则
对 β 求导

$$= \frac{\rho_0^2 \alpha J'_n(\alpha \rho_0) J'_n(\alpha \rho_0)}{2\alpha}$$

此时 $\beta = \alpha$

$$= \frac{\rho_0^2 \alpha J'_n(\alpha \rho_0) J'_n(\alpha \rho_0)}{2\alpha}$$

$$= \frac{\rho_0^2}{2} [J'_n(\alpha \rho_0)]^2$$

$\alpha = \sqrt{\lambda_m} = \frac{\mu_m^{(n)}}{\rho_0}$

平方模

$$\int_0^{\rho_0} \rho R_\alpha^2(\rho) d\rho = \frac{\rho_0^2}{2} \left[J'_n \left(\mu_m^{(n)} \right) \right]^2$$

3.2 Bessel函数

Bessel方程特征值问题

例1 求如下Bessel方程的特征值问题

$$\begin{cases} \rho^2 R''(\rho) + \rho R'(\rho) + (\lambda \rho^2 - 4) R(\rho) = 0, & (0 < \rho < \rho_0) \\ R(\rho_0) = 0, & |R(0)| < +\infty \end{cases}$$

解 变量代换 $x = \sqrt{\lambda} \rho, \rho = \frac{x}{\sqrt{\lambda}}$

各阶导数

$$\begin{aligned} y(x) &= R(\rho) = R\left(\frac{x}{\sqrt{\lambda}}\right) \\ y'(x) &= R'(\rho) \frac{d\rho}{dx} = \frac{1}{\sqrt{\lambda}} R'(\rho) \\ y''(x) &= \frac{1}{\sqrt{\lambda}} R''(\rho) \frac{d\rho}{dx} = \frac{1}{\lambda} R''(\rho) \end{aligned}$$

代入原方程

$$\begin{aligned} \rho^2 R''(\rho) + \rho R'(\rho) + (\lambda \rho^2 - 4) R(\rho) &= 0 \\ \frac{x^2}{\lambda} \lambda y''(x) + \frac{x}{\sqrt{\lambda}} \sqrt{\lambda} y'(x) + \left(\lambda \frac{x^2}{\lambda} - 2^2 \right) y(x) &= 0 \\ x^2 y''(x) + x y'(x) + (x^2 - 2^2) y(x) &= 0 \end{aligned}$$

通解 $y(x) = C_1 J_2(x) + C_2 N_2(x)$

还原变量 $R(\rho) = C_1 J_2(\sqrt{\lambda} \rho) + C_2 N_2(\sqrt{\lambda} \rho)$

由自然边界 $|R(0)| < +\infty \rightarrow C_2 = 0$

由外缘1齐 $R(\rho_0) = C_1 J_2(\sqrt{\lambda} \rho_0) = 0$

为取得非零解, $\sqrt{\lambda} \rho_0$ 应取 $J_2(x)$ 的正零点

$$\sqrt{\lambda} \rho_0 = \mu_m^{(2)}, \quad (m \geq 1)$$

得特征值/特征函数

$$\lambda_m = \left(\frac{\mu_m^{(2)}}{\rho_0} \right)^2, \quad R_m(\rho) = J_2\left(\frac{\mu_m^{(2)}}{\rho_0} \rho \right), \quad (m \geq 1)$$